Additional file 2

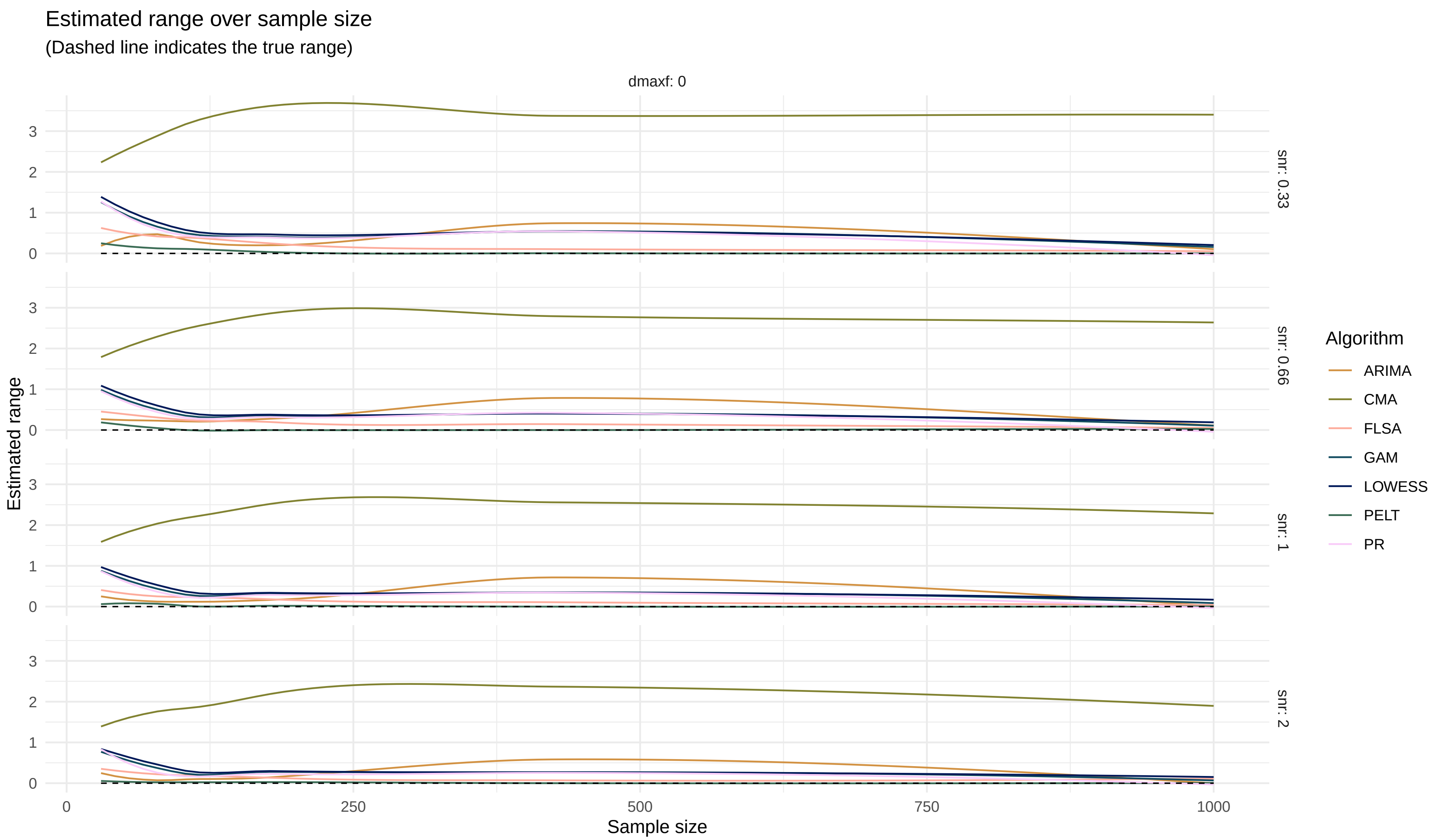
for
Assessing systematic measurement changes in single-wave study data: a simulation study

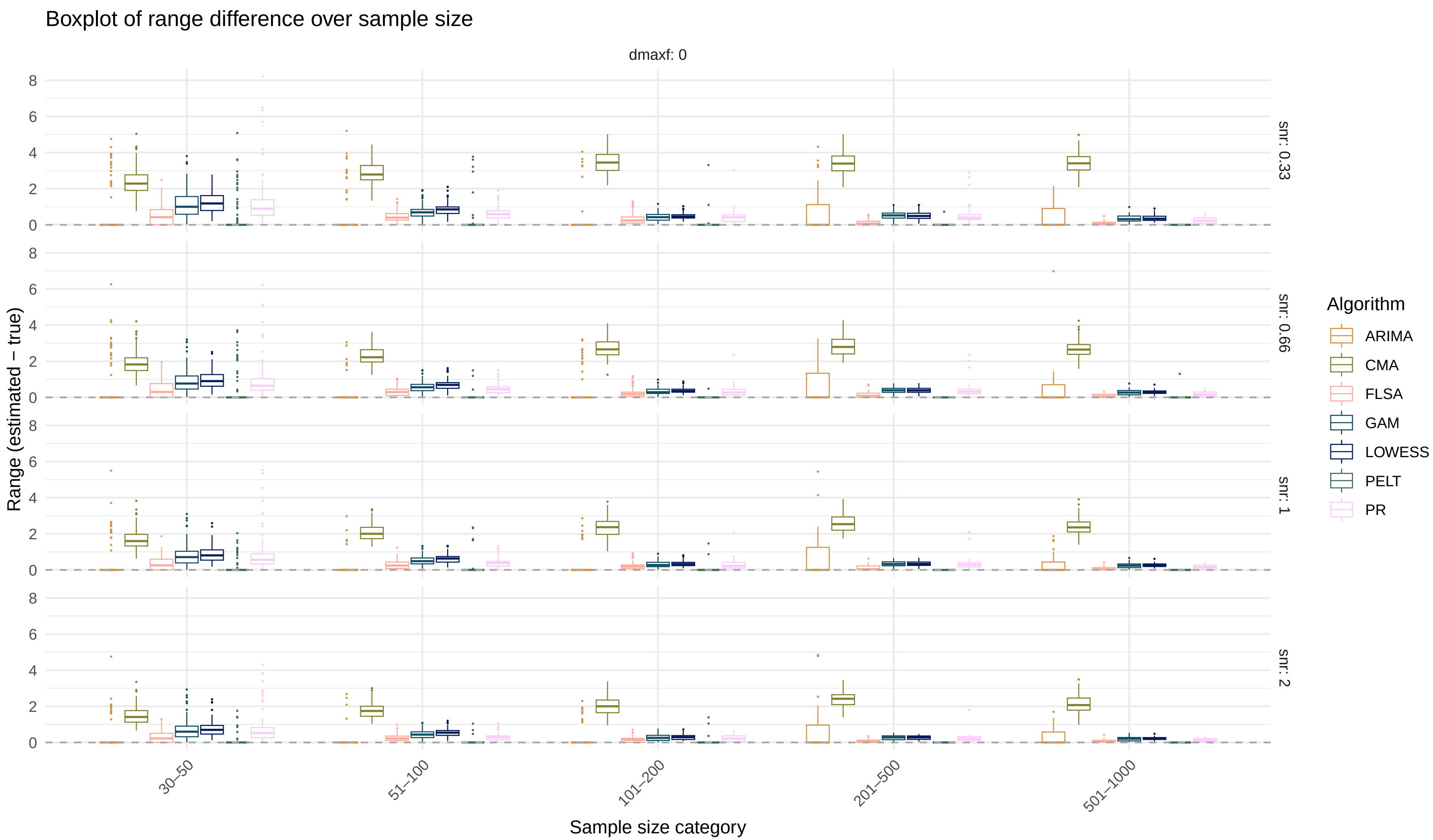
Contents

Normal distribution

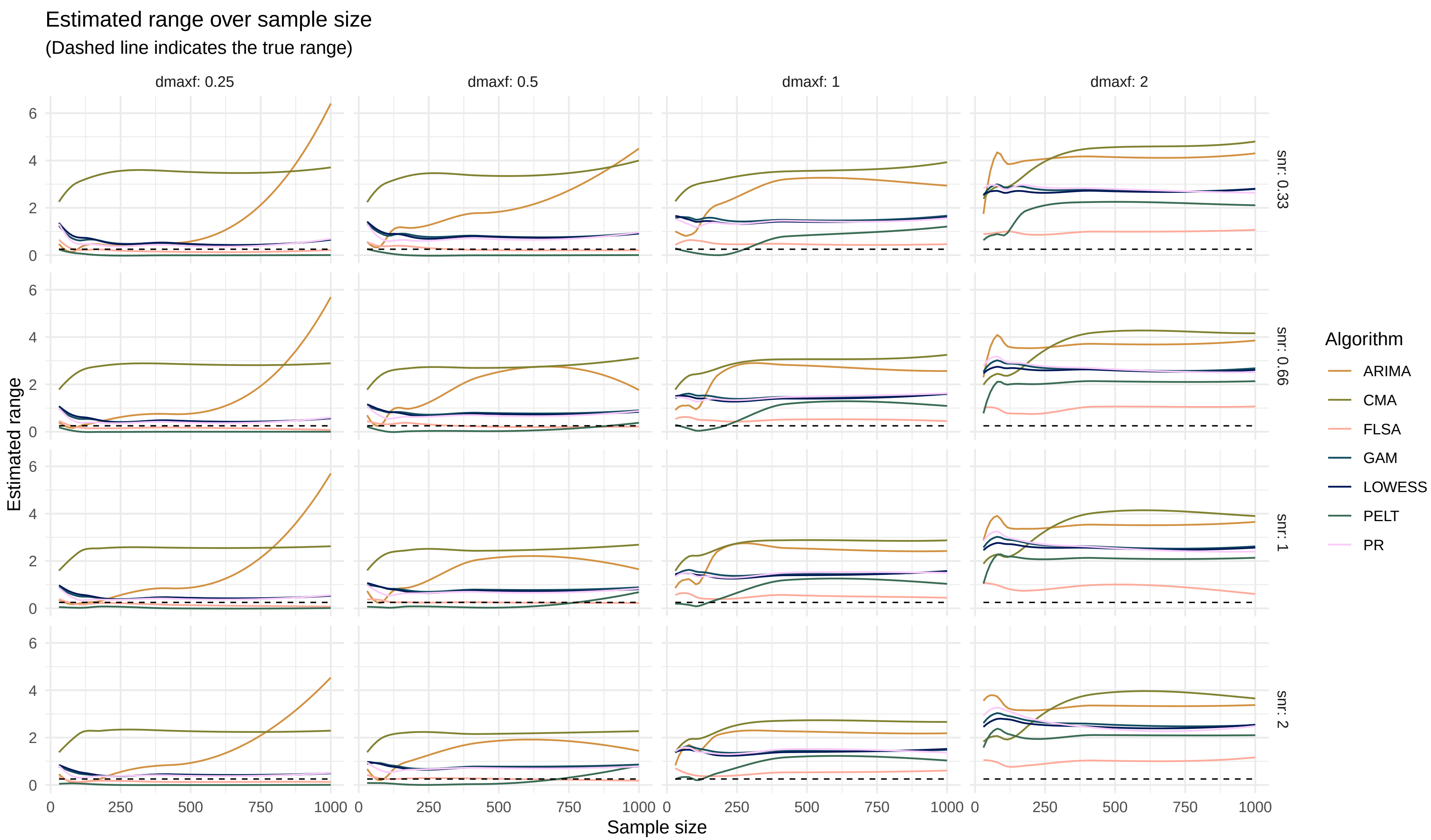
Estimand: Range

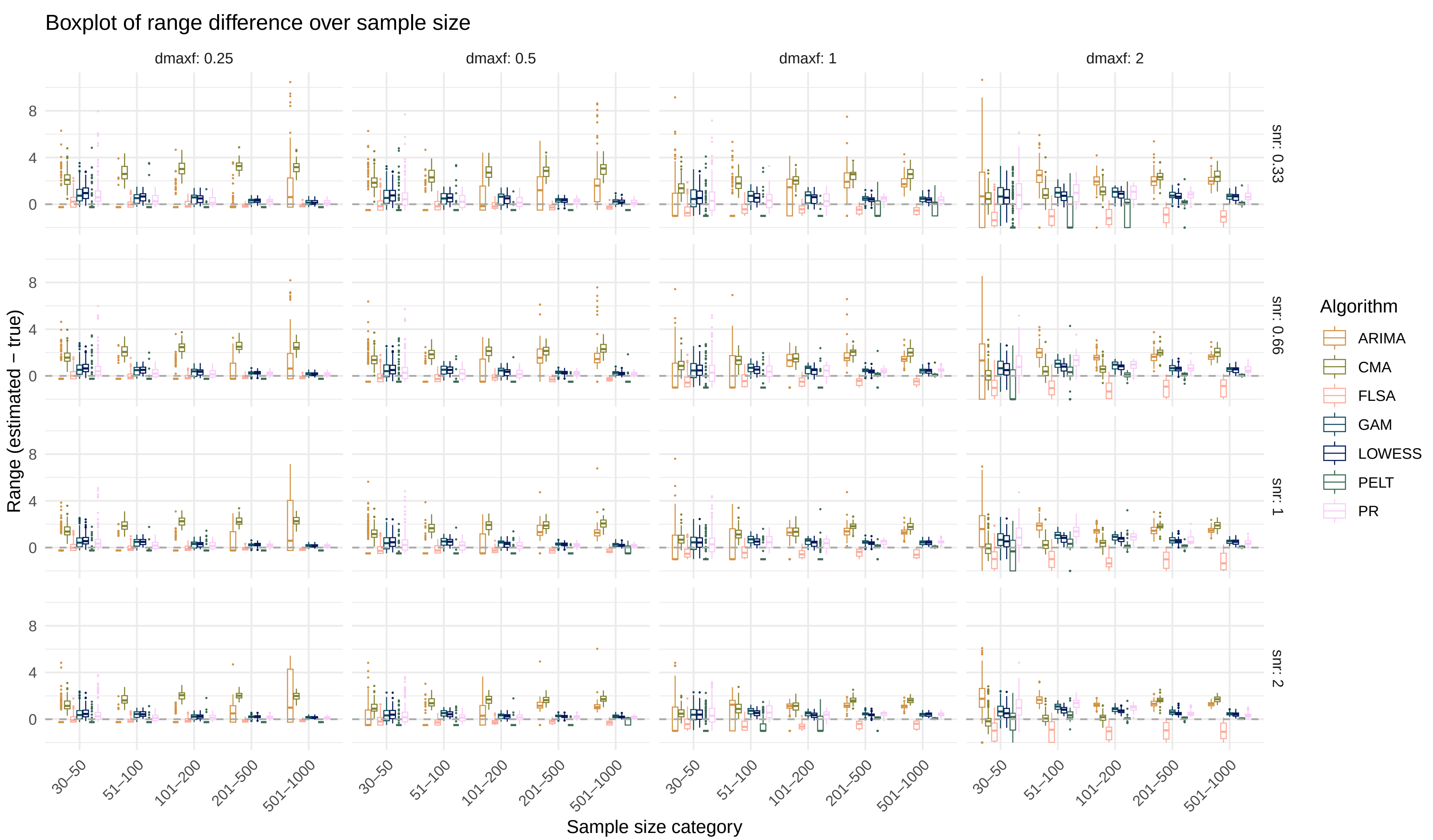
Systematic change pattern: No change



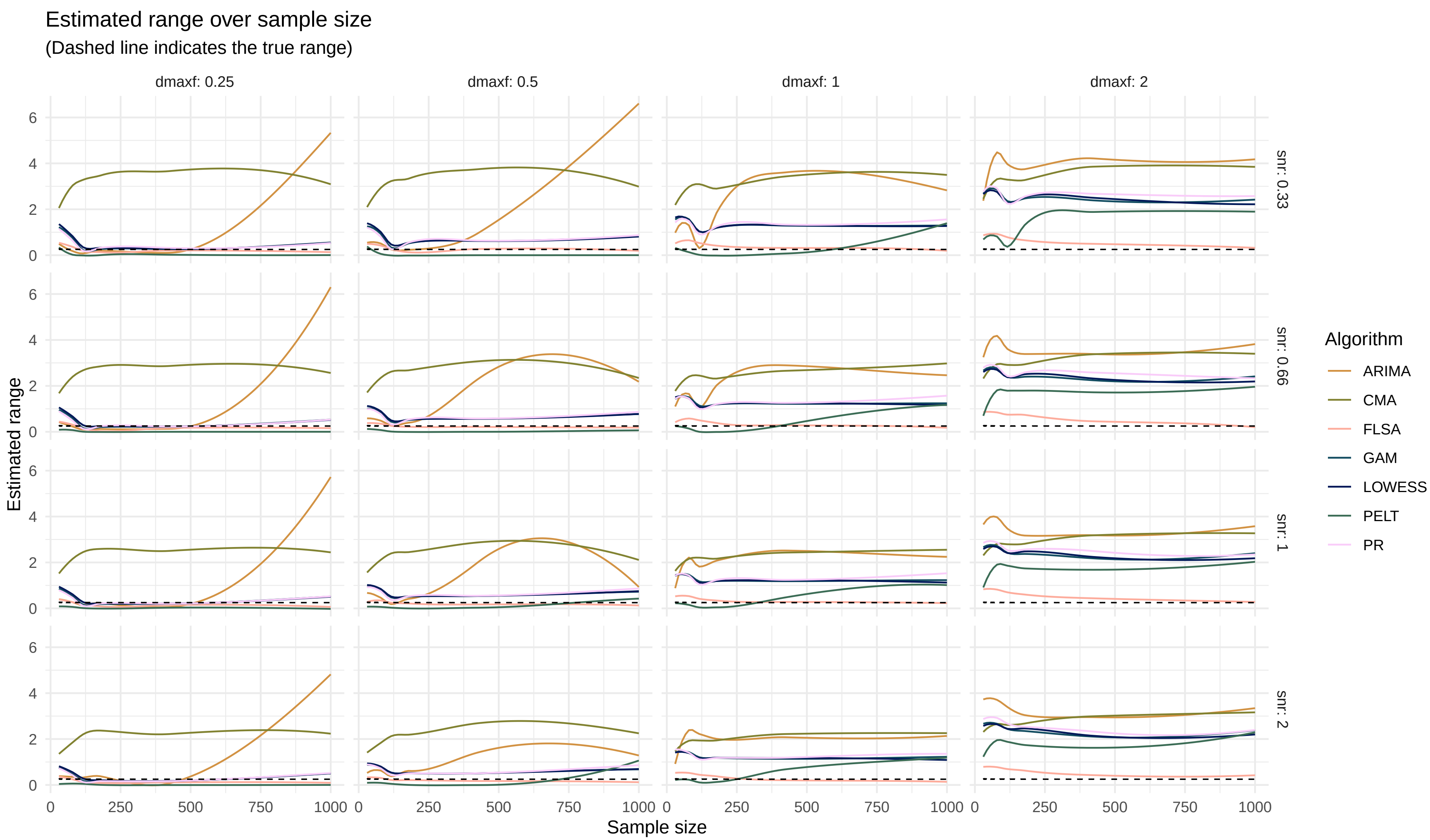

Boxplot of range difference over sample size
dmaxf: 0
Range (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

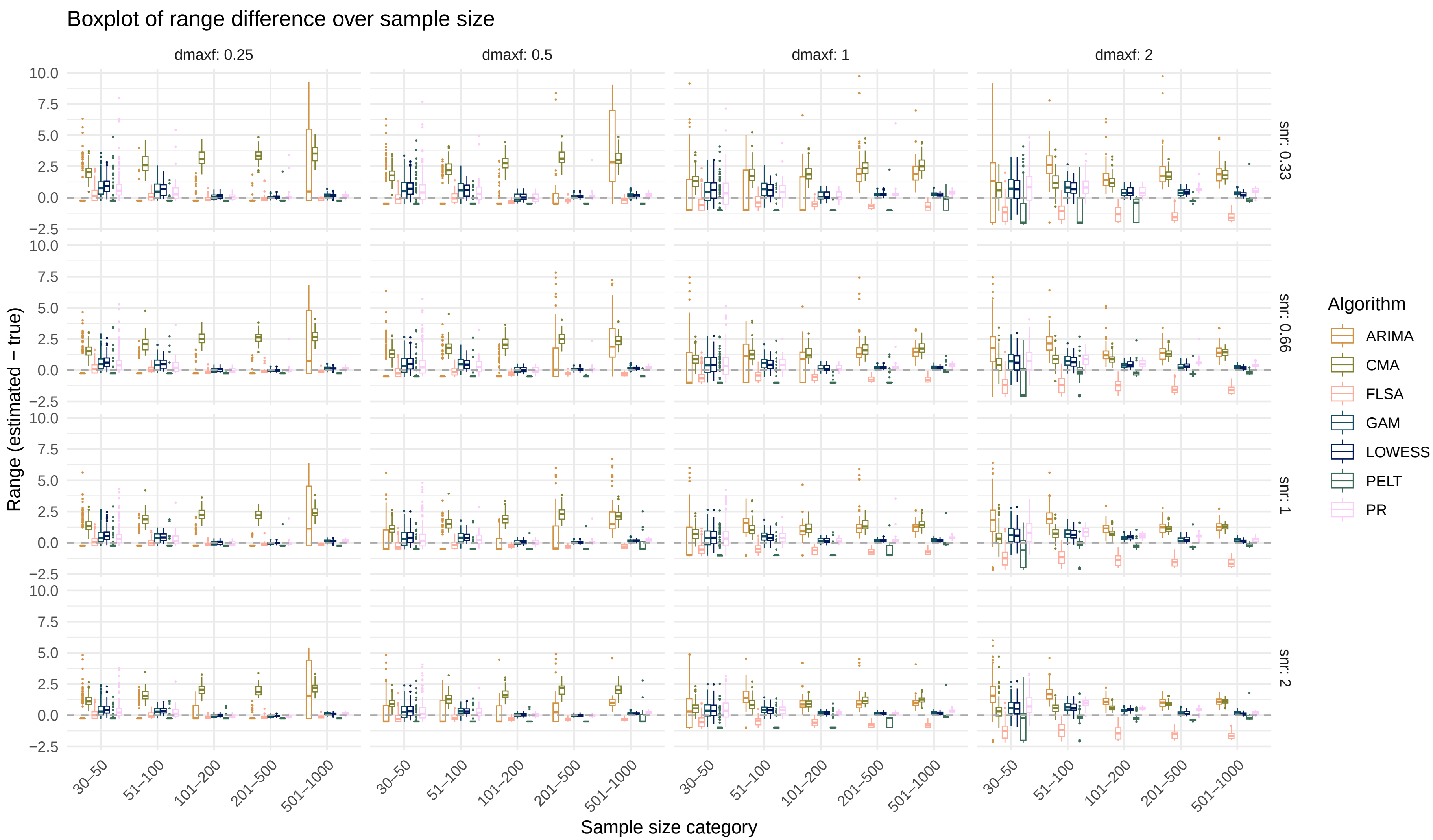
Systematic change pattern: Jump



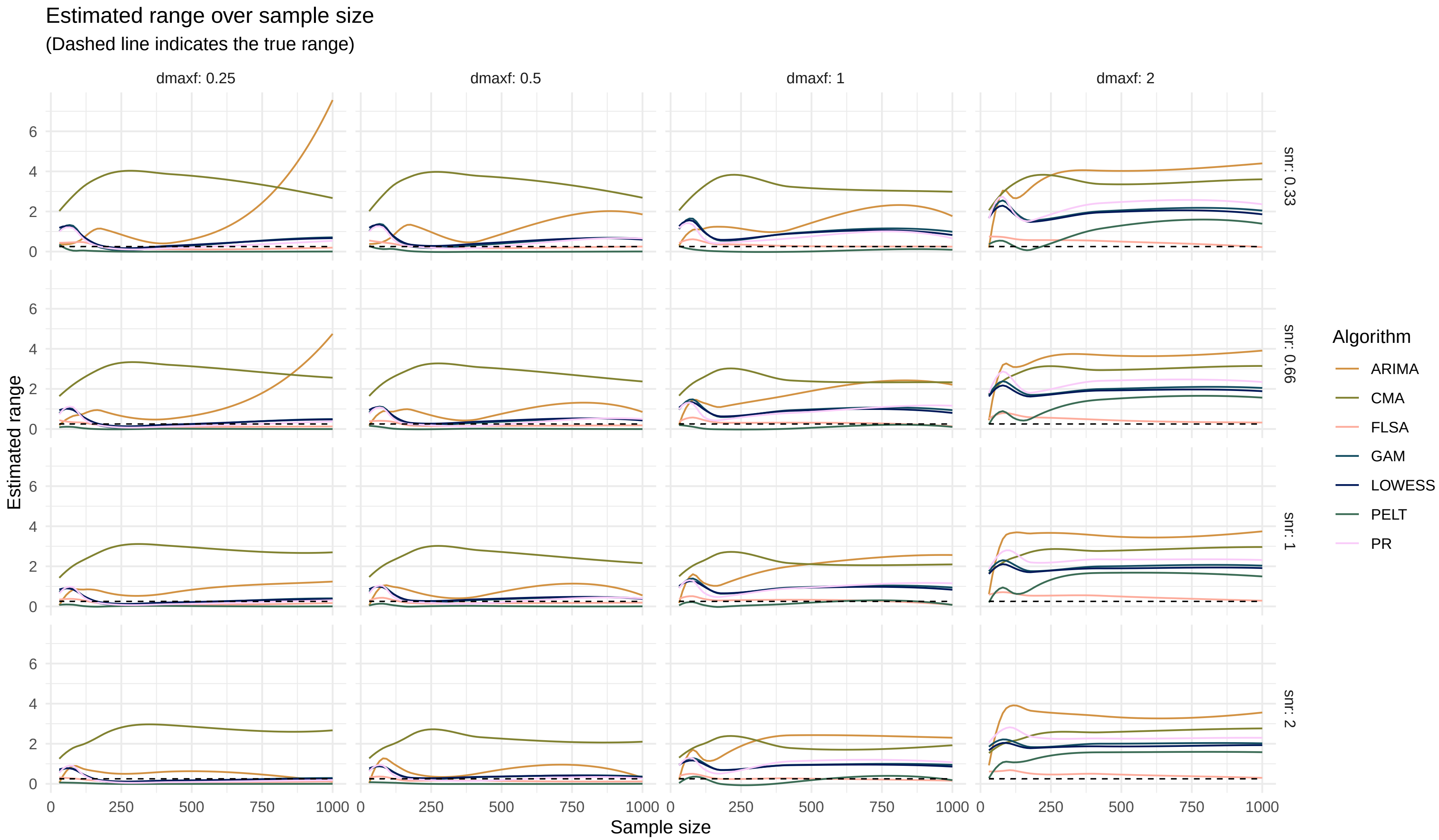

Boxplot of range difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Range (estimated – true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

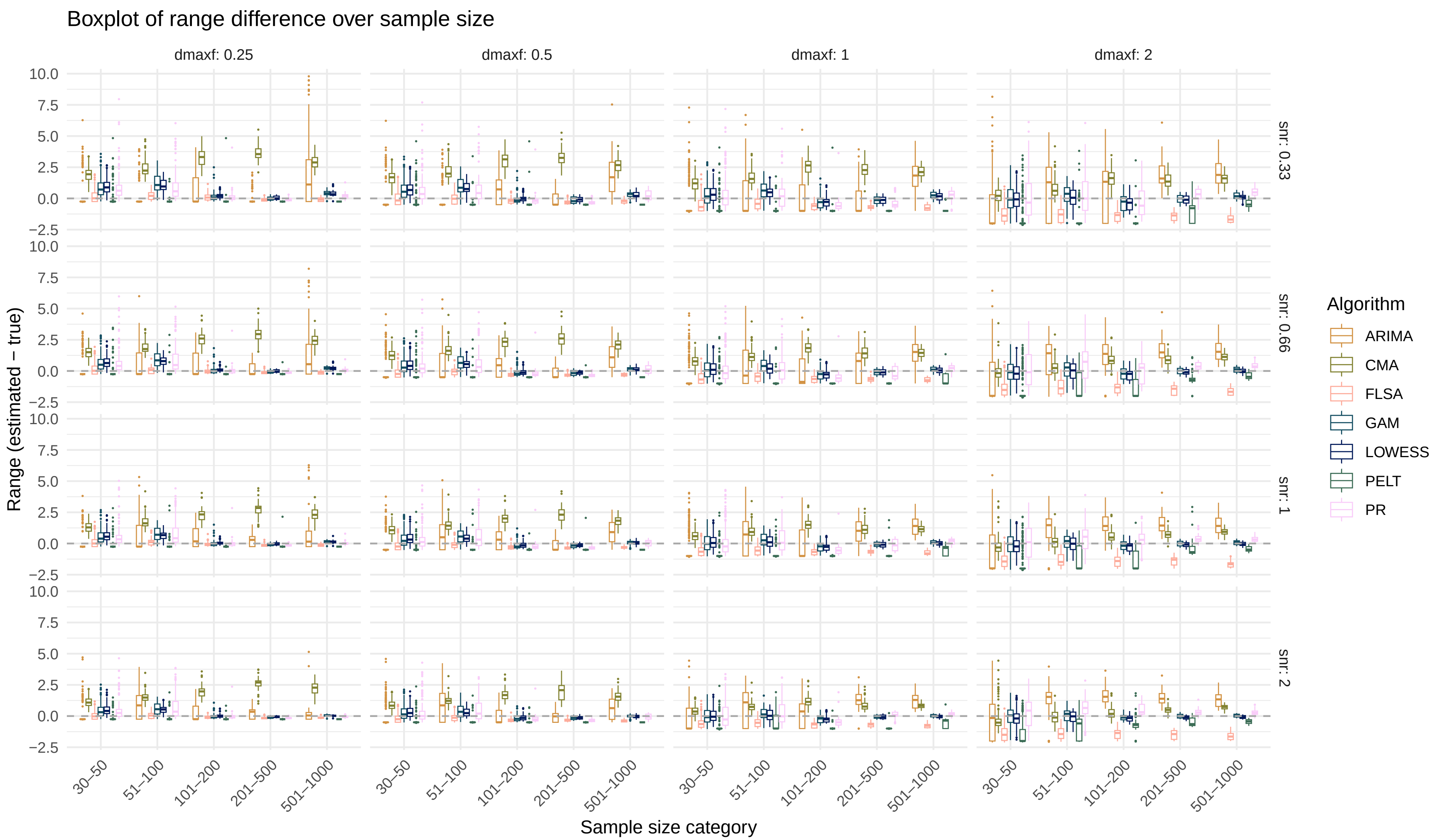
Systematic change pattern: Linear




Boxplot of range difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Range (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

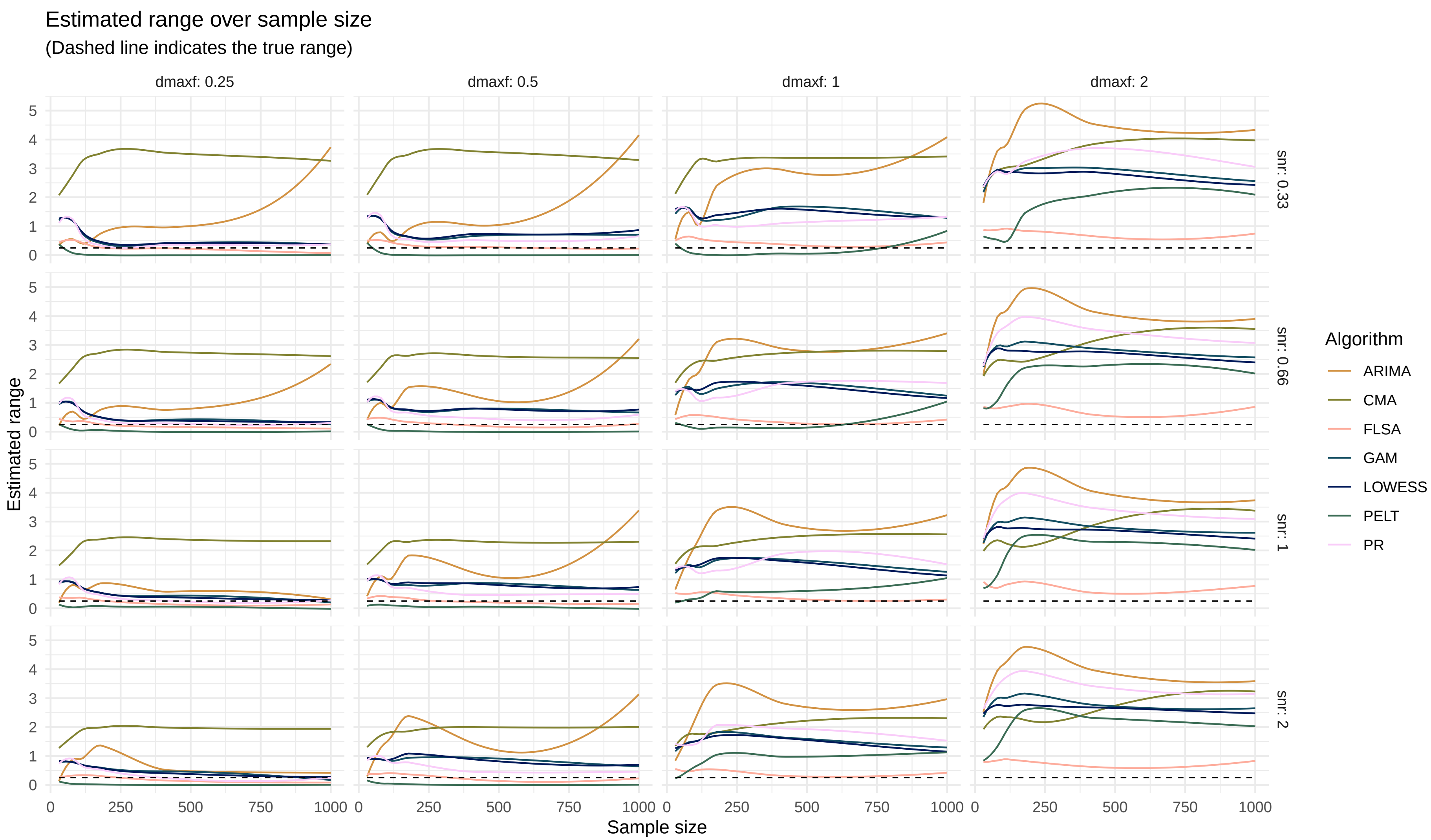
Systematic change pattern: Quadratic

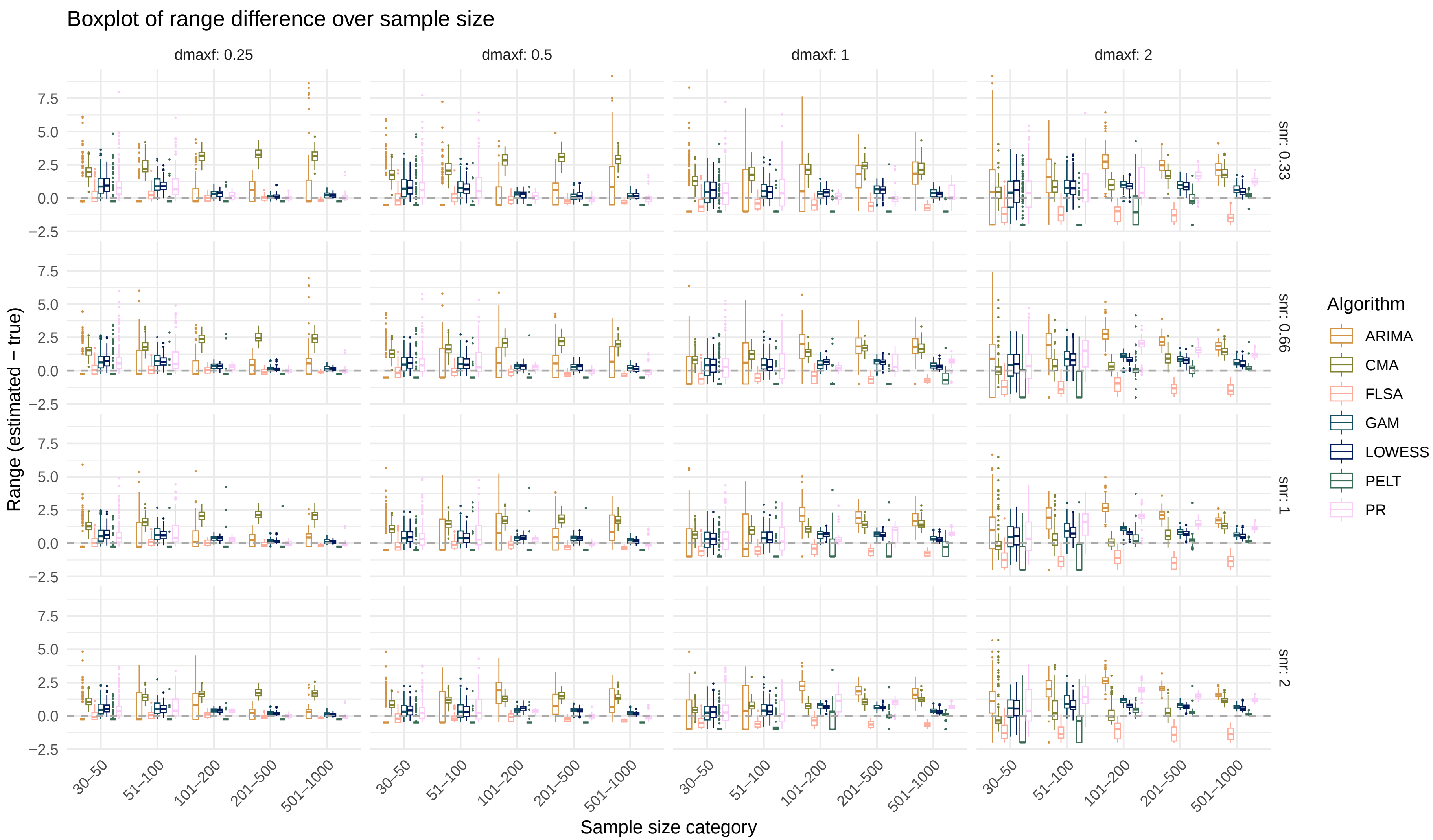




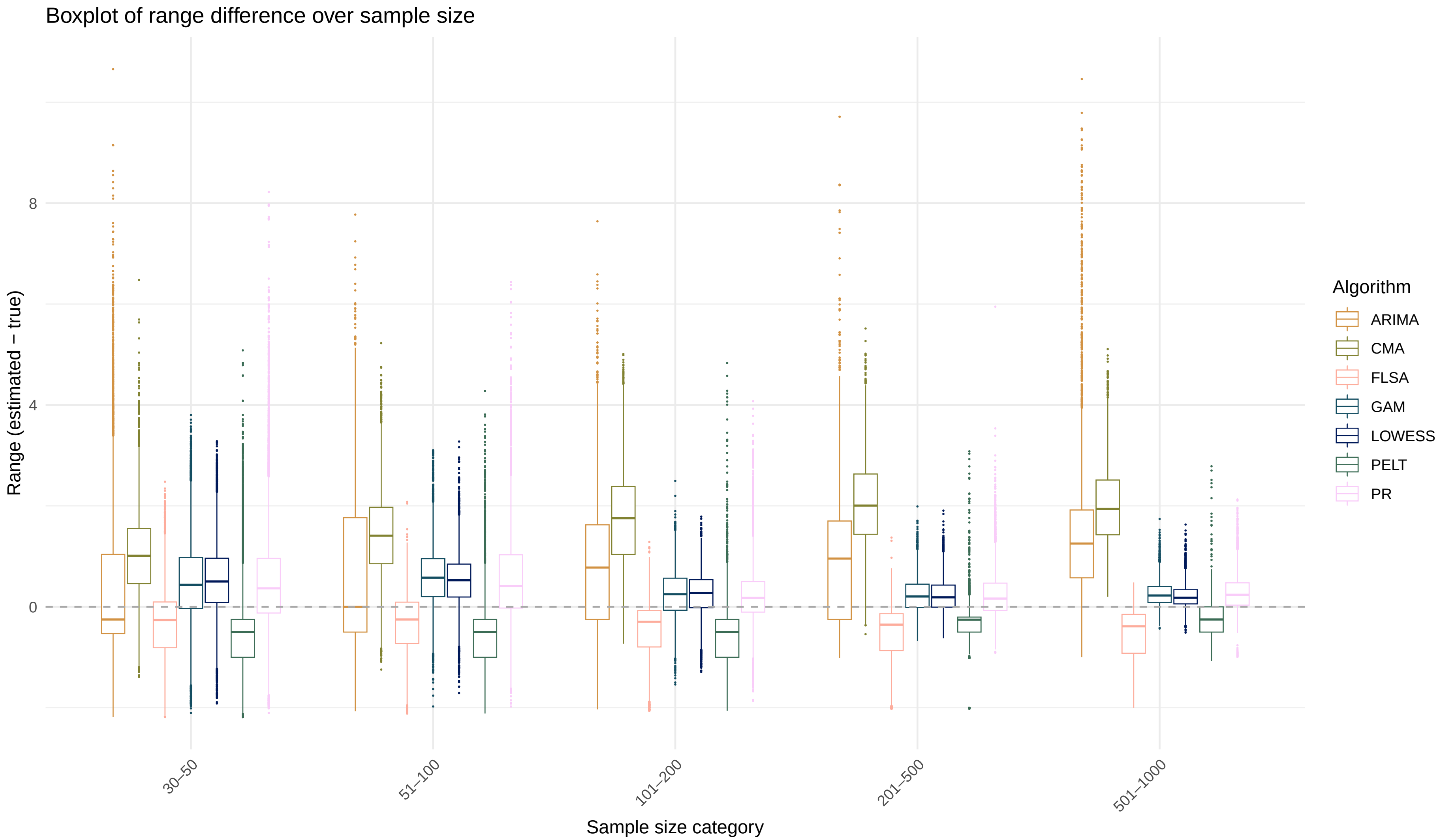
Boxplot of range difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Range (estimated – true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

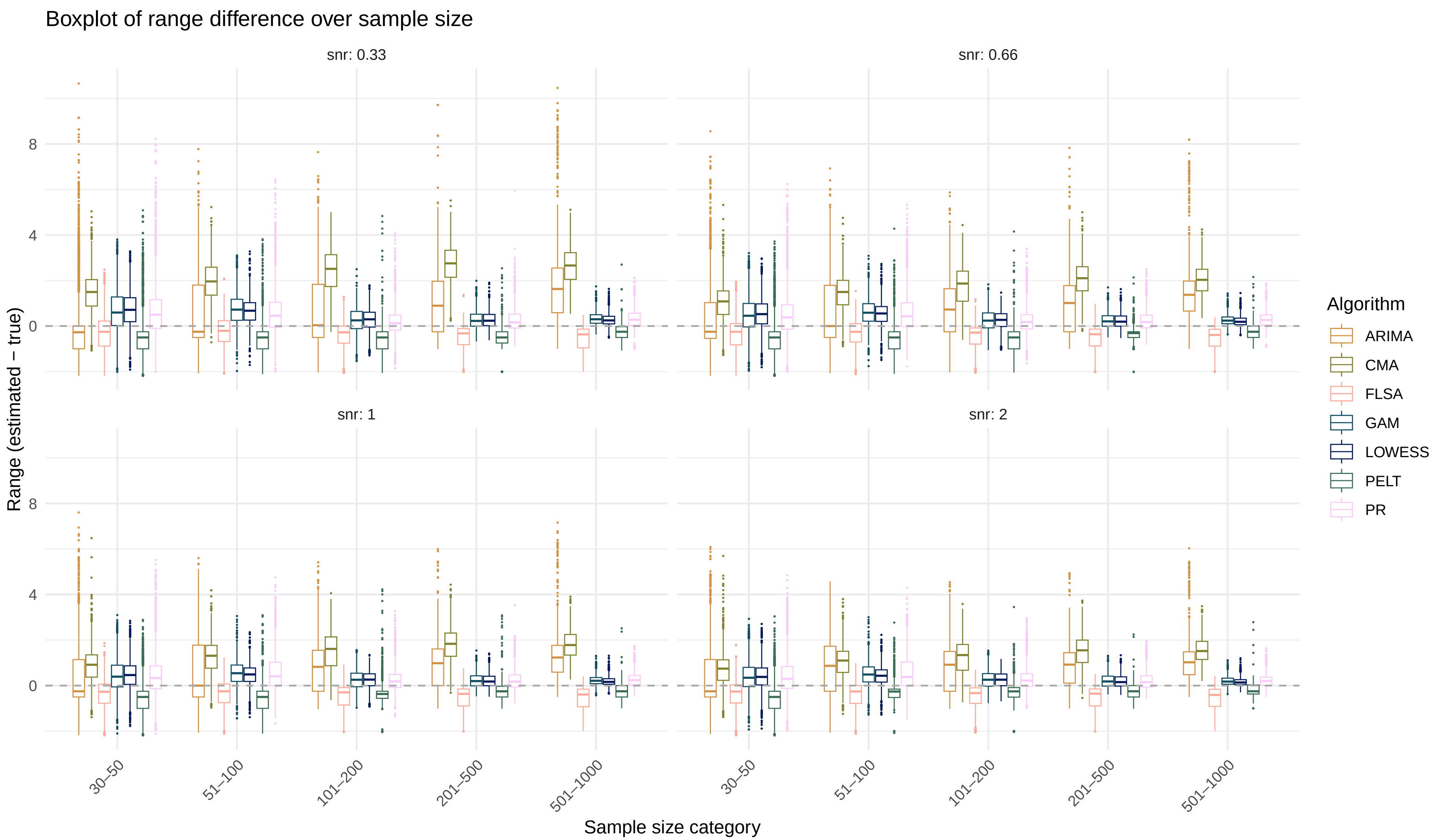
Systematic change pattern: Recurrent




Boxplot of range difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Range (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

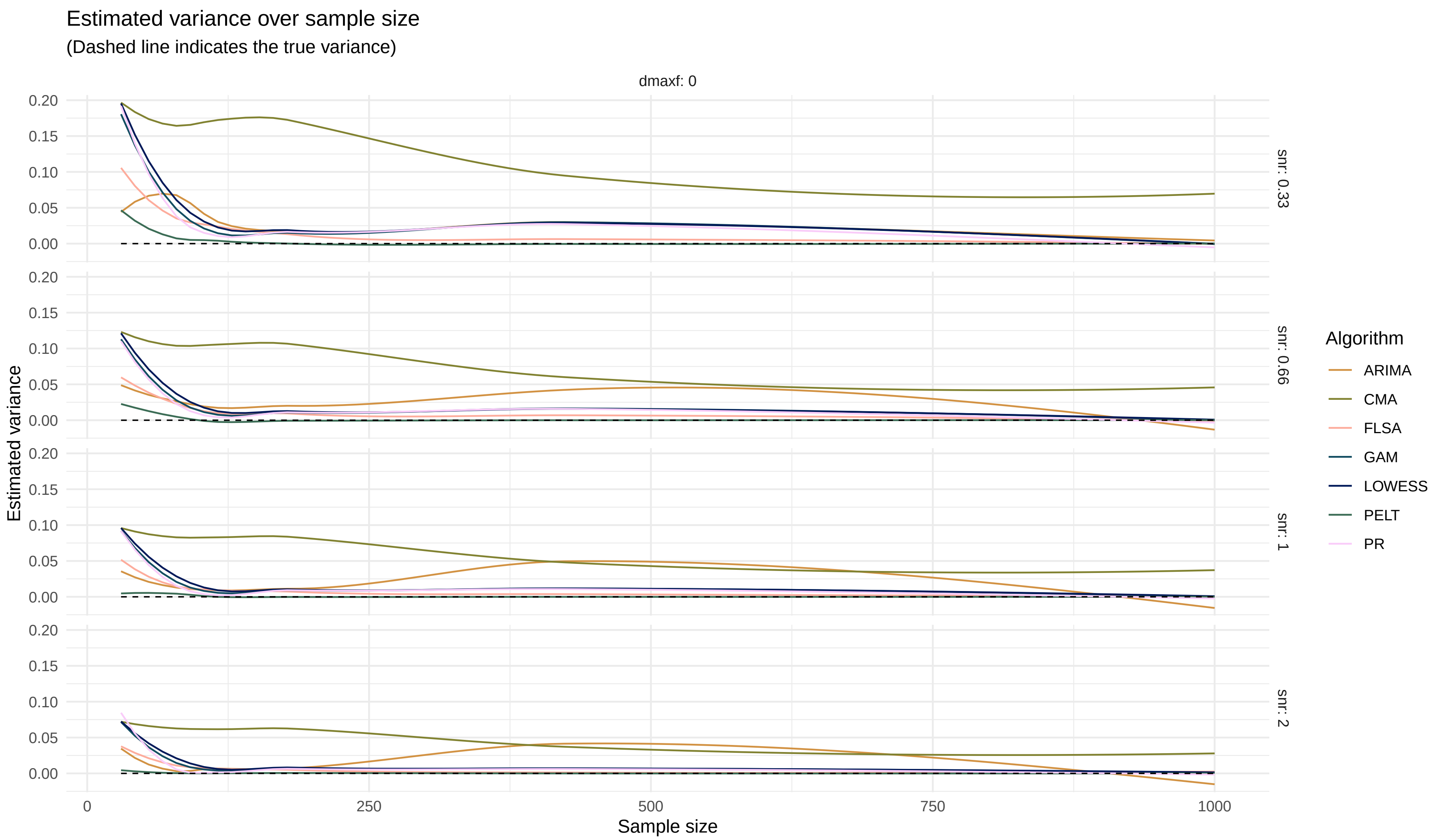
Systematic change pattern: All patterns combined

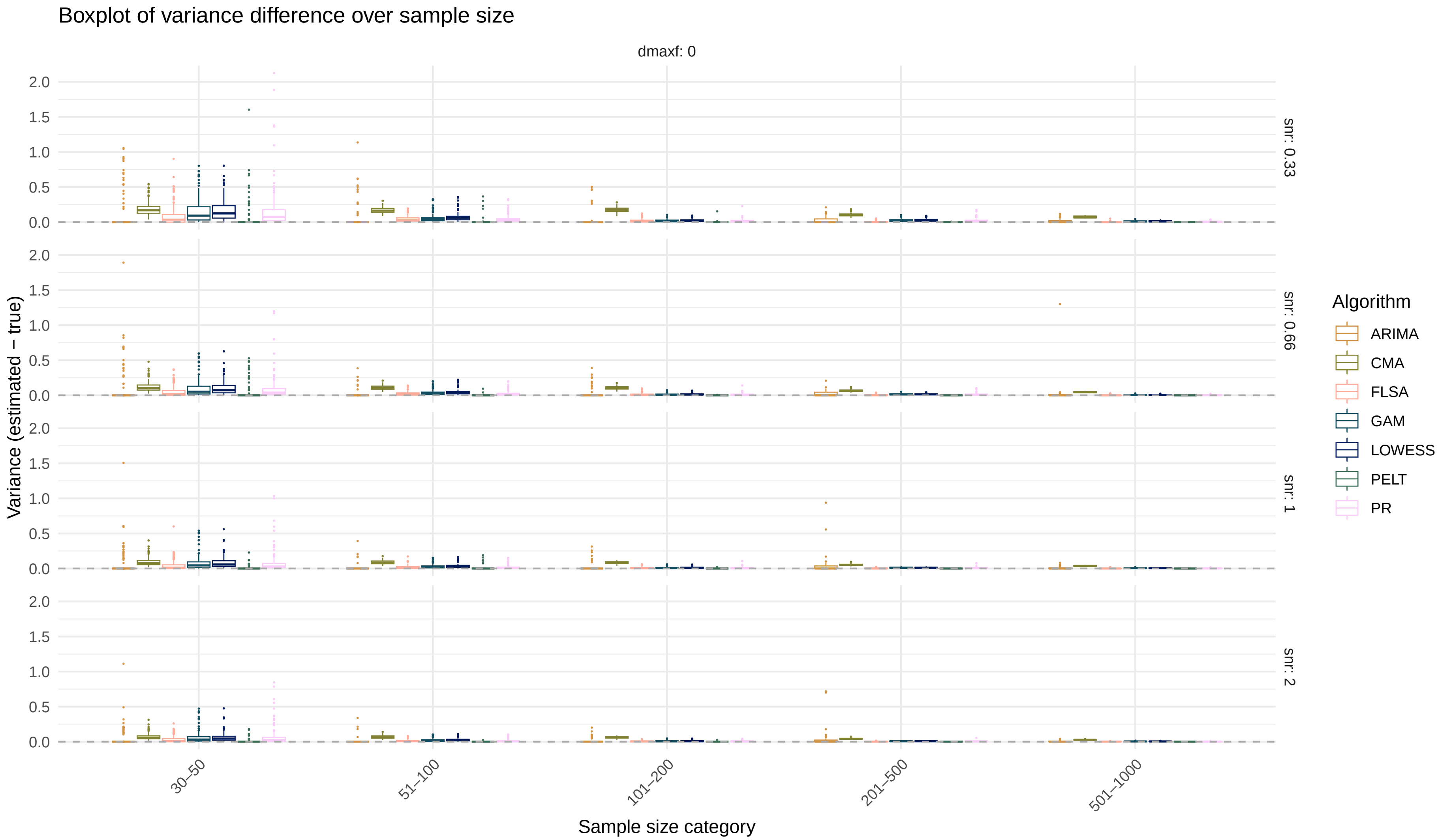



Boxplot of range difference over sample size
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Range (estimated – true)
8
4
0
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

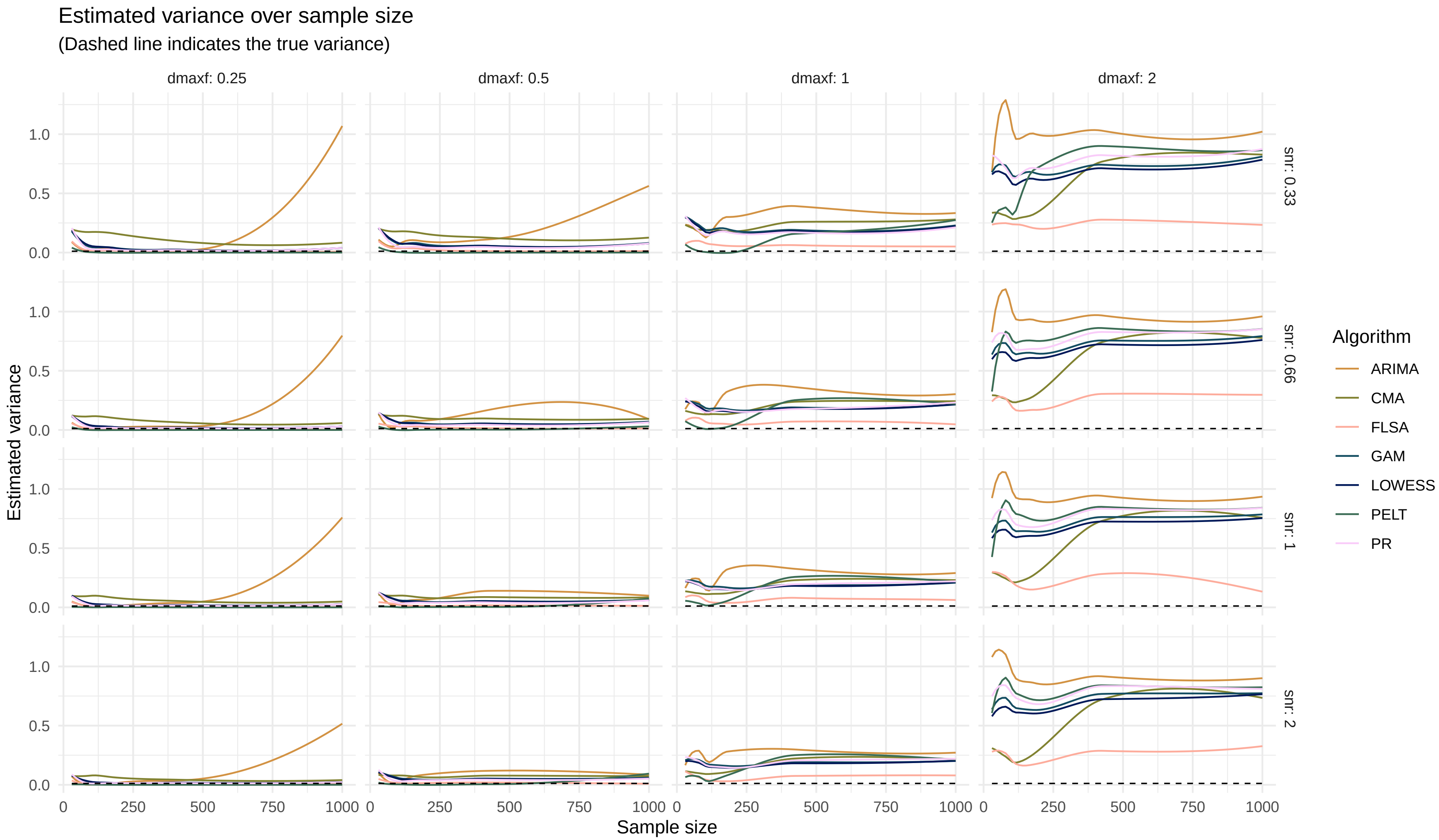
Estimand: Variance

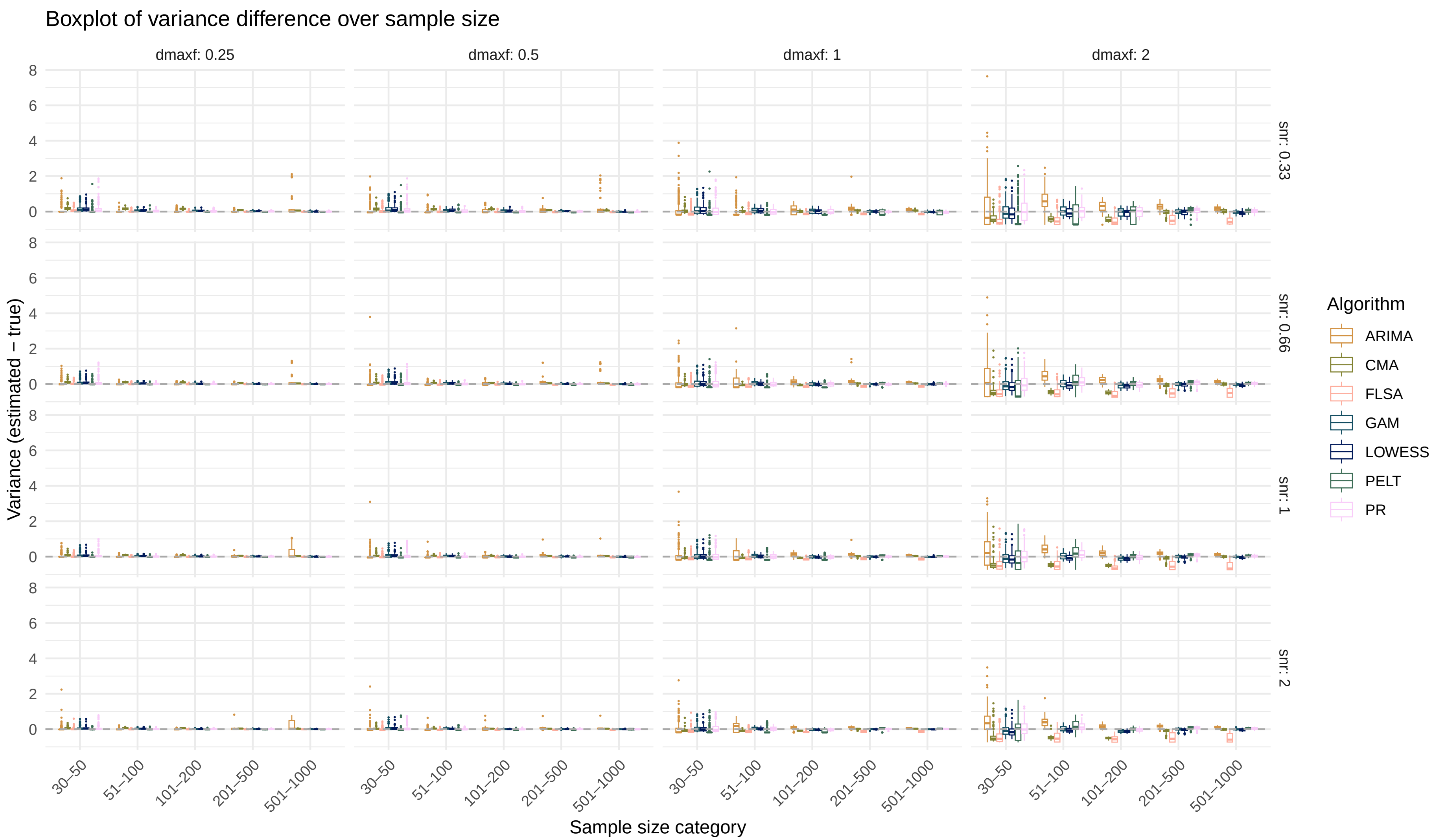
Systematic change pattern: No change



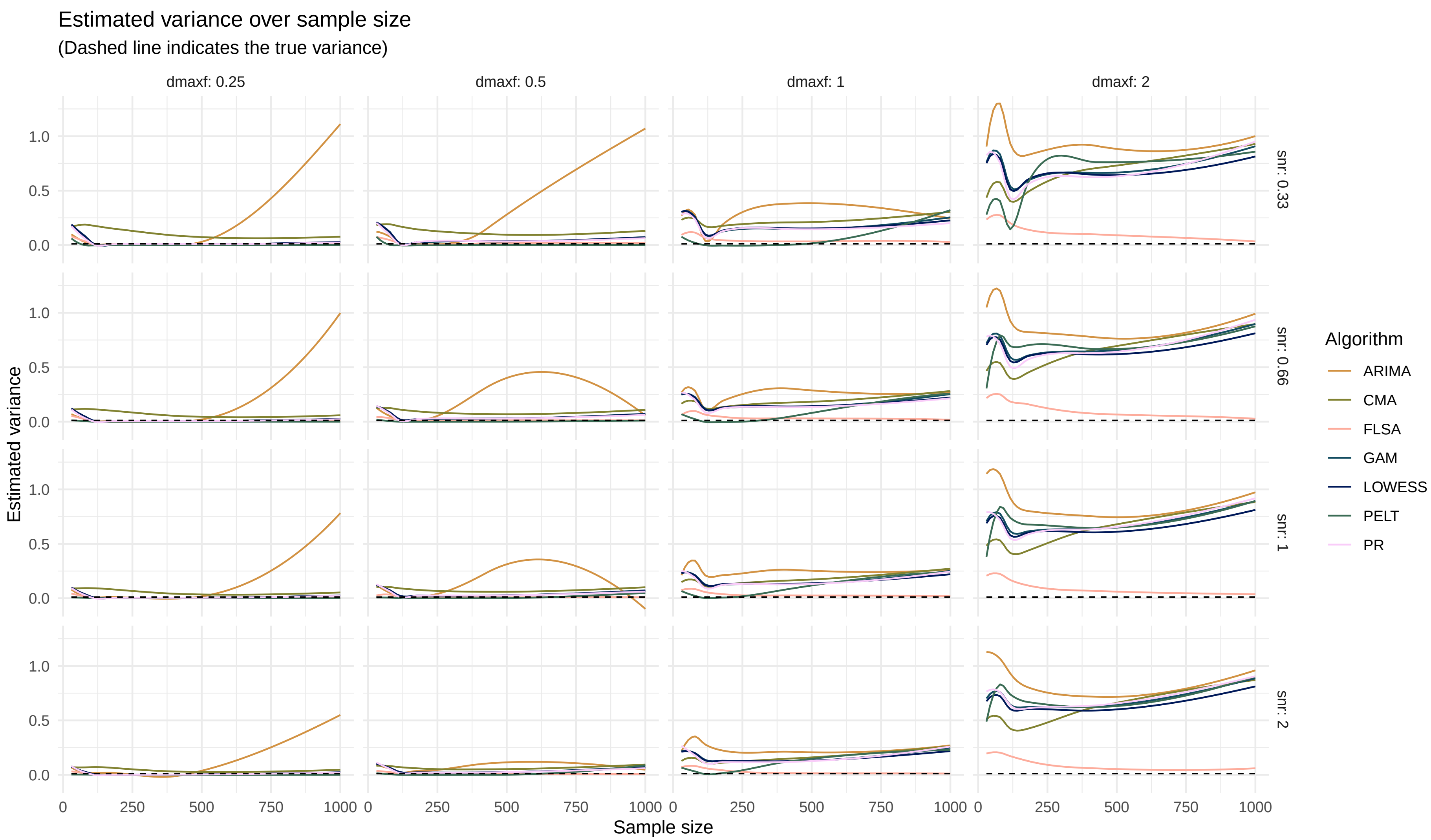

Boxplot of variance difference over sample size
dmaxf: 0
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Variance (estimated − true)
2.0
1.5
1.0
0.5
0.0
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

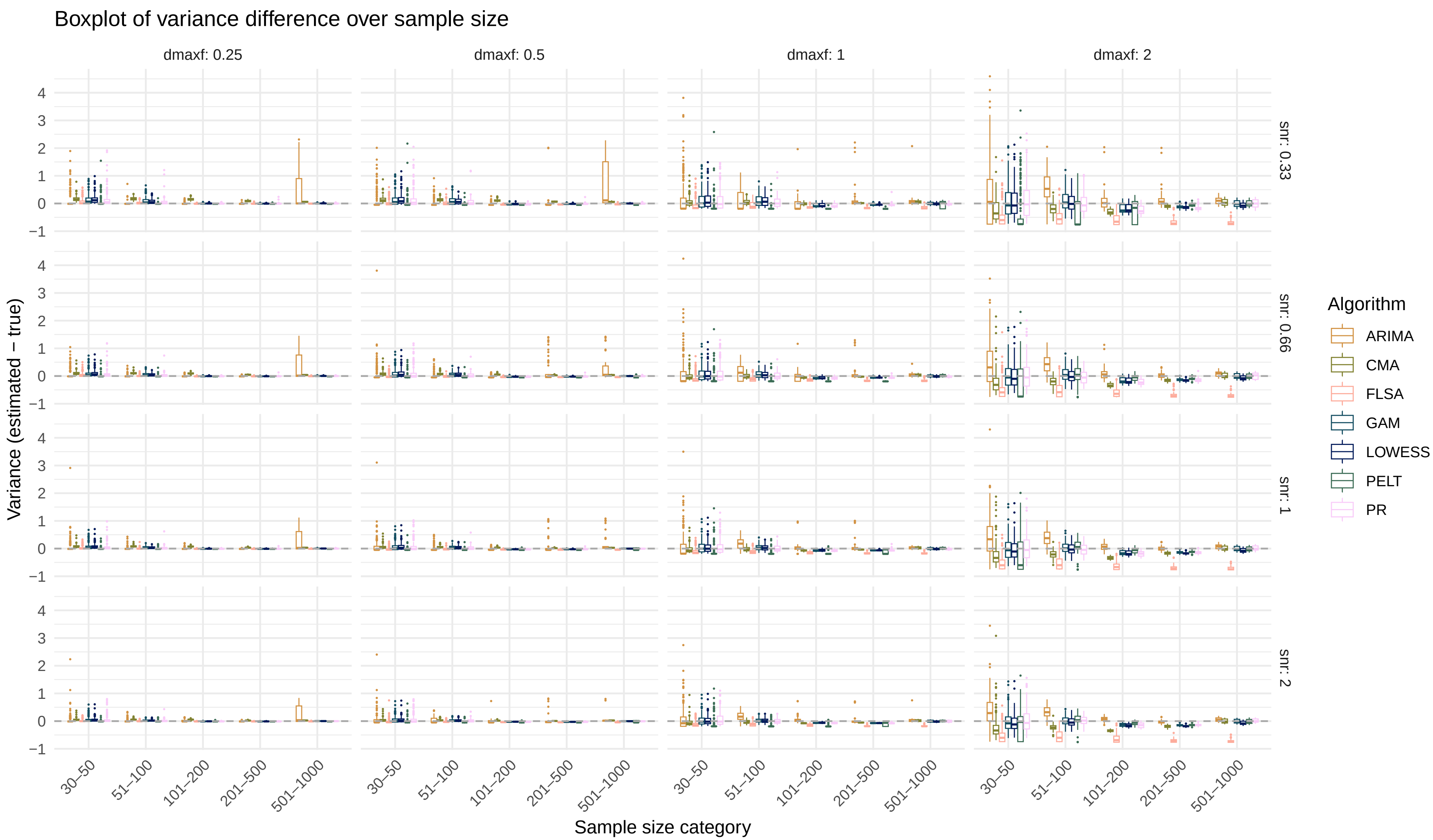
Systematic change pattern: Jump



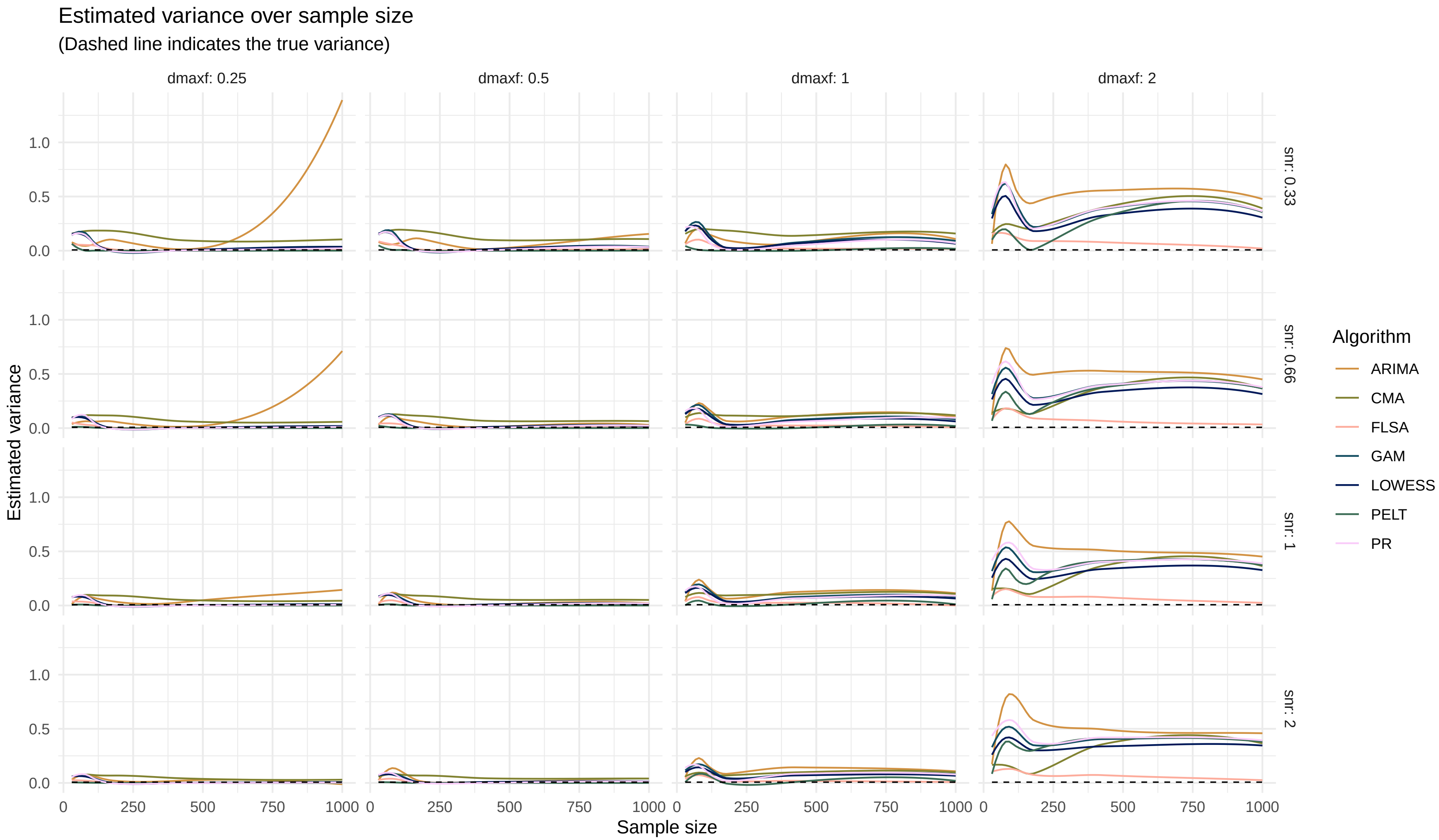

Boxplot of variance difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Variance (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

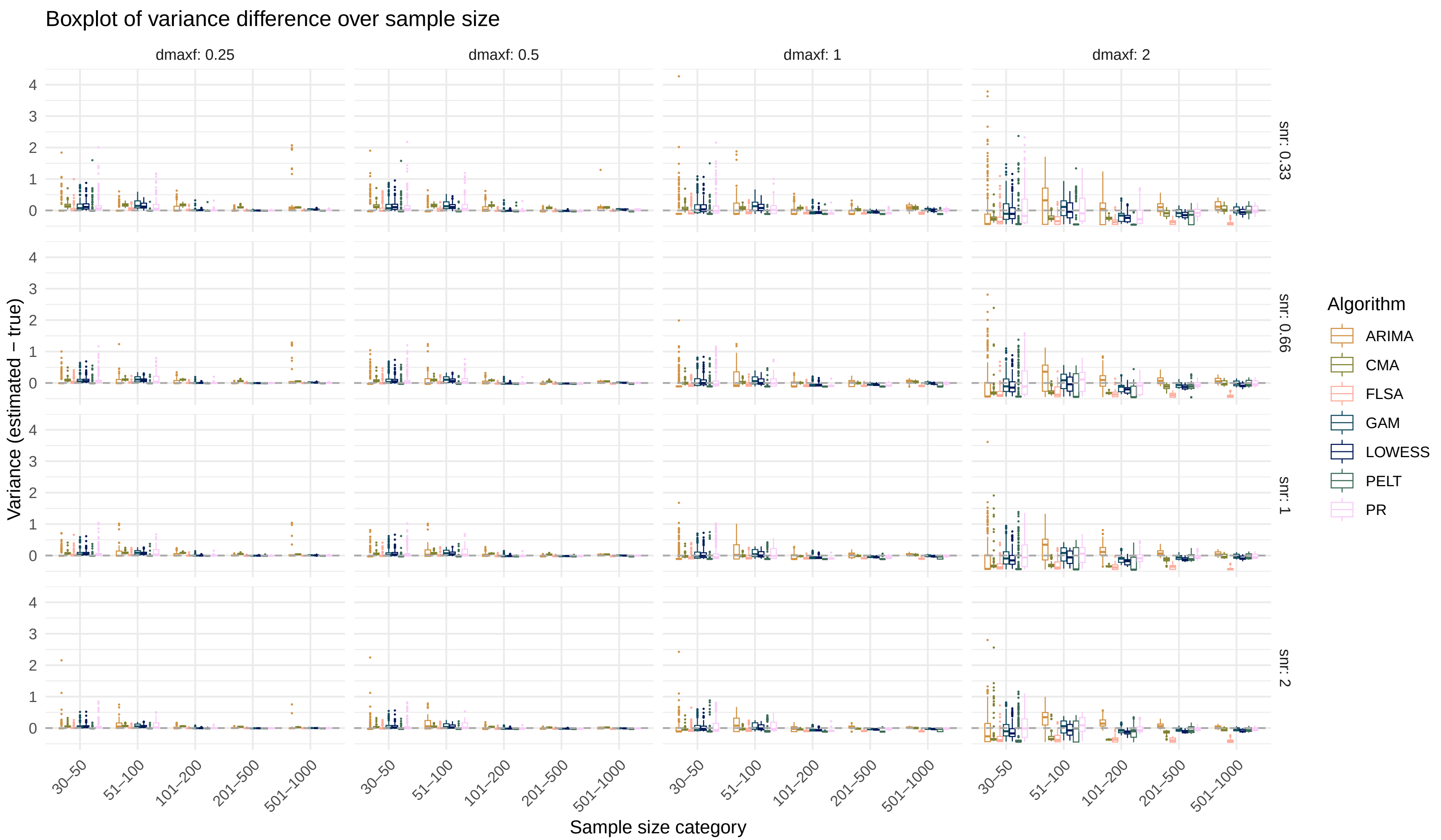
Systematic change pattern: Linear




Boxplot of variance difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Variance (estimated – true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

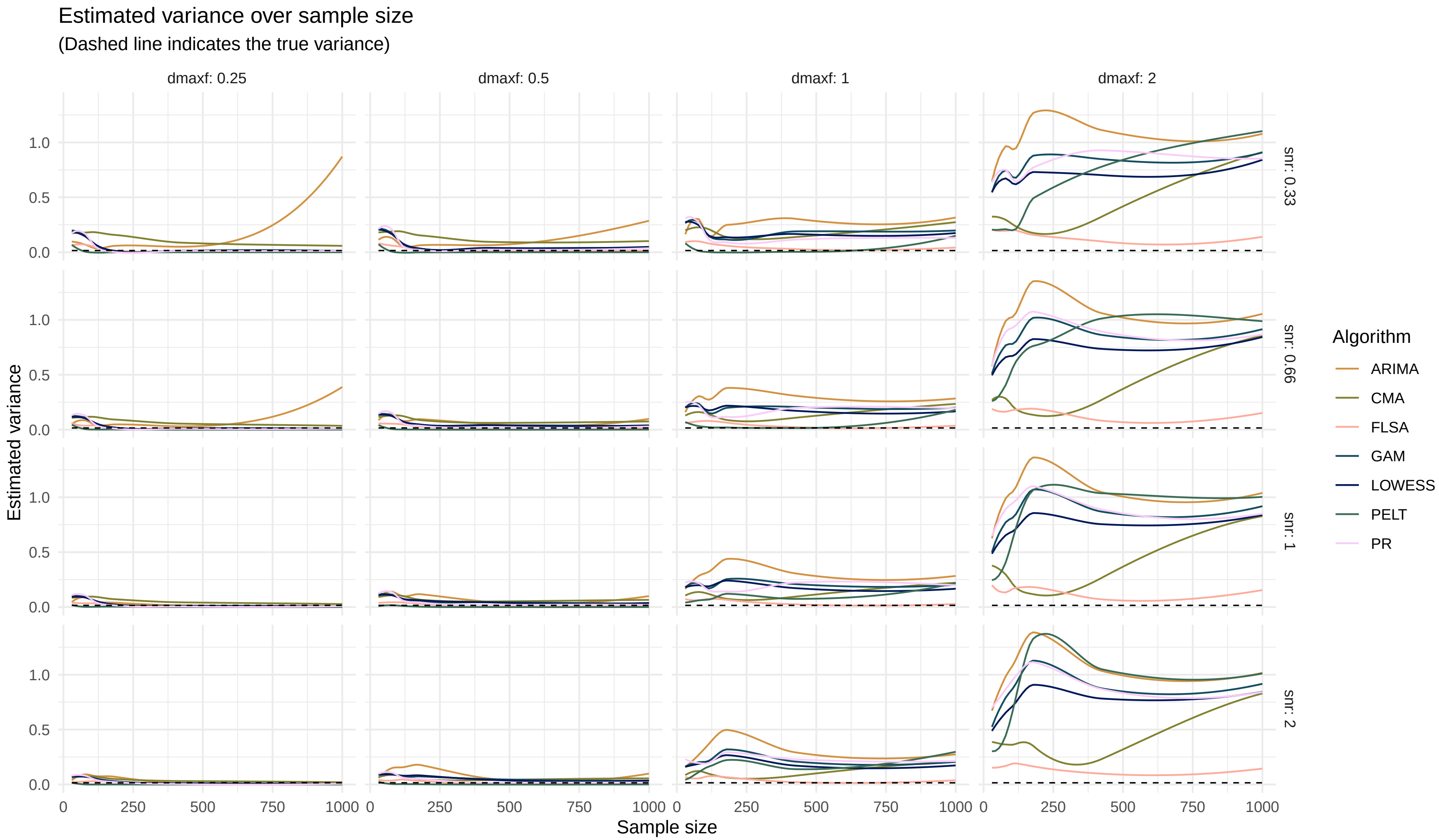
Systematic change pattern: Quadratic

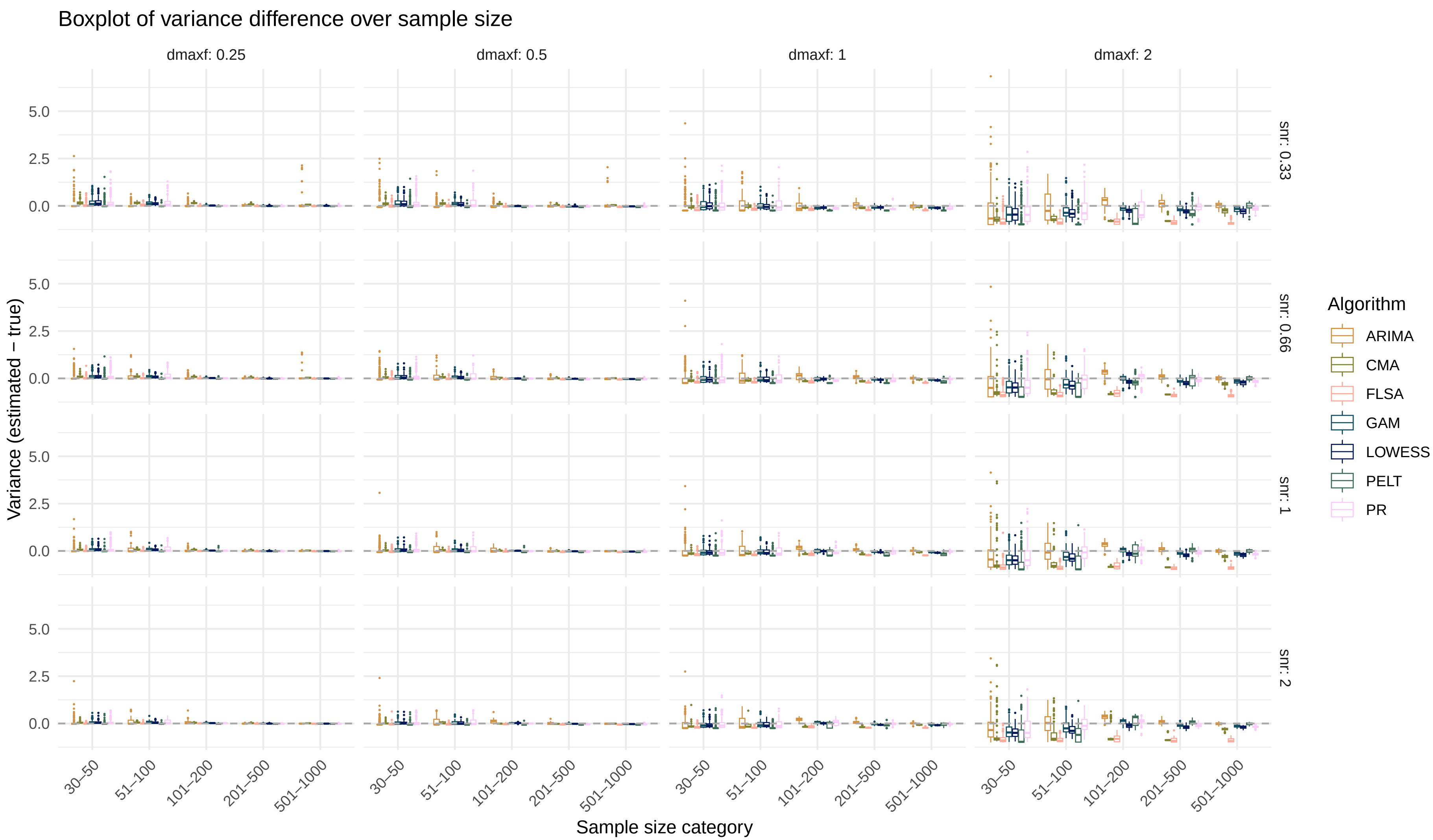




Boxplot of variance difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Variance (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

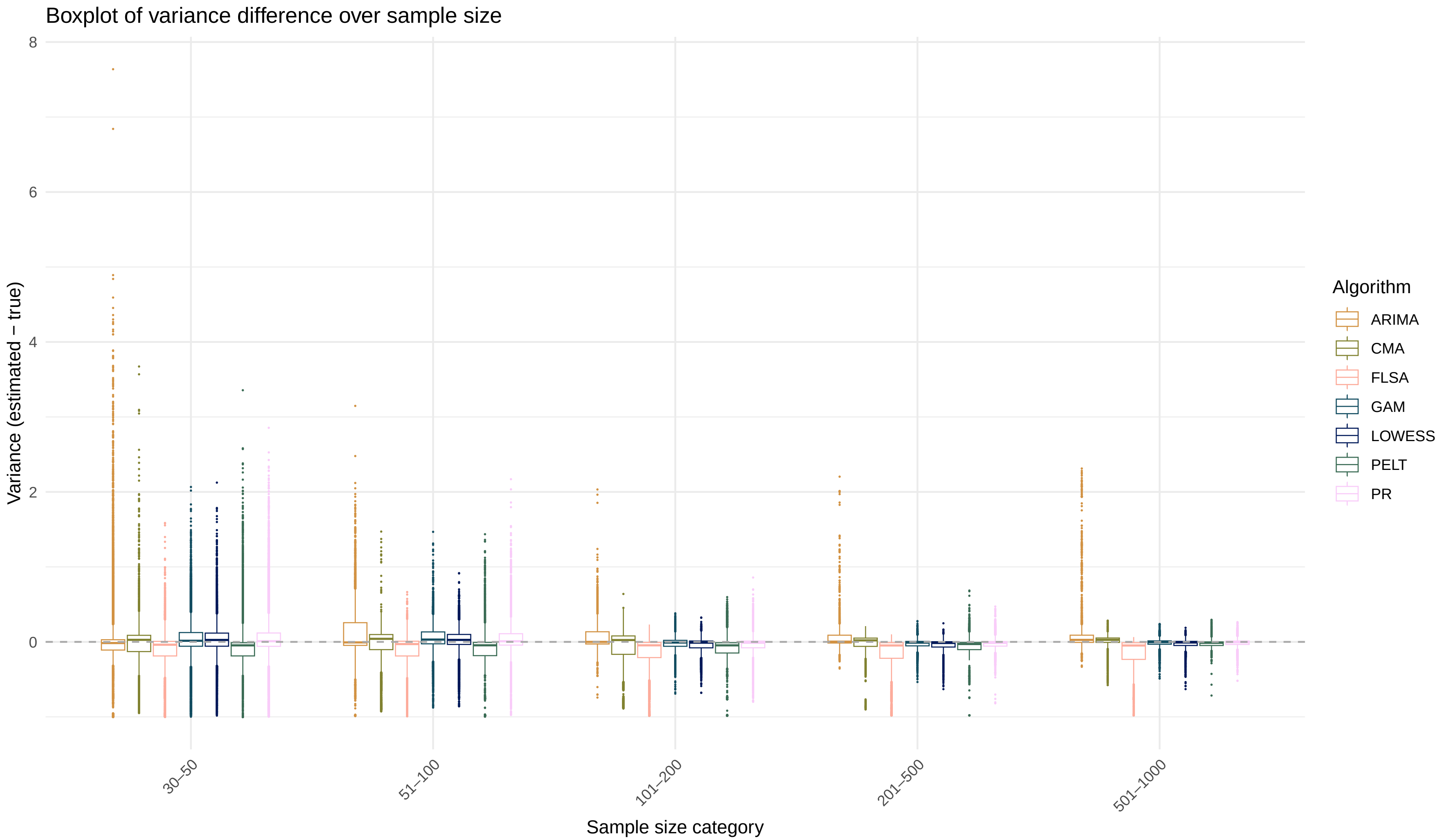
Systematic change pattern: Recurrent

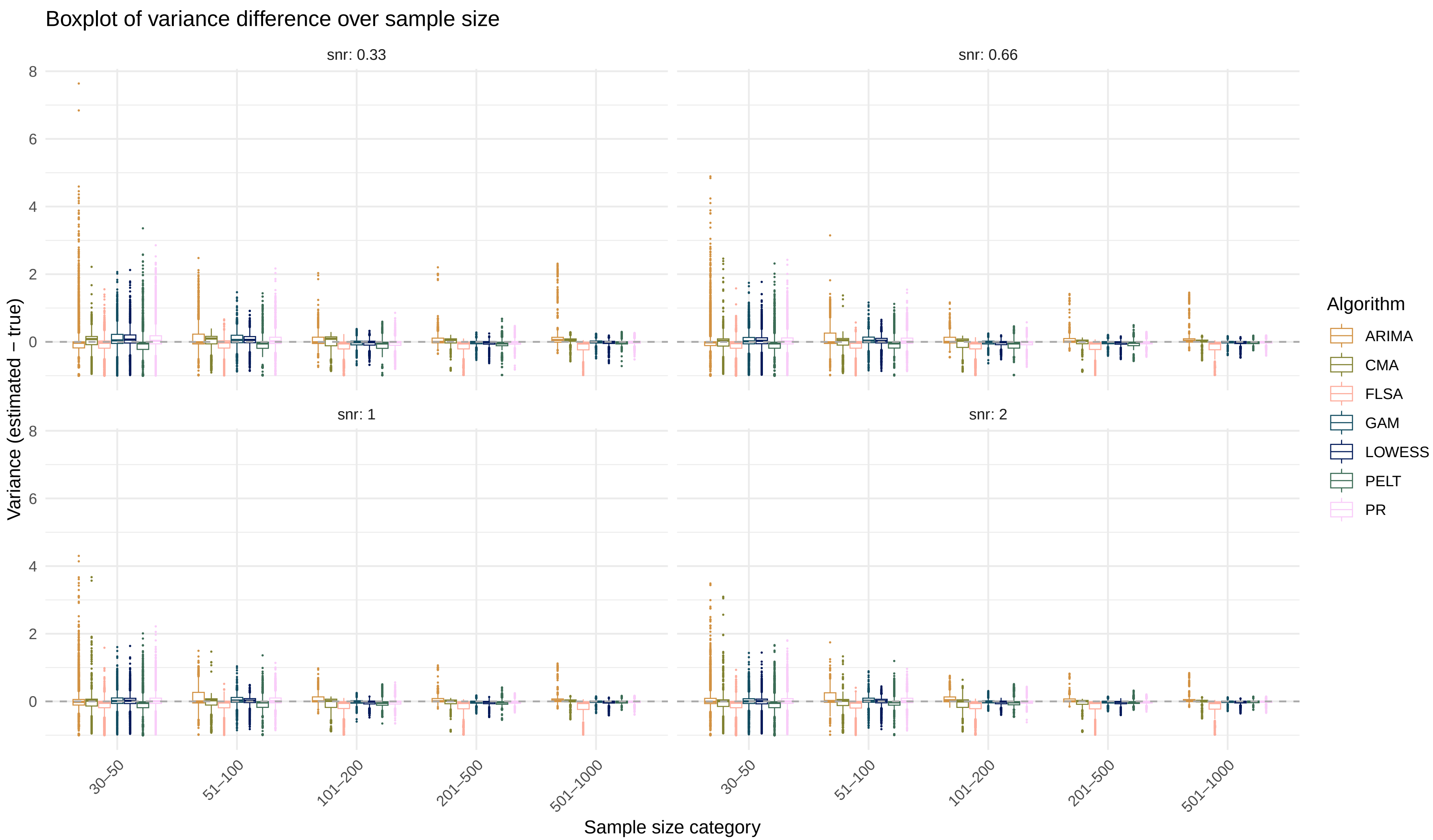




Boxplot of variance difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
5.0
2.5
0.0
Variance (estimated – true)
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

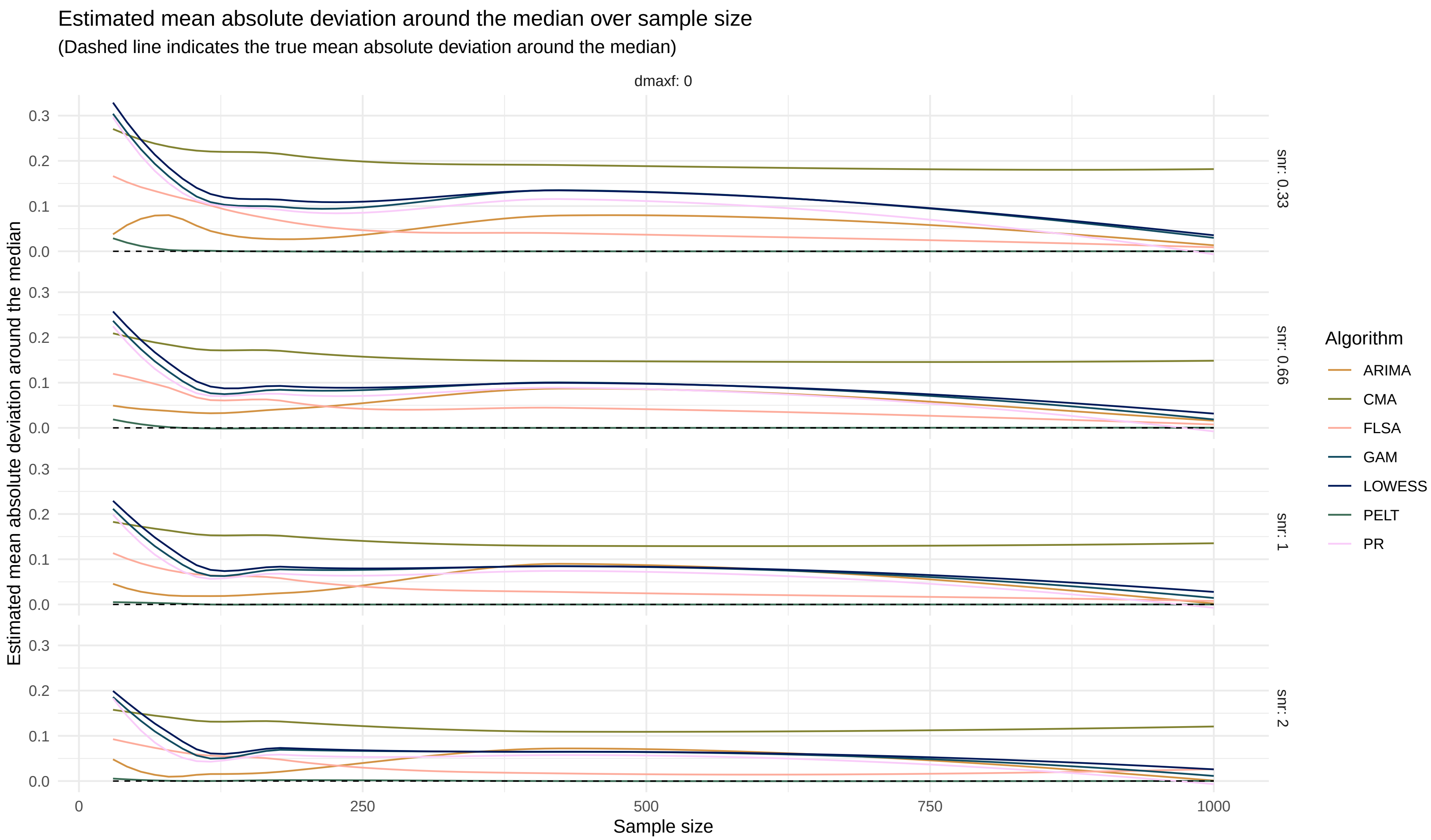
Systematic change pattern: All patterns combined

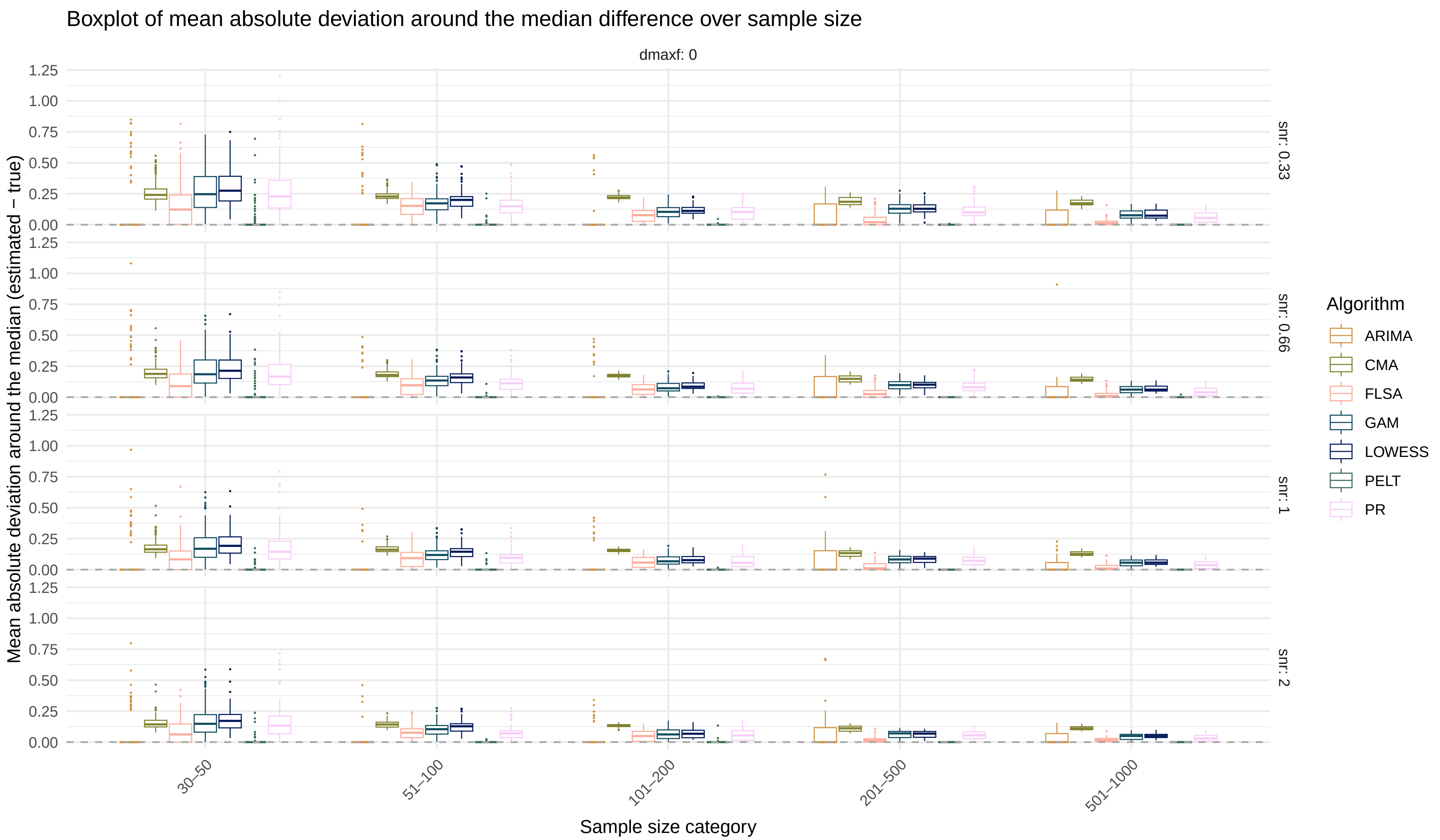



Boxplot of variance difference over sample size
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Variance (estimated – true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

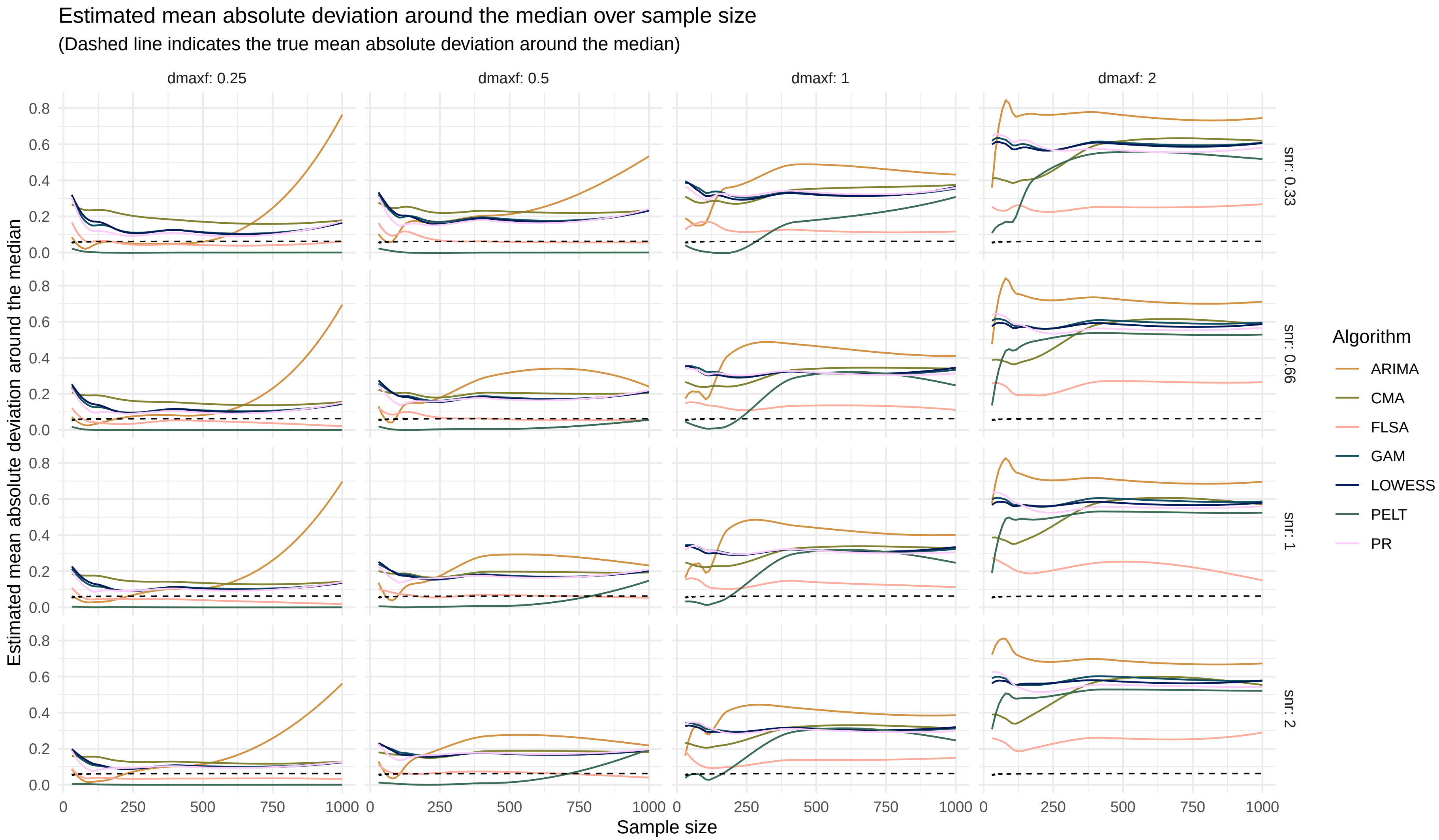
Estimand: Mean absolute deviation around the median

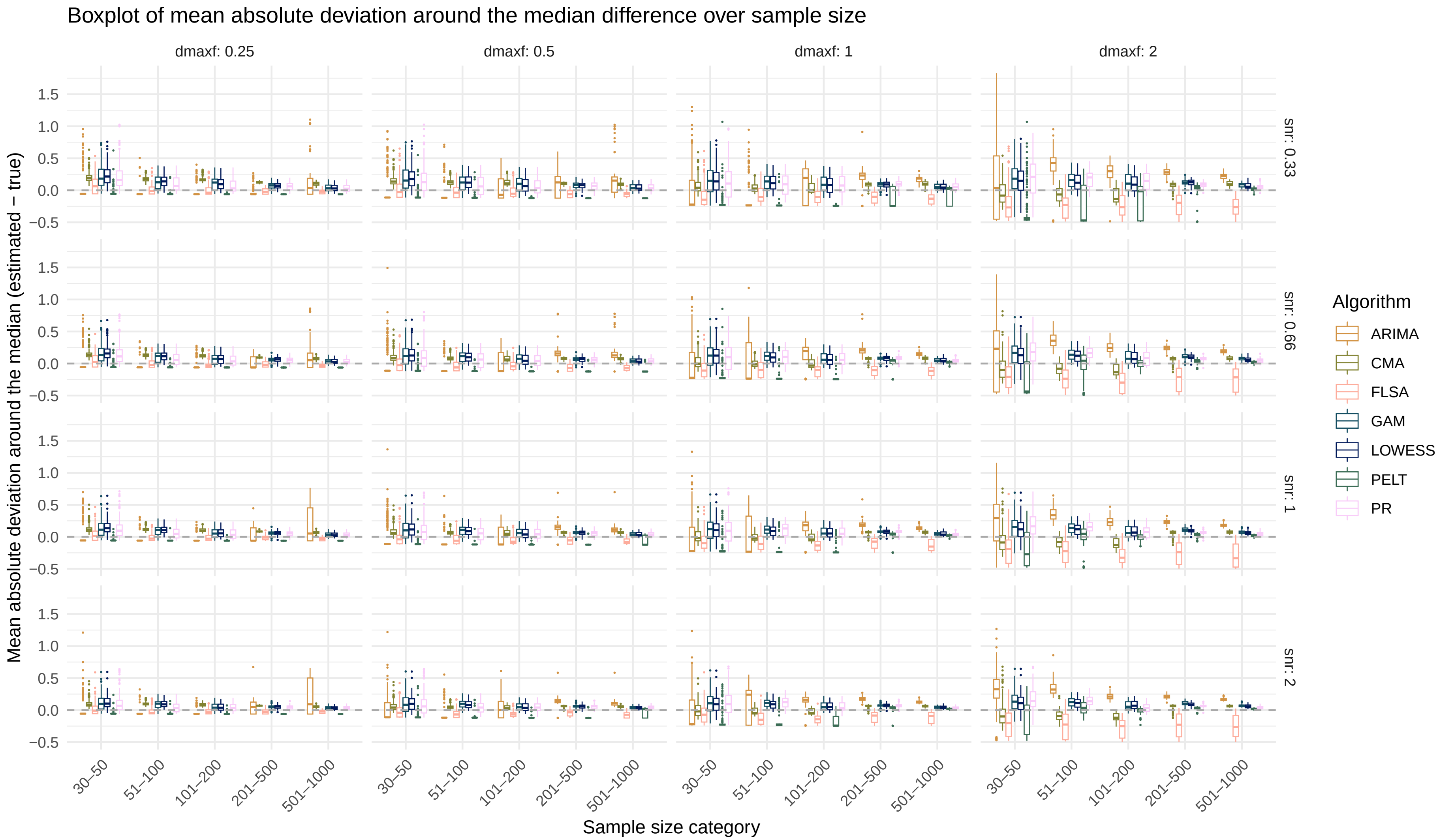
Systematic change pattern: No change



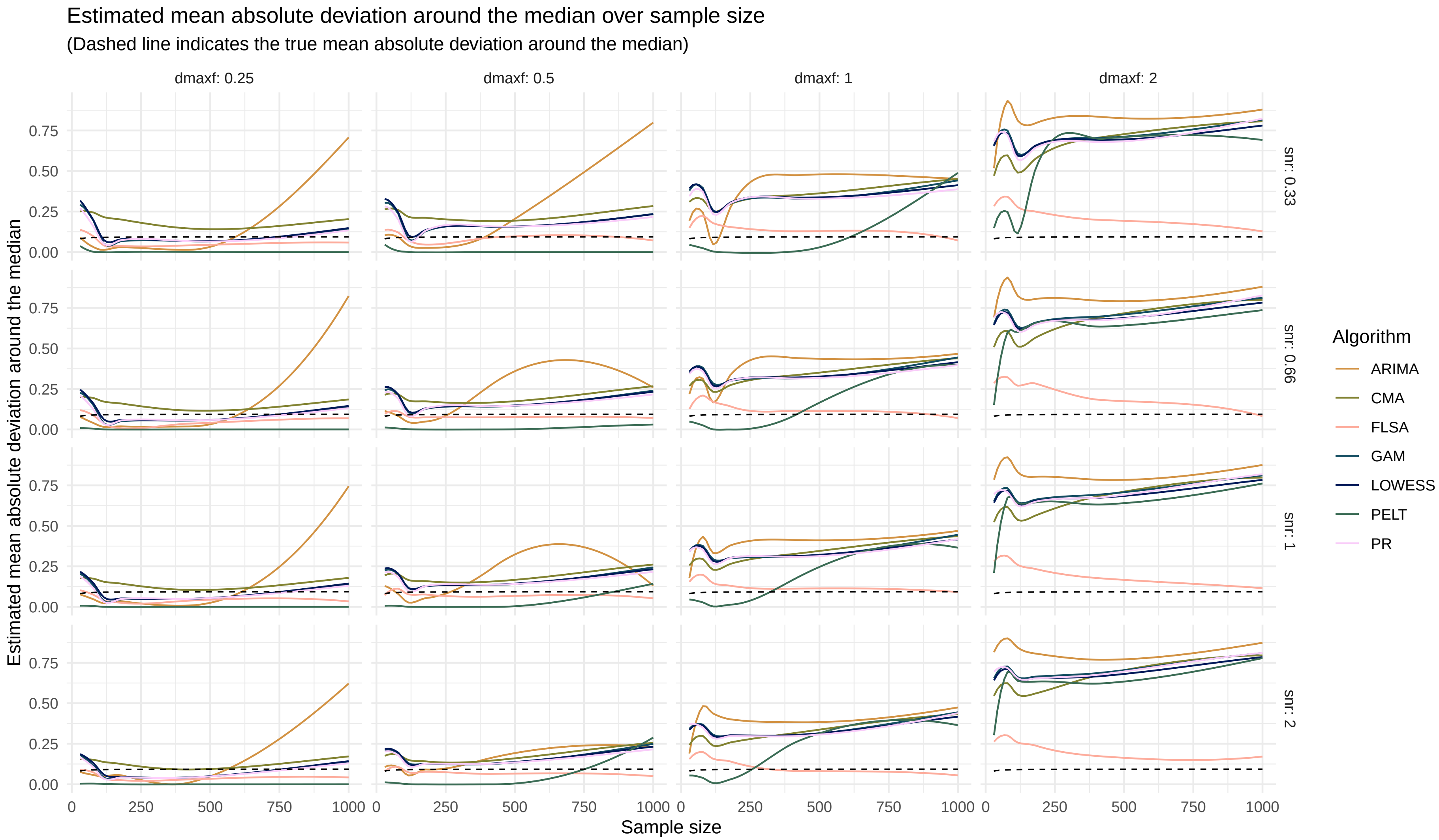

Boxplot of mean absolute deviation around the median difference over sample size
dmaxf: 0
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Mean absolute deviation around the median (estimated − true)
0.00
0.25
0.50
0.75
1.00
1.25
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

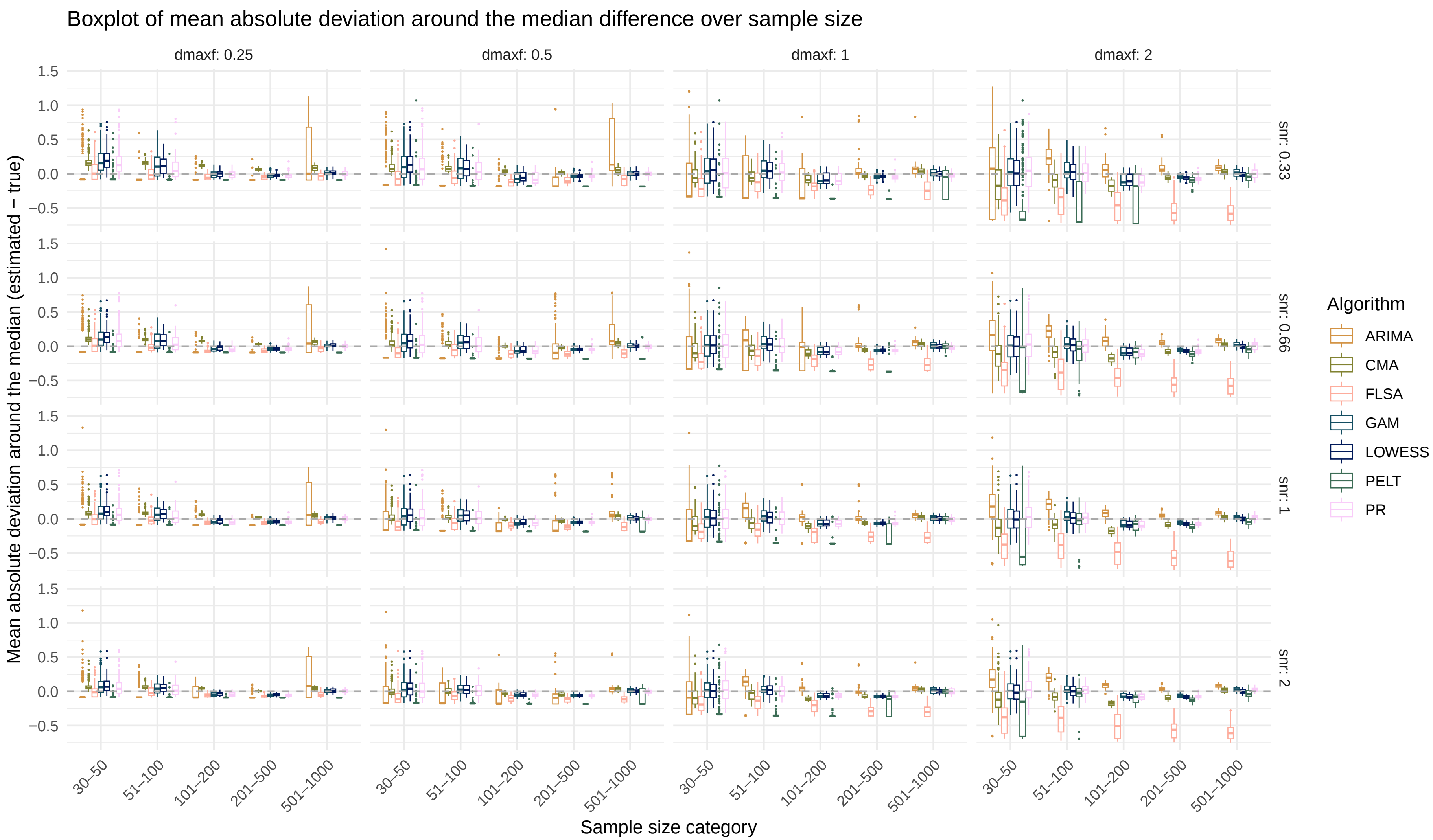
Systematic change pattern: Jump




Boxplot of mean absolute deviation around the median difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Mean absolute deviation around the median (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

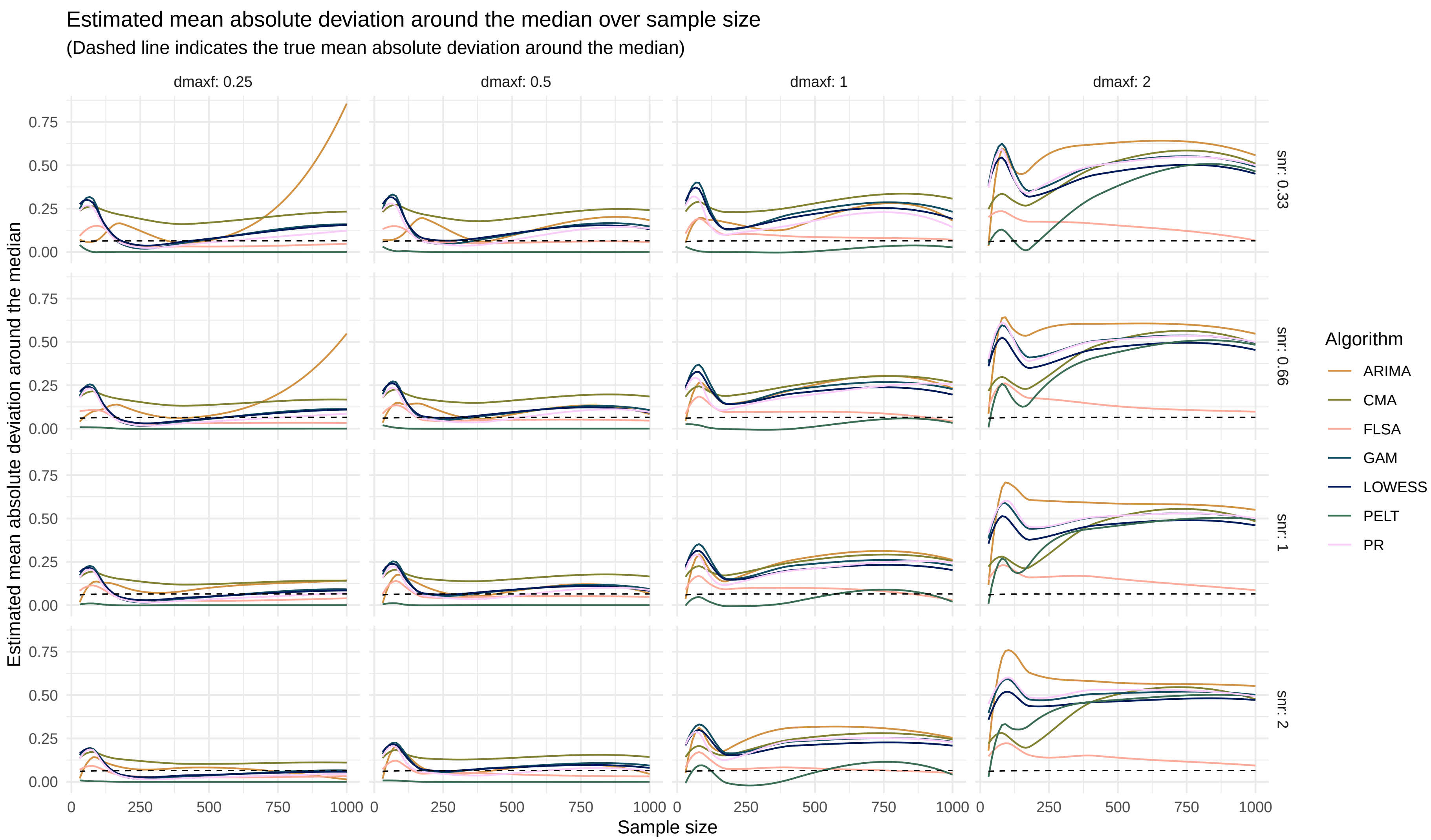
Systematic change pattern: Linear

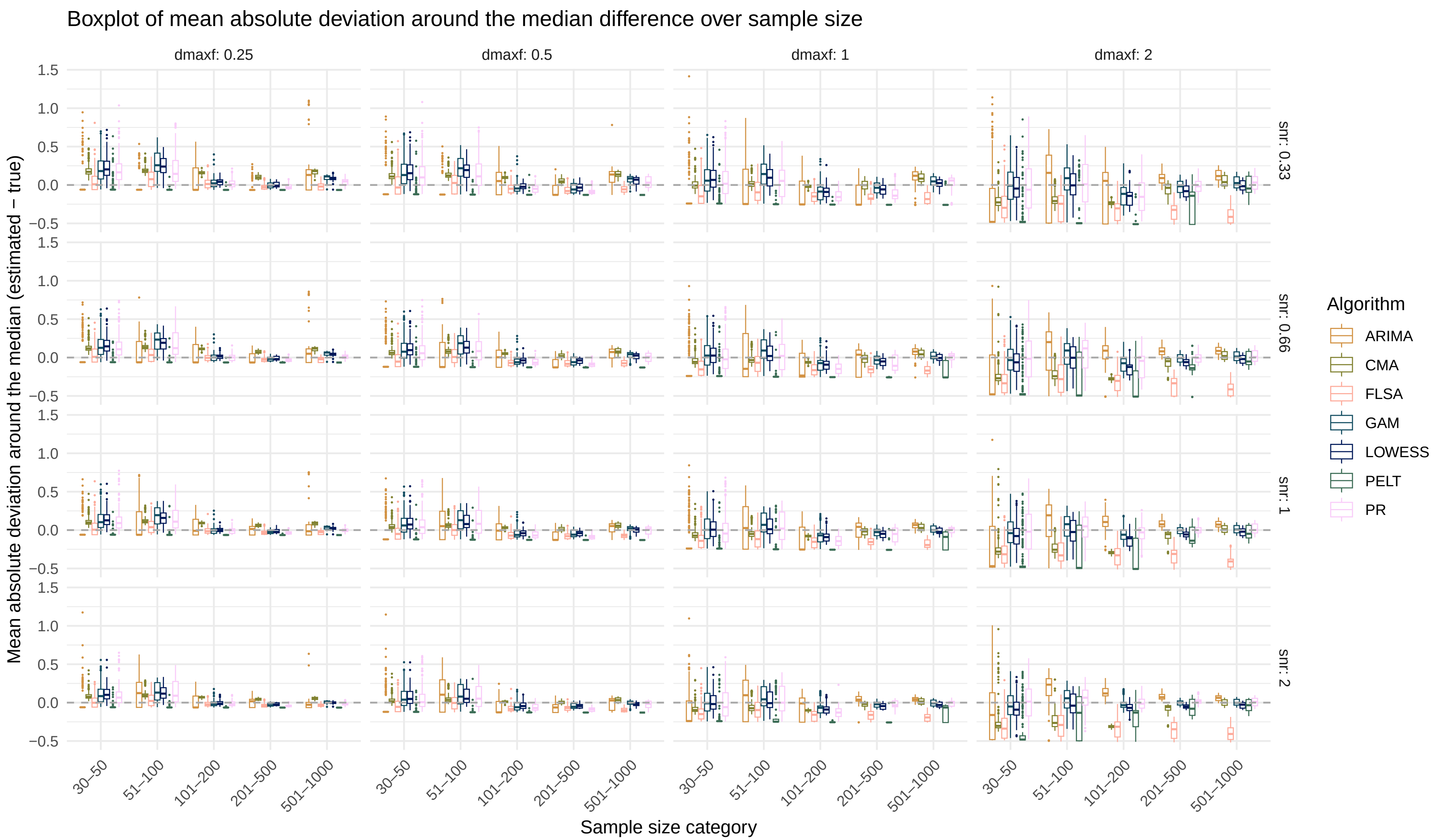




Boxplot of mean absolute deviation around the median difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Mean absolute deviation around the median (estimated – true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

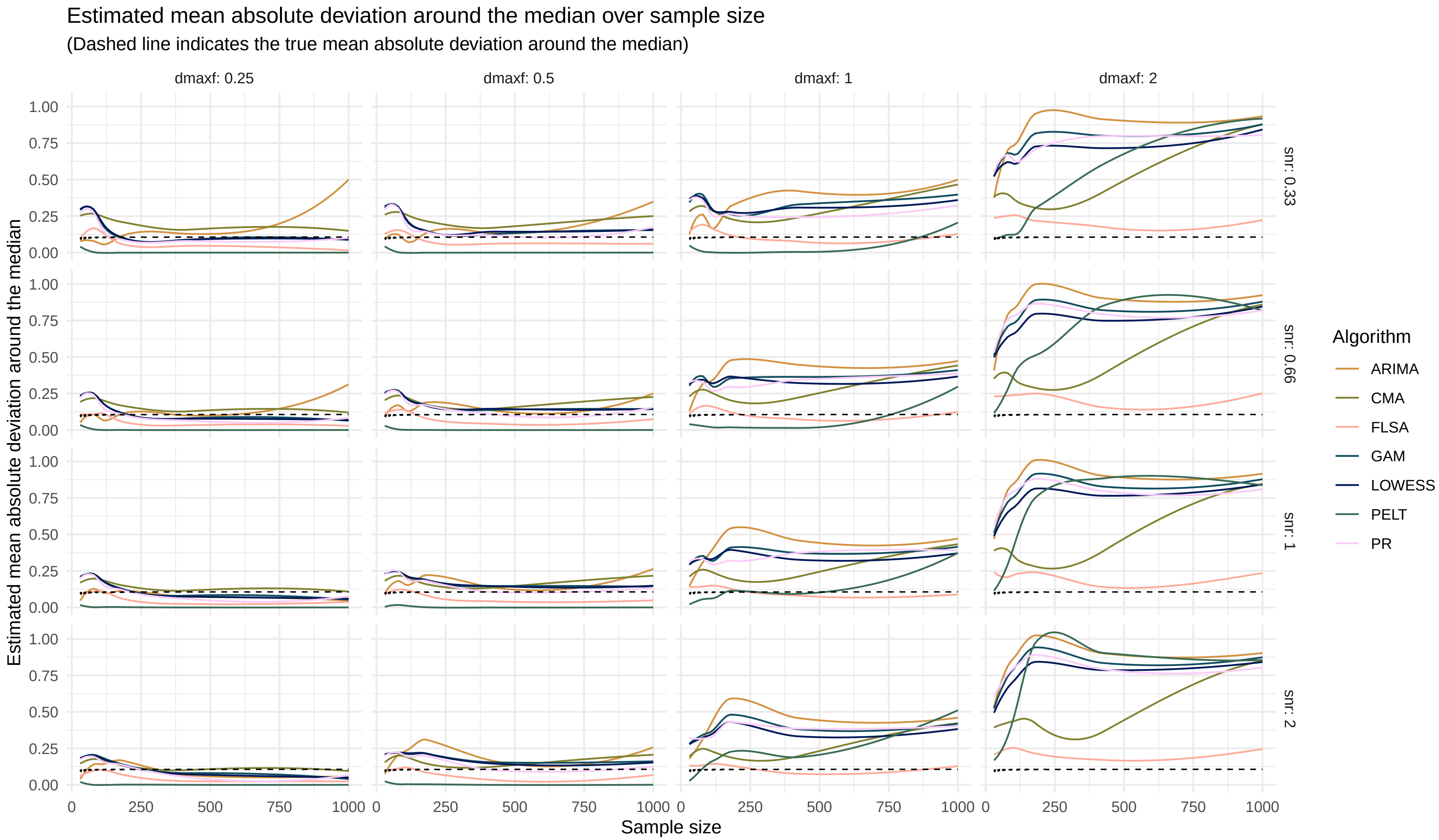
Systematic change pattern: Quadratic

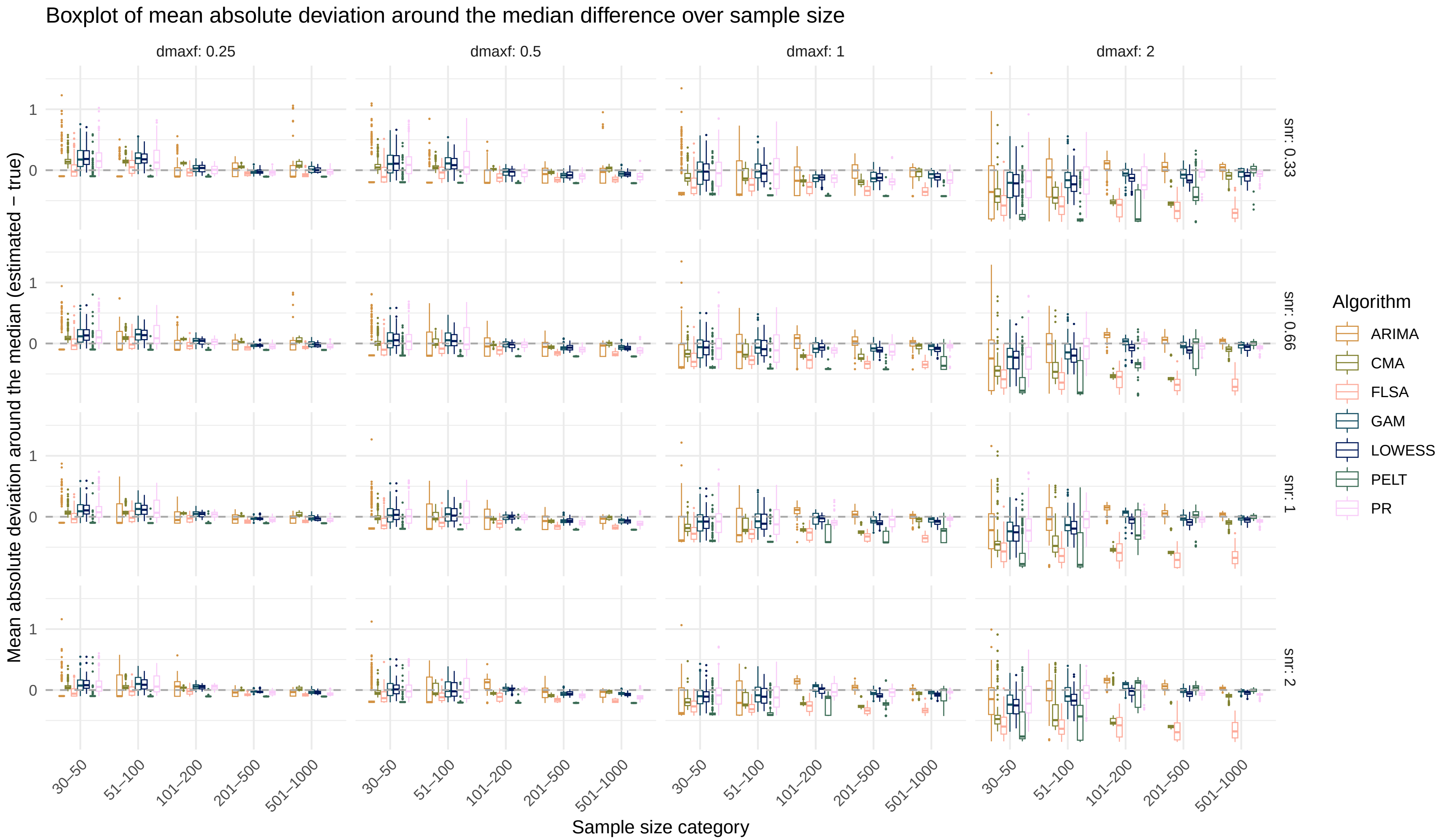




Boxplot of mean absolute deviation around the median difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Mean absolute deviation around the median (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

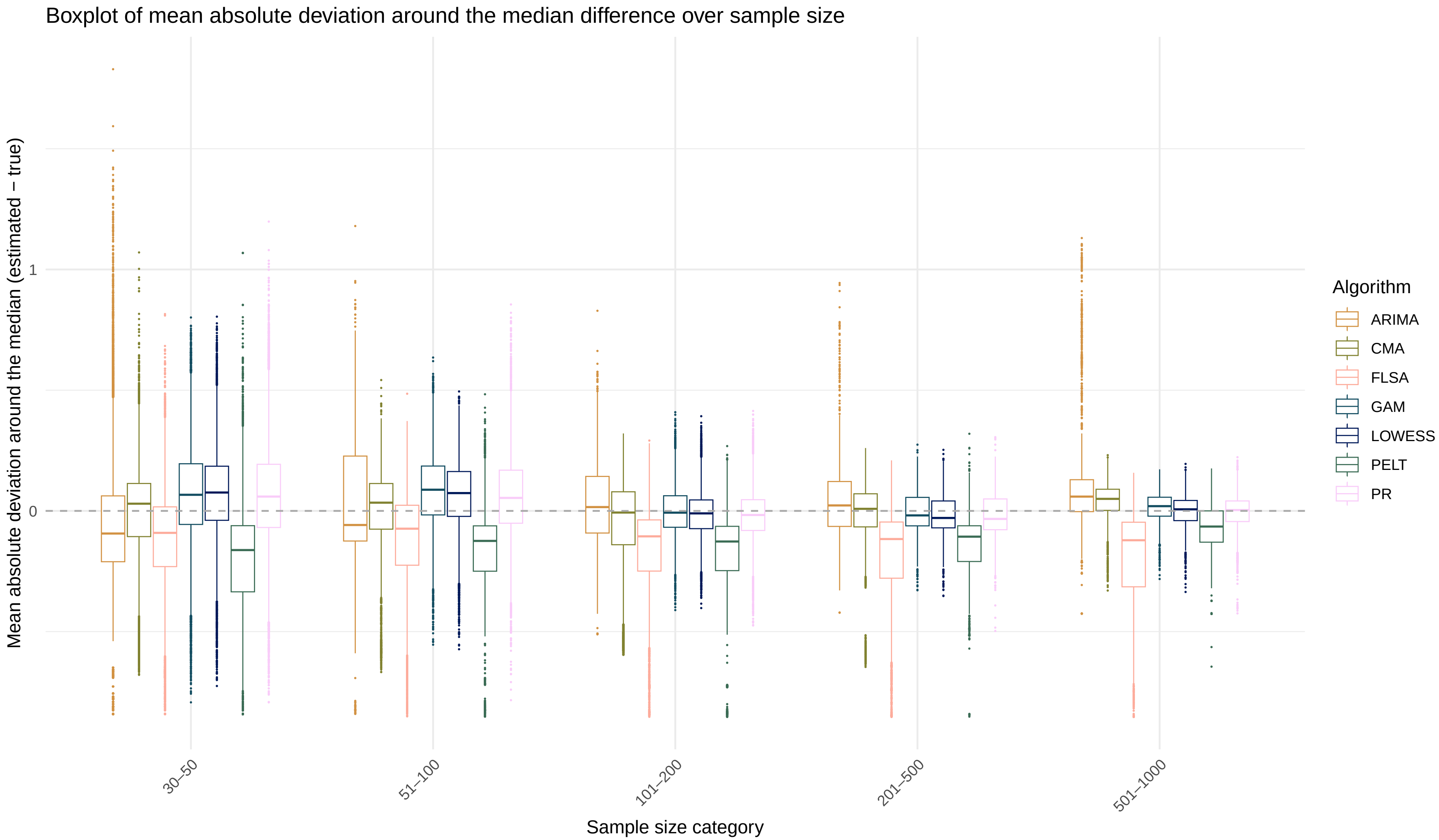
Systematic change pattern: Recurrent

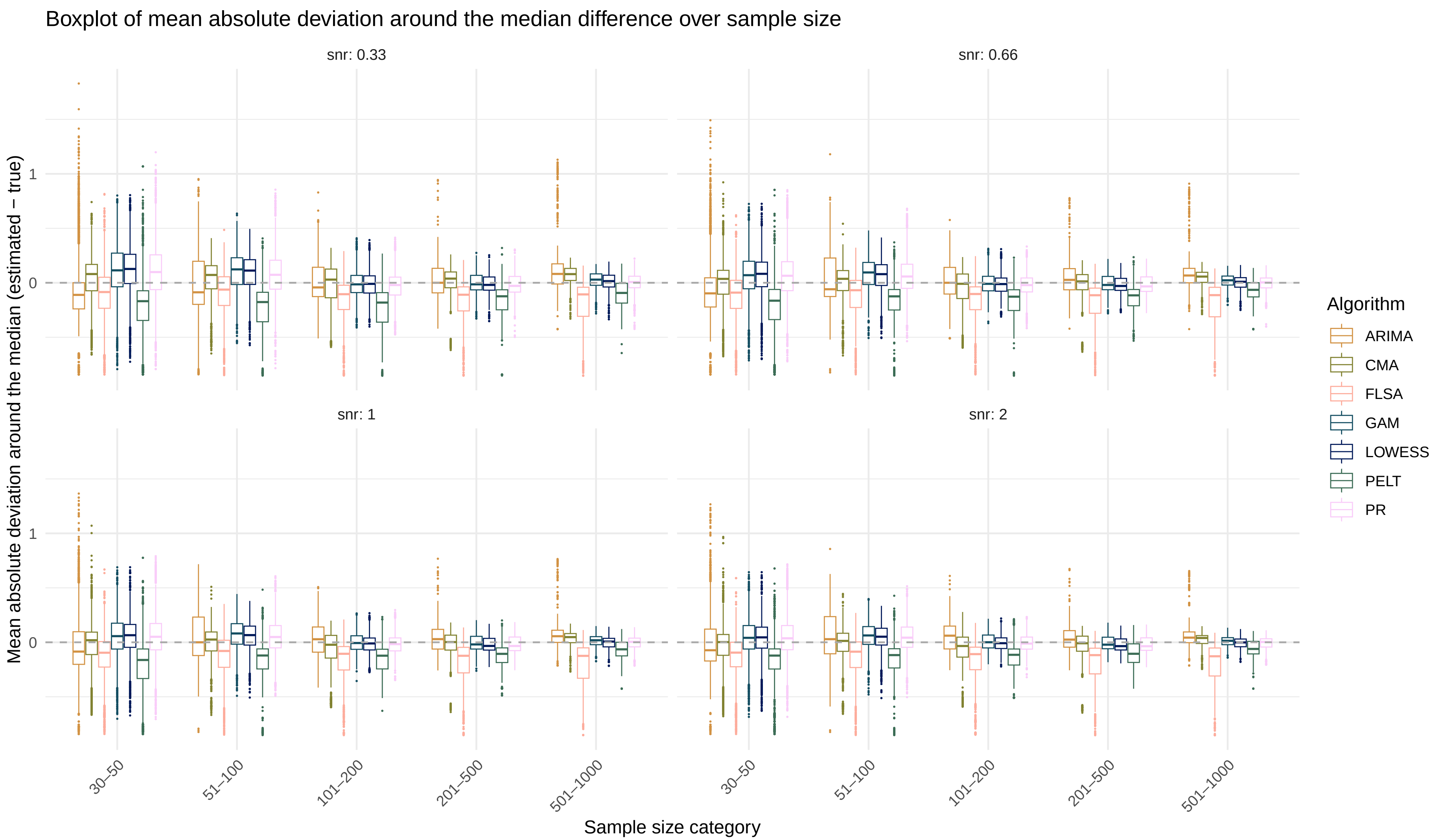




Boxplot of mean absolute deviation around the median difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Mean absolute deviation around the median (estimated − true)
1
0
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

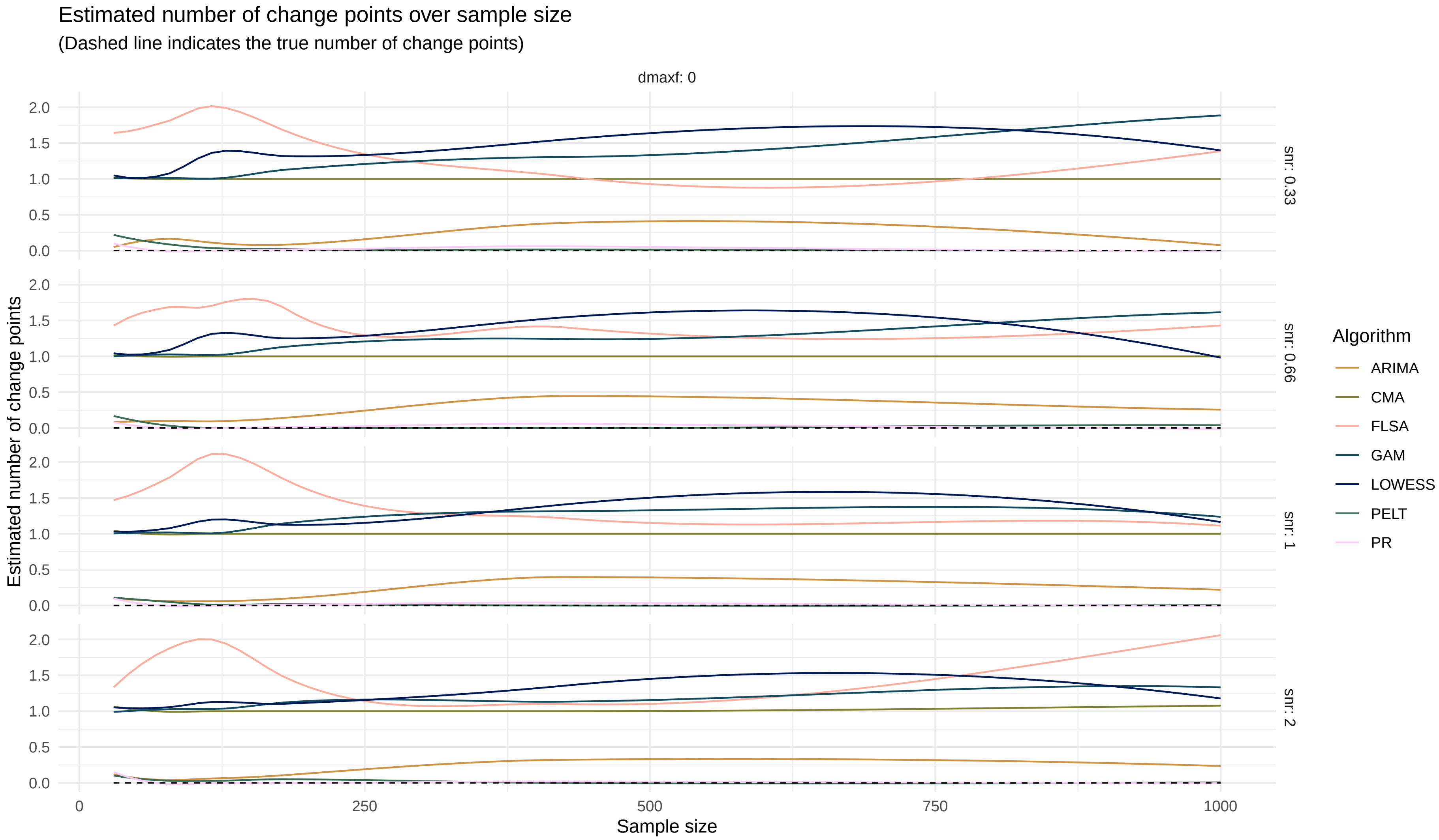
Systematic change pattern: All patterns combined

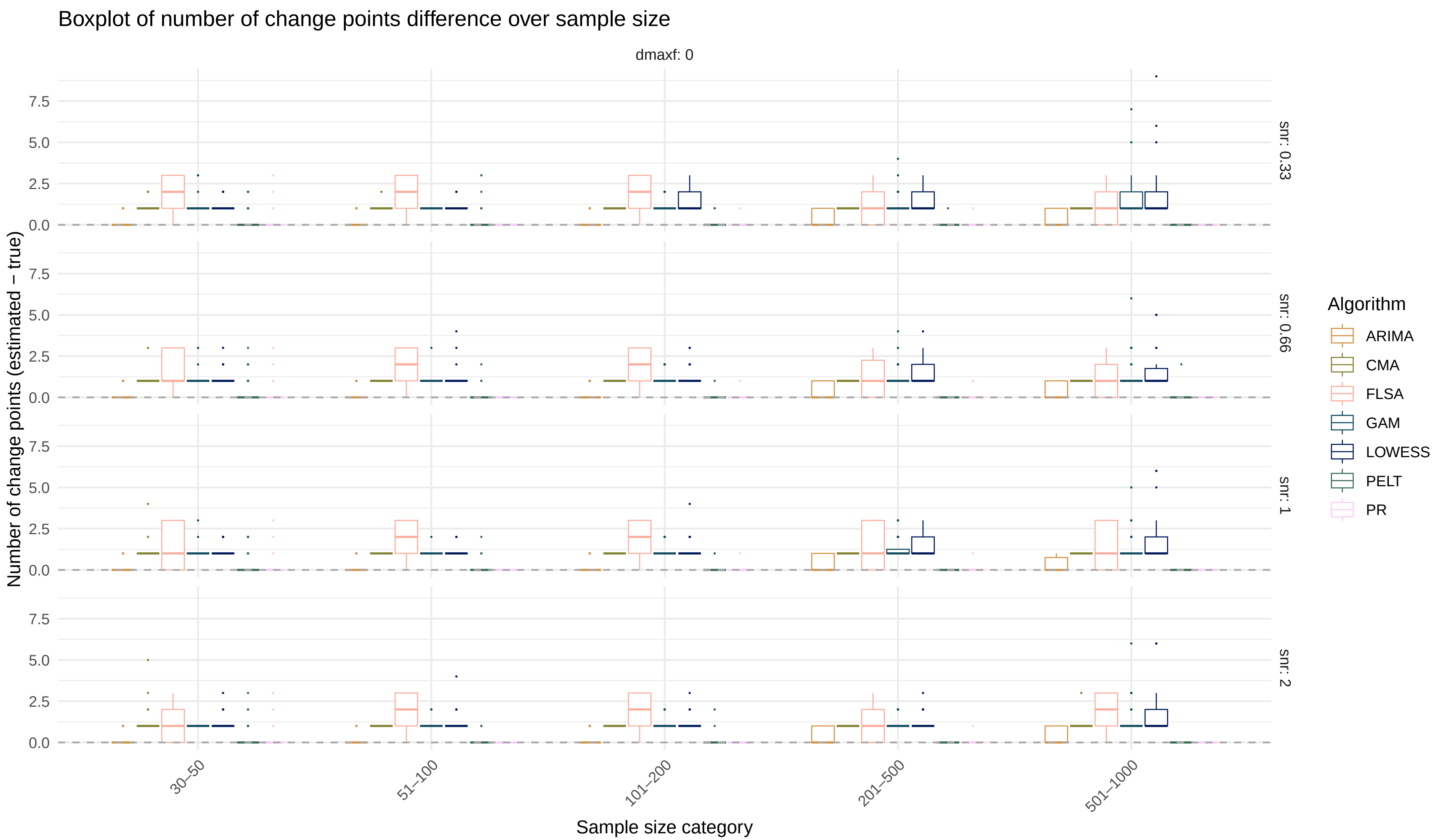



Boxplot of mean absolute deviation around the median difference over sample size
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Mean absolute deviation around the median (estimated − true)
1
0
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

Estimand: Number of change points

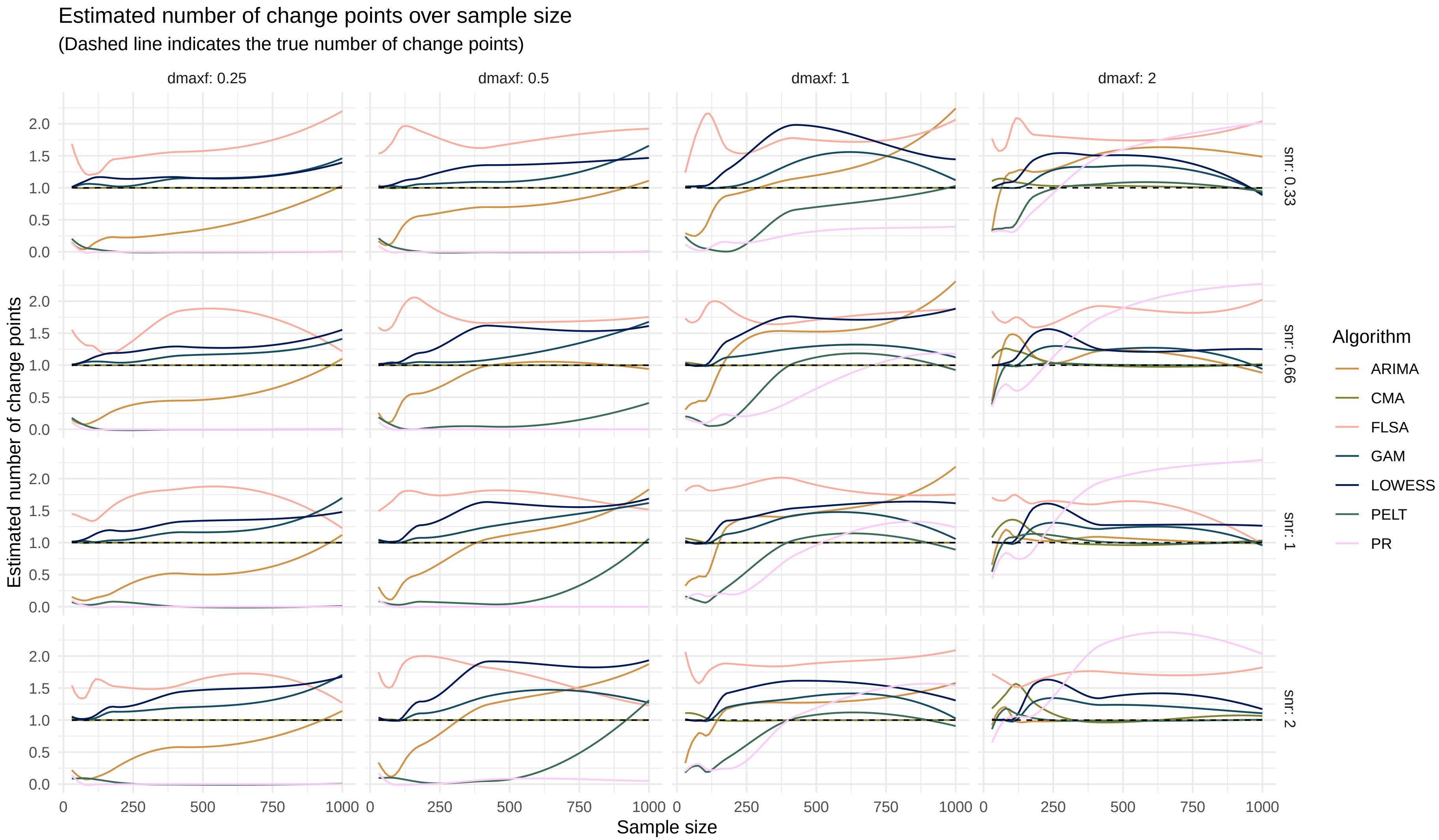
Systematic change pattern: No change

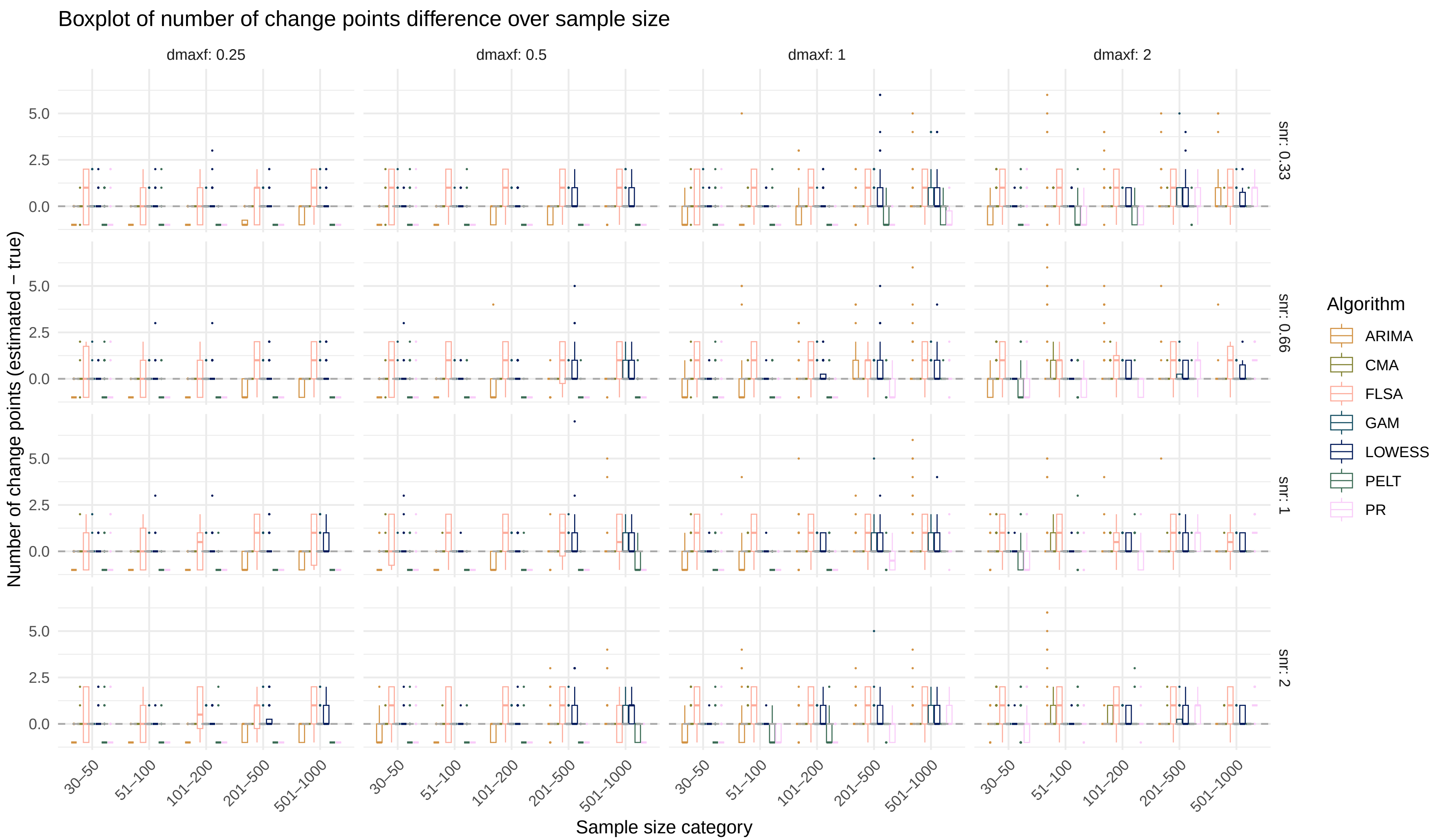




Boxplot of number of change points difference over sample size
dmaxf: 0
Number of change points (estimated − true)
7.5
5.0
2.5
0.0
snr: 0.33
snr: 0.66
snr: 1
snr: 2
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

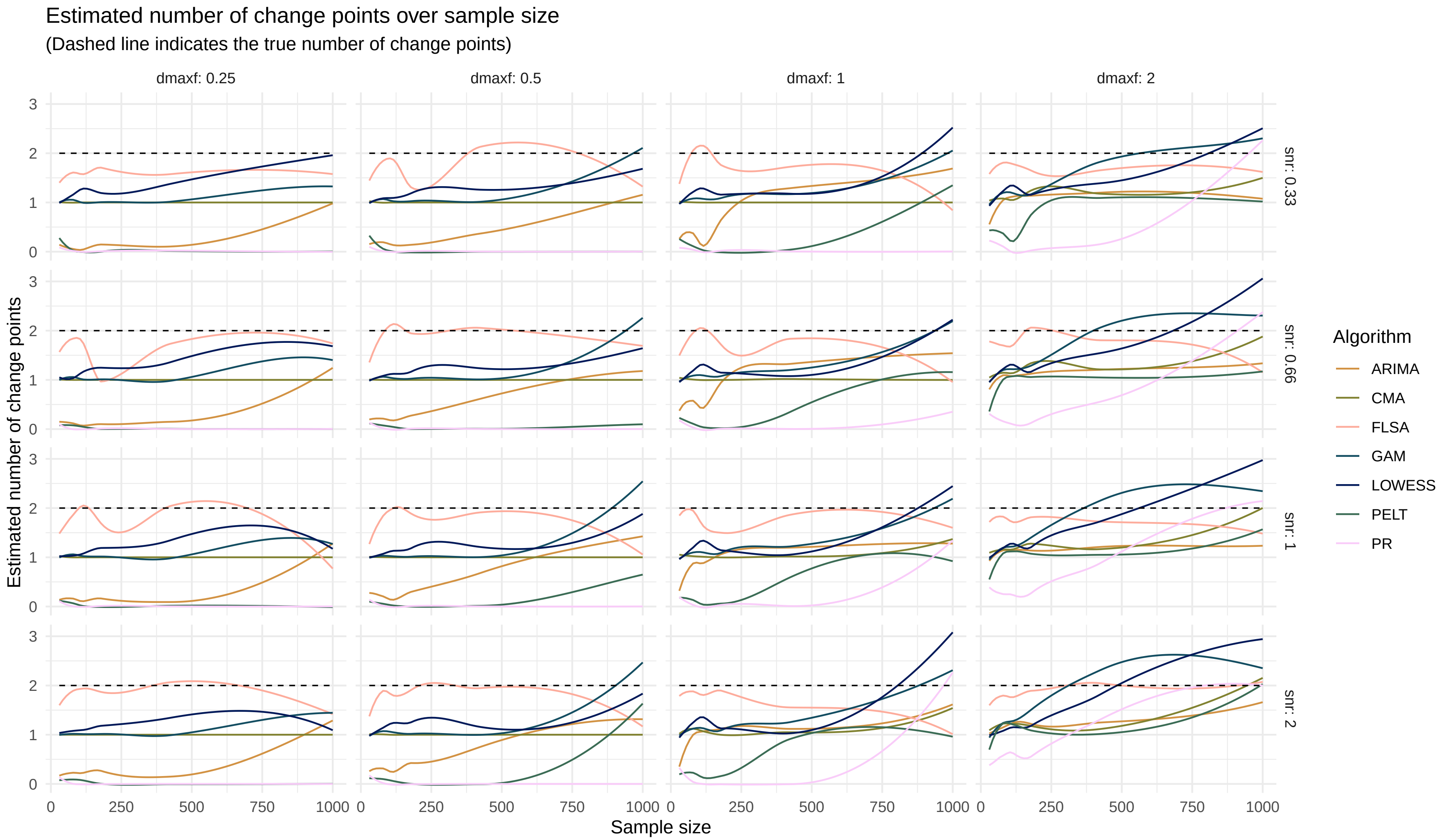
Systematic change pattern: Jump

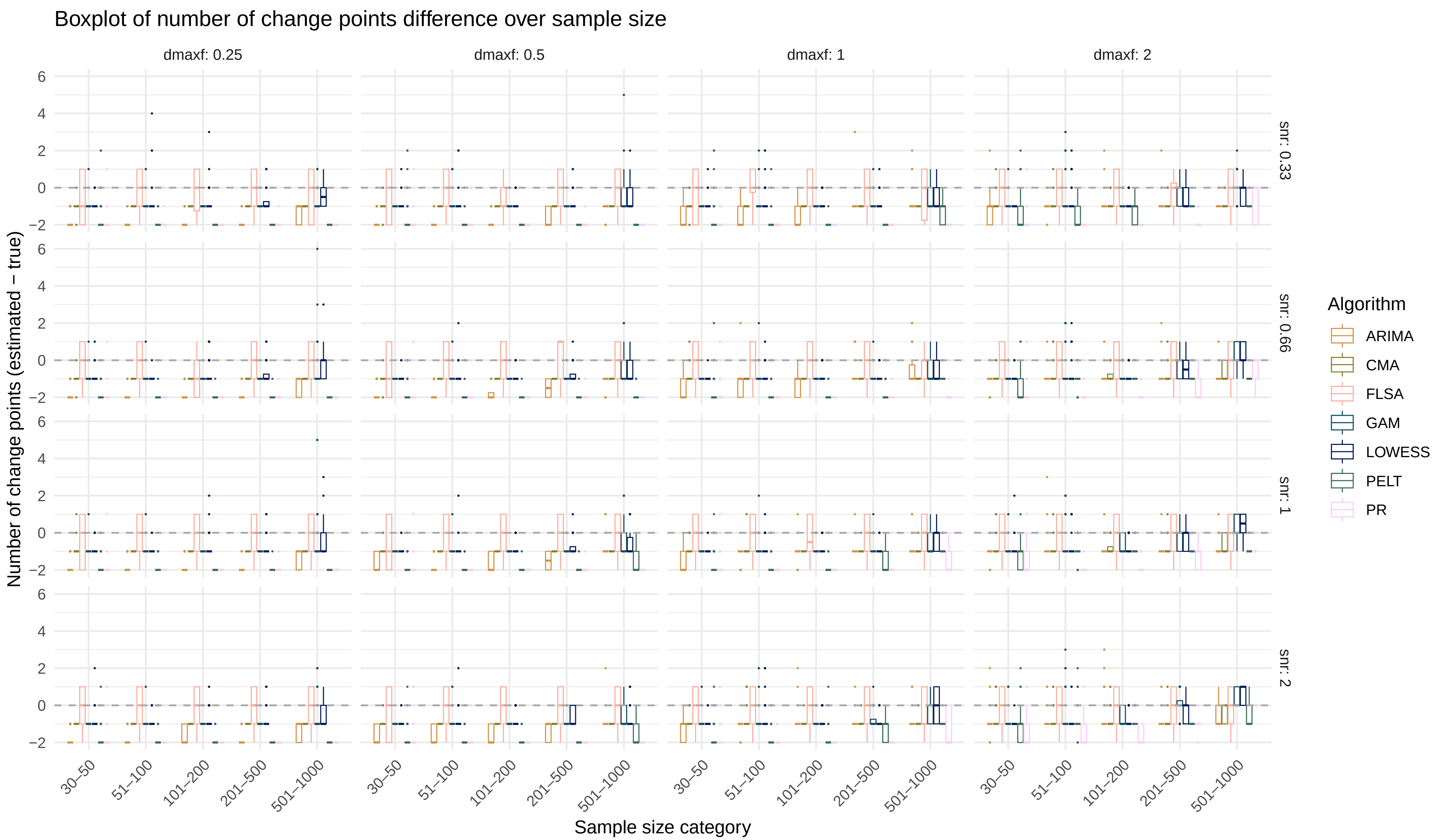




Boxplot of number of change points difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Number of change points (estimated – true)
5.0
2.5
0.0
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

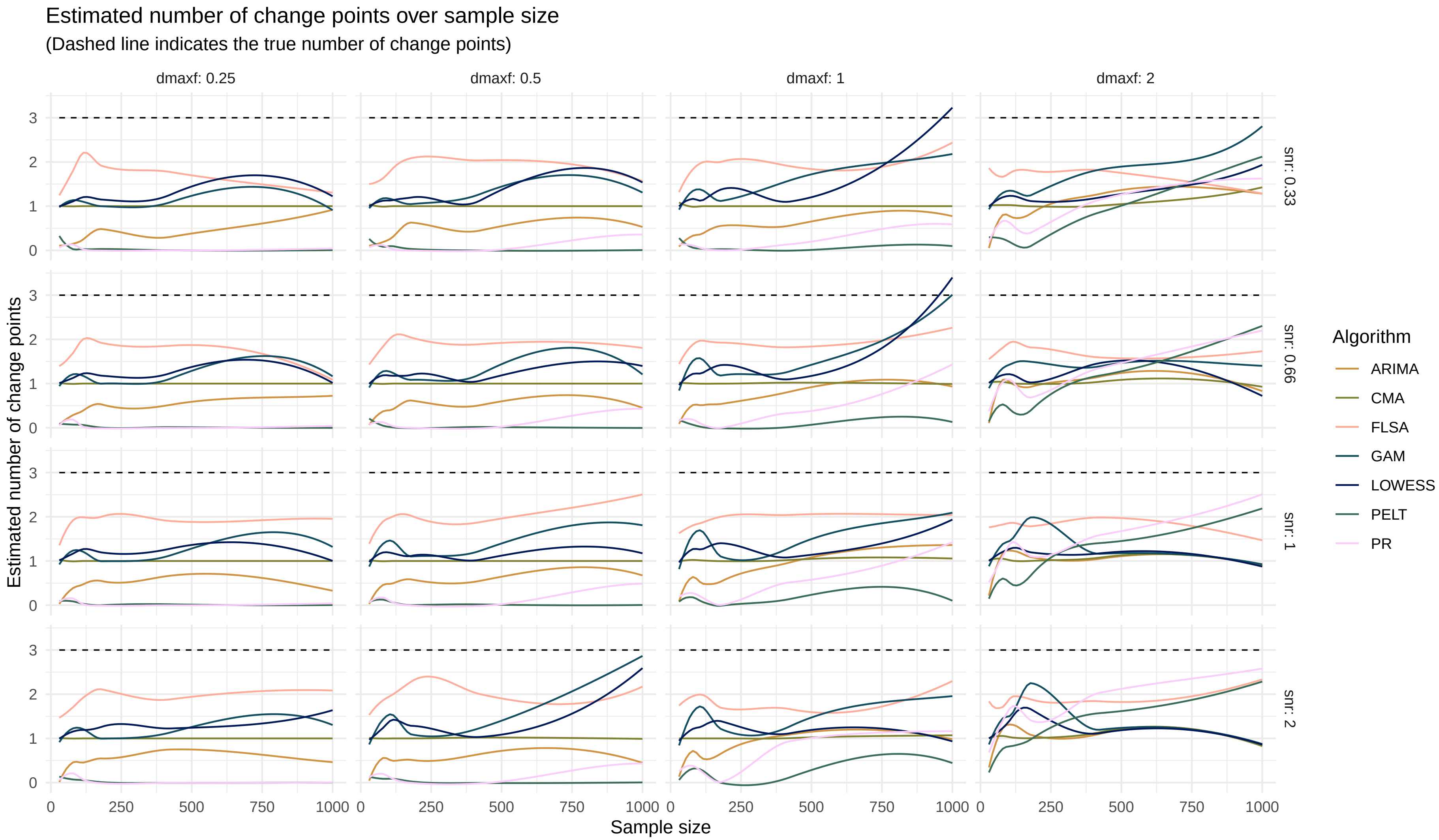
Systematic change pattern: Linear

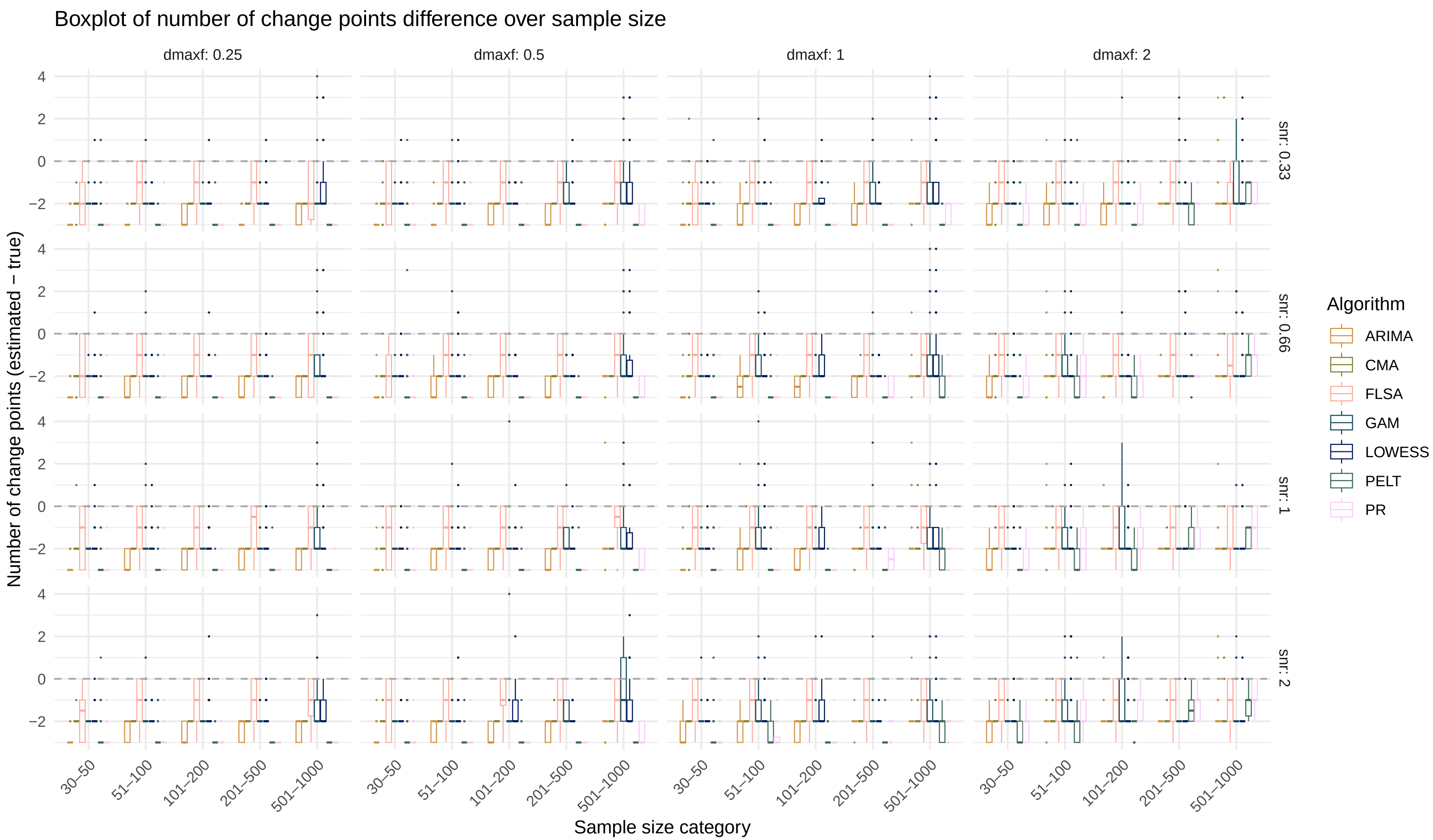




Boxplot of number of change points difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Number of change points (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

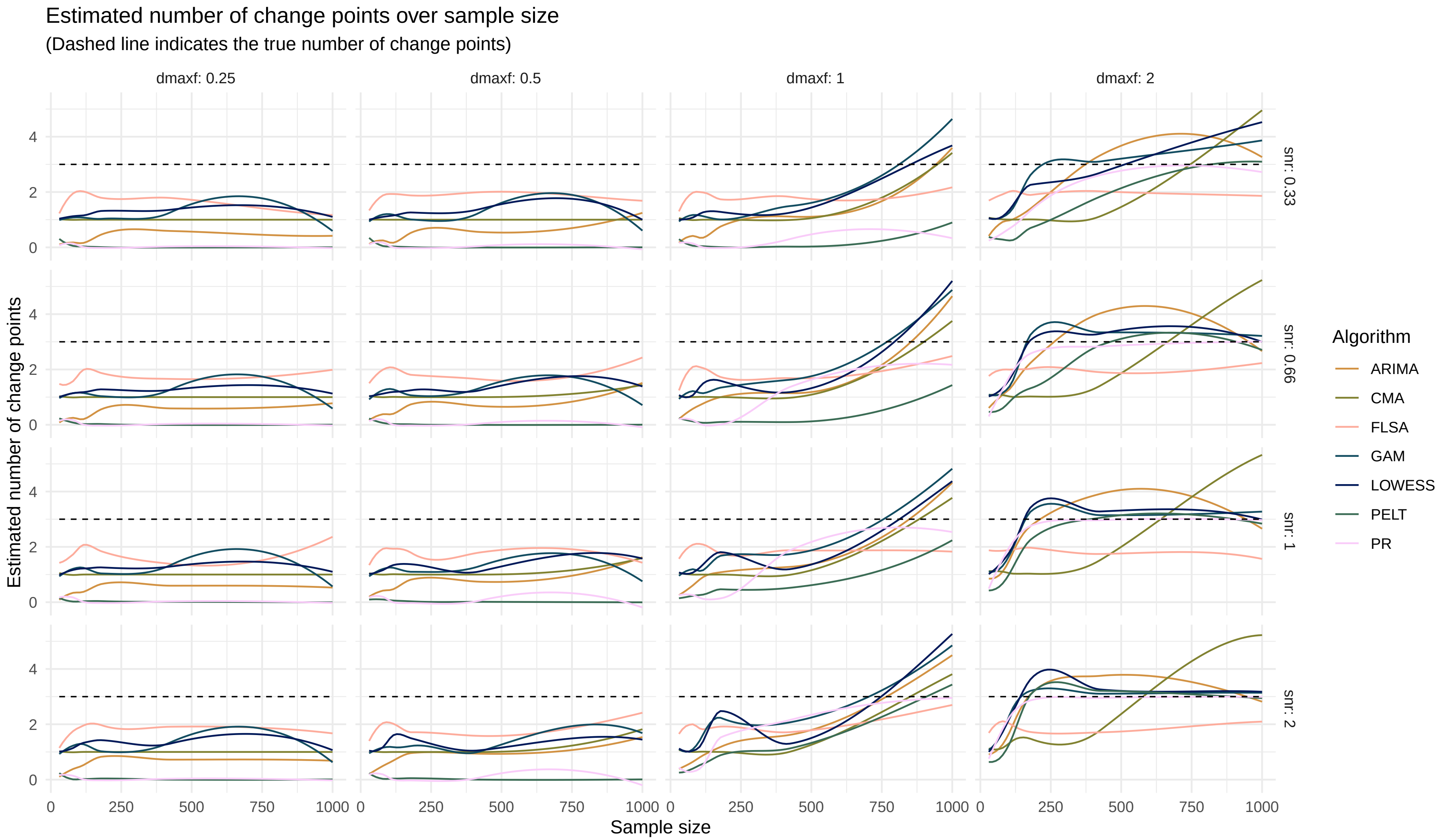
Systematic change pattern: Quadratic

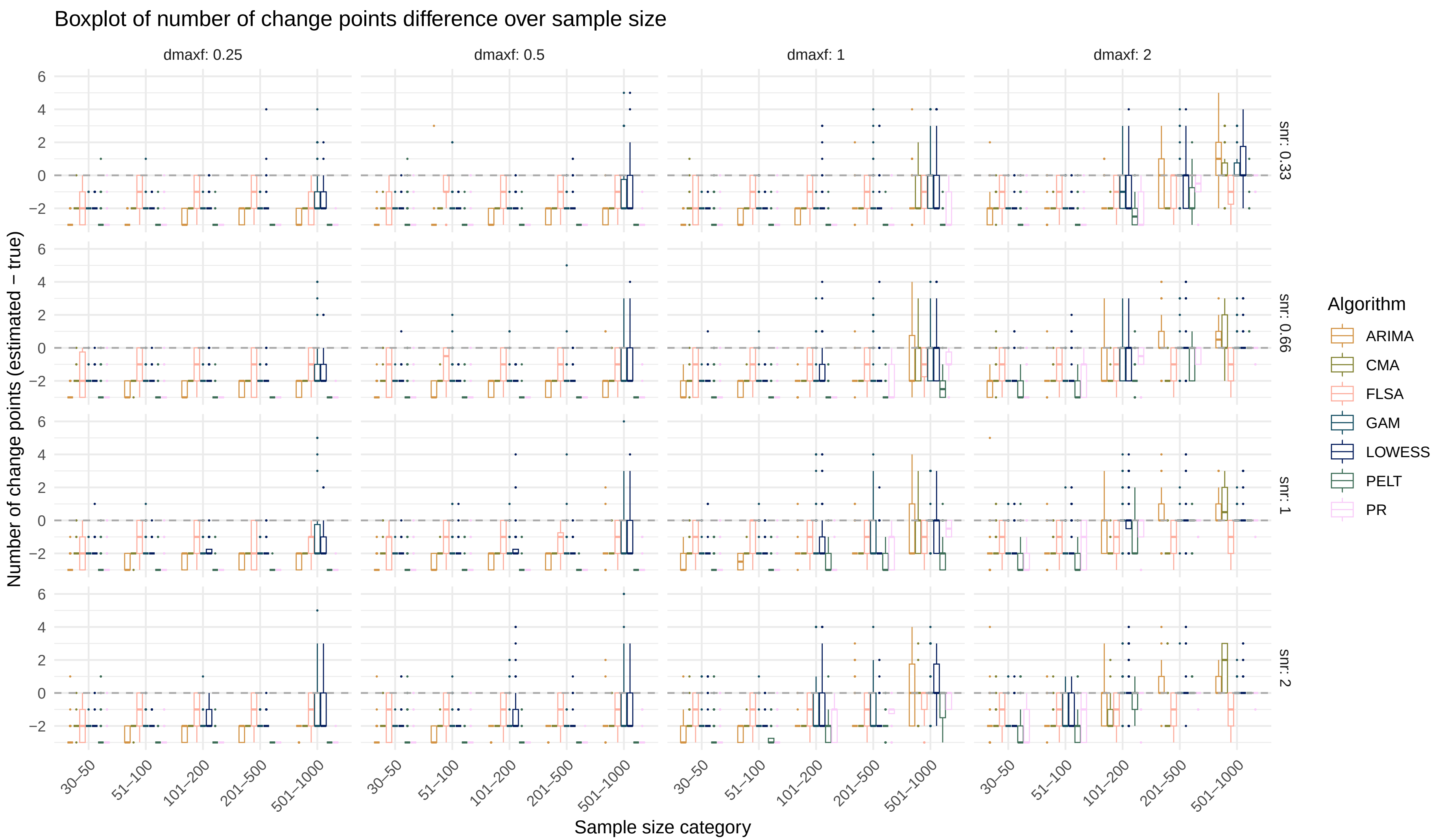




Boxplot of number of change points difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Number of change points (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

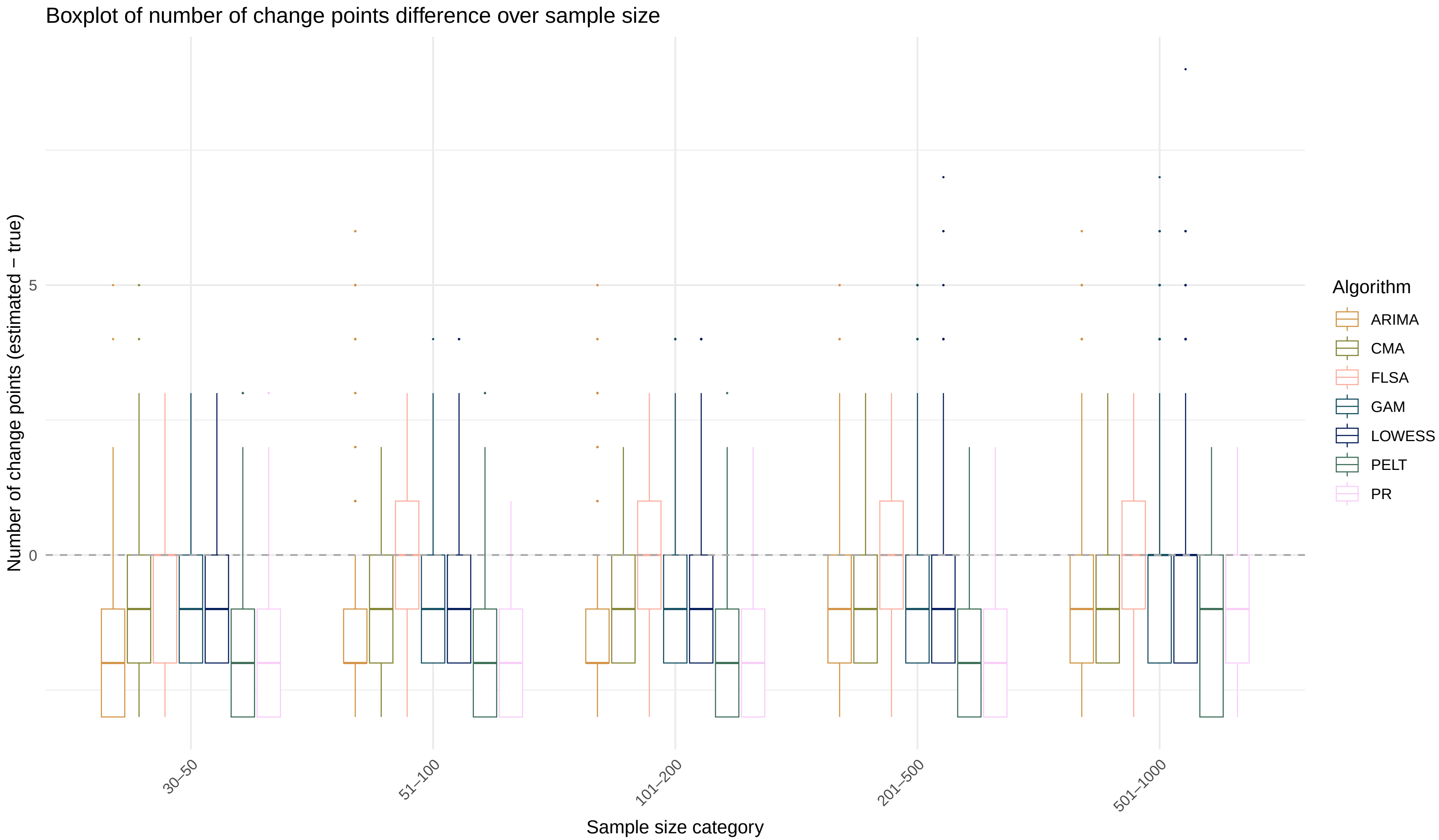
Systematic change pattern: Recurrent

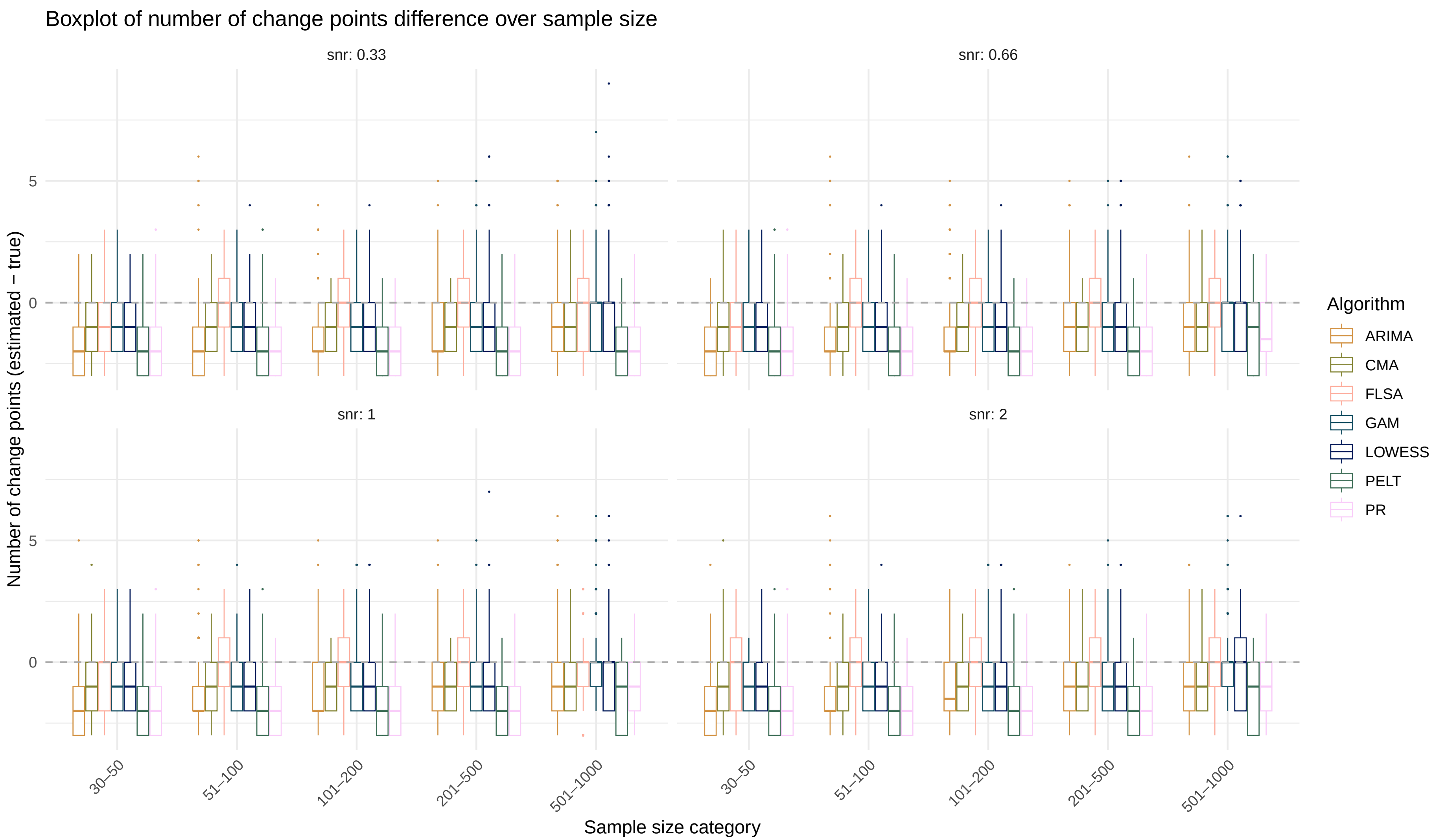




Boxplot of number of change points difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Number of change points (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

Systematic change pattern: All patterns combined

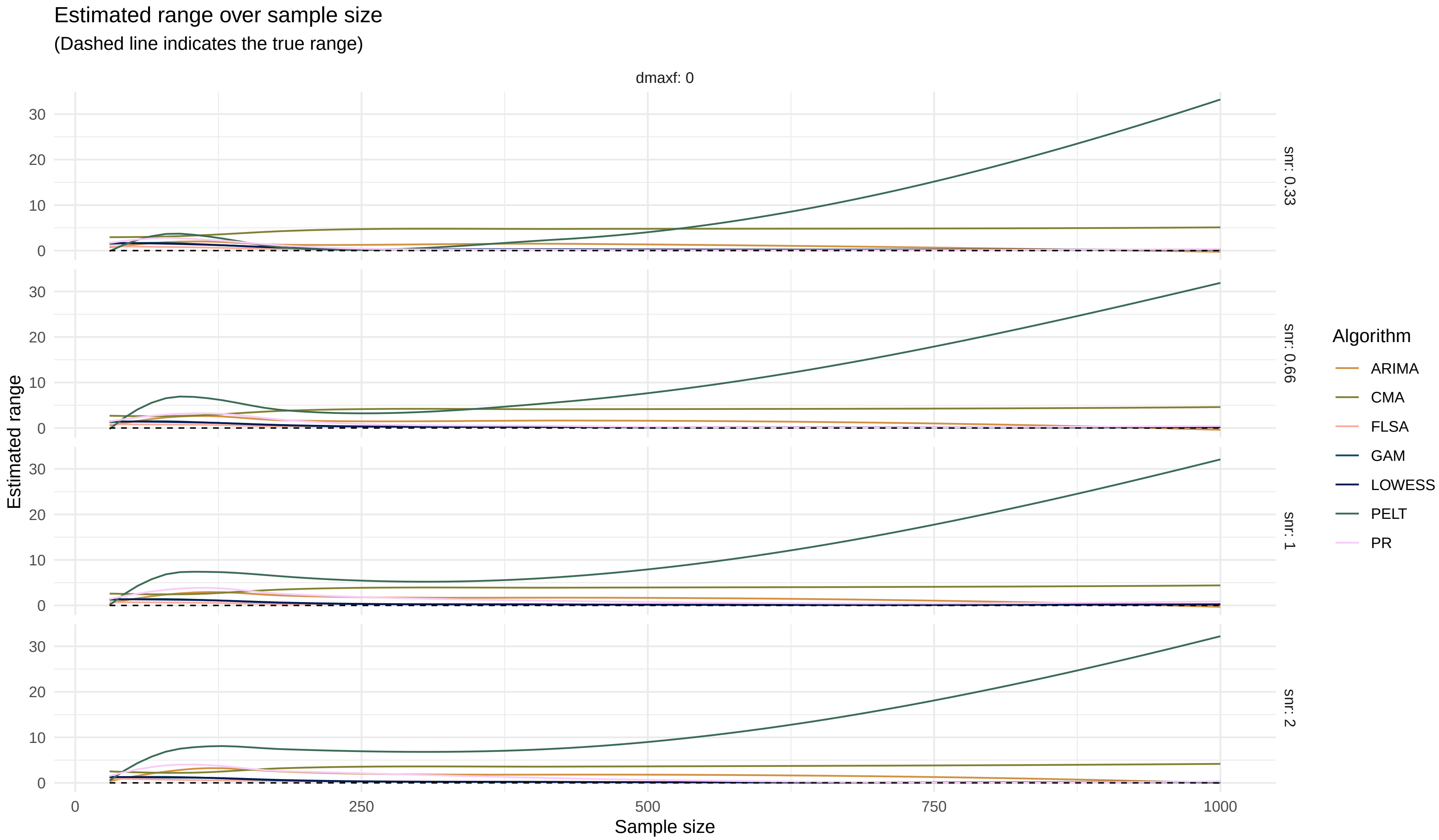


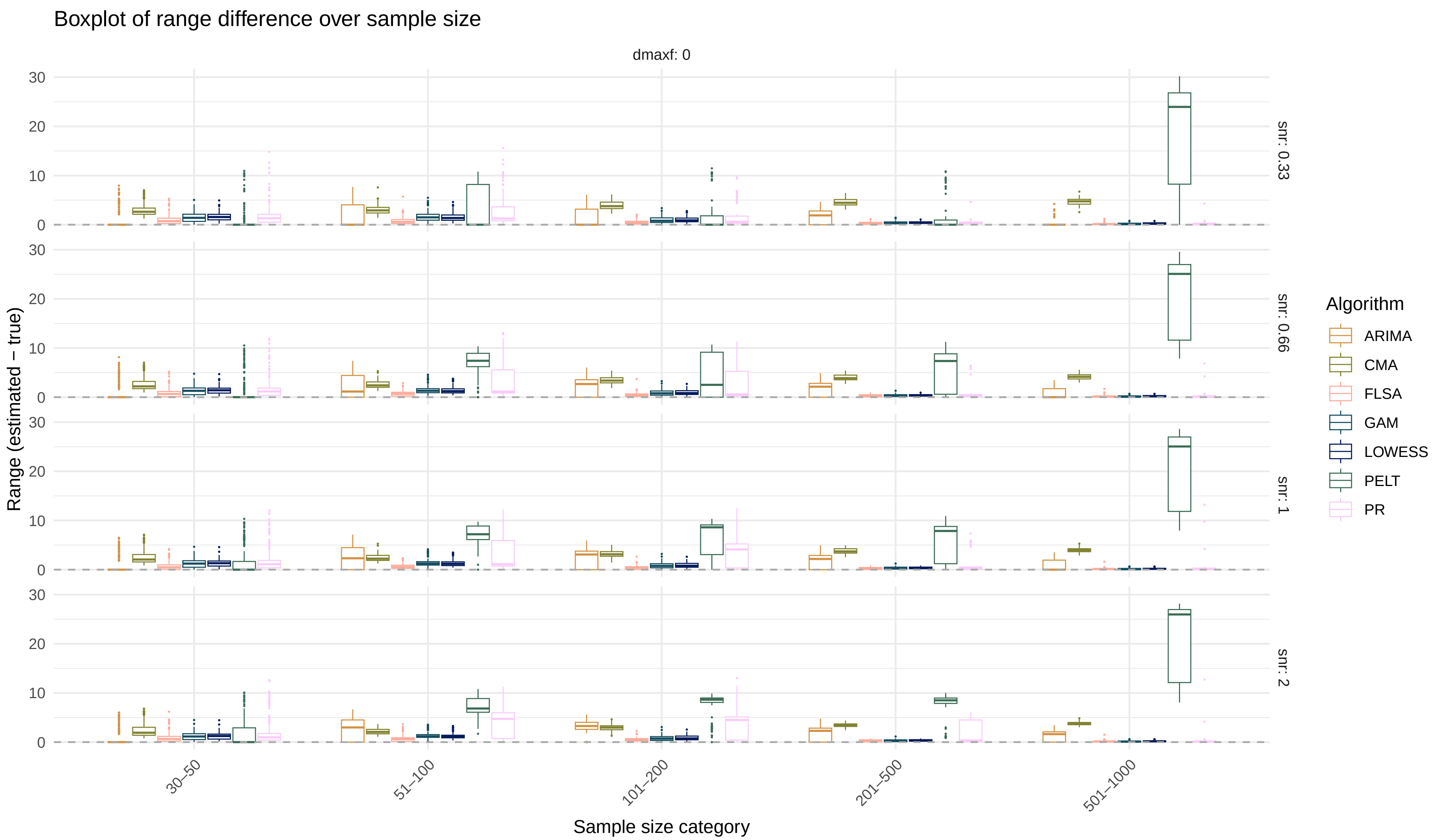

Boxplot of number of change points difference over sample size
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Number of change points (estimated − true)
5
0
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

Log-normal distribution

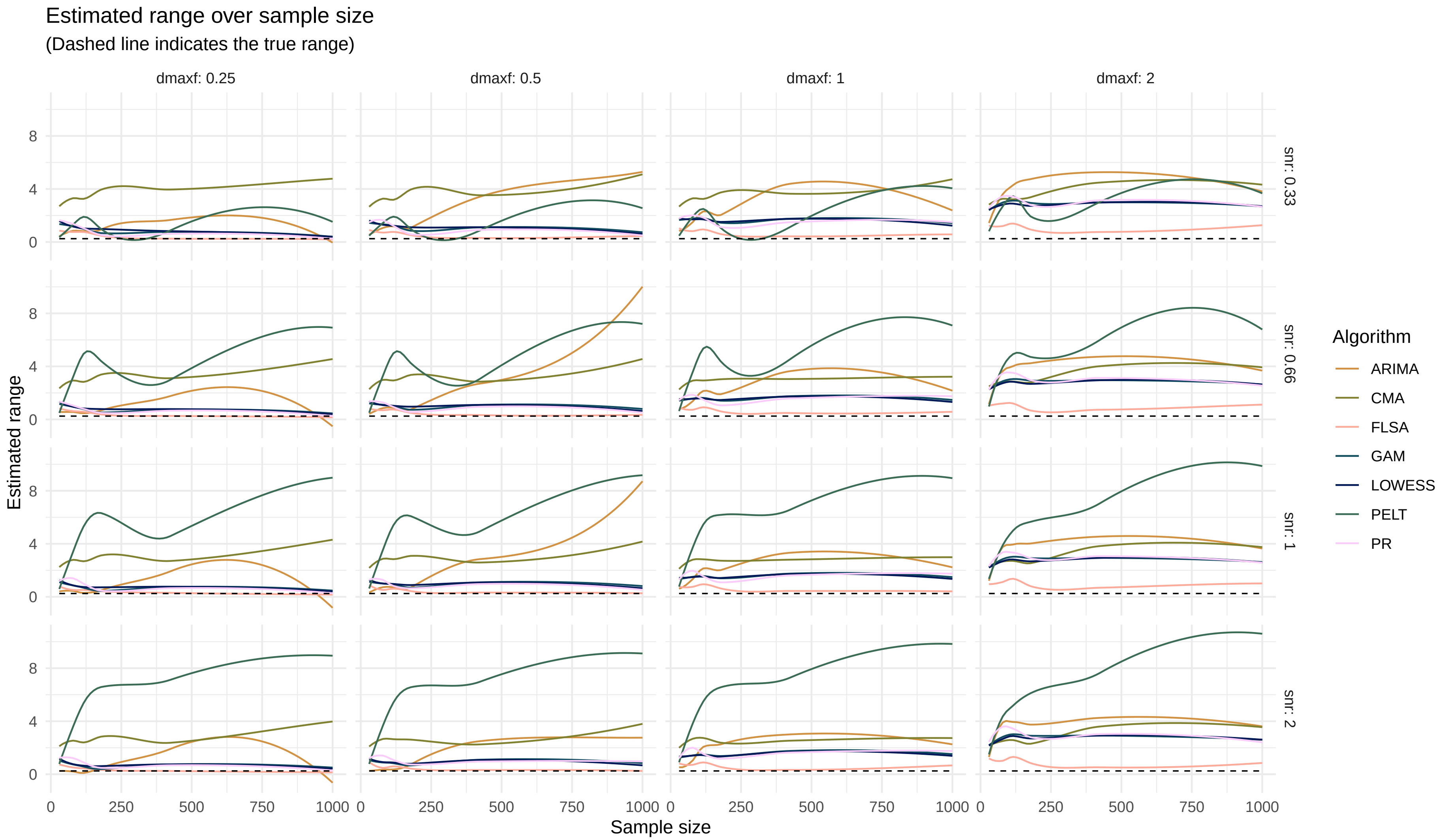
Estimand: Range

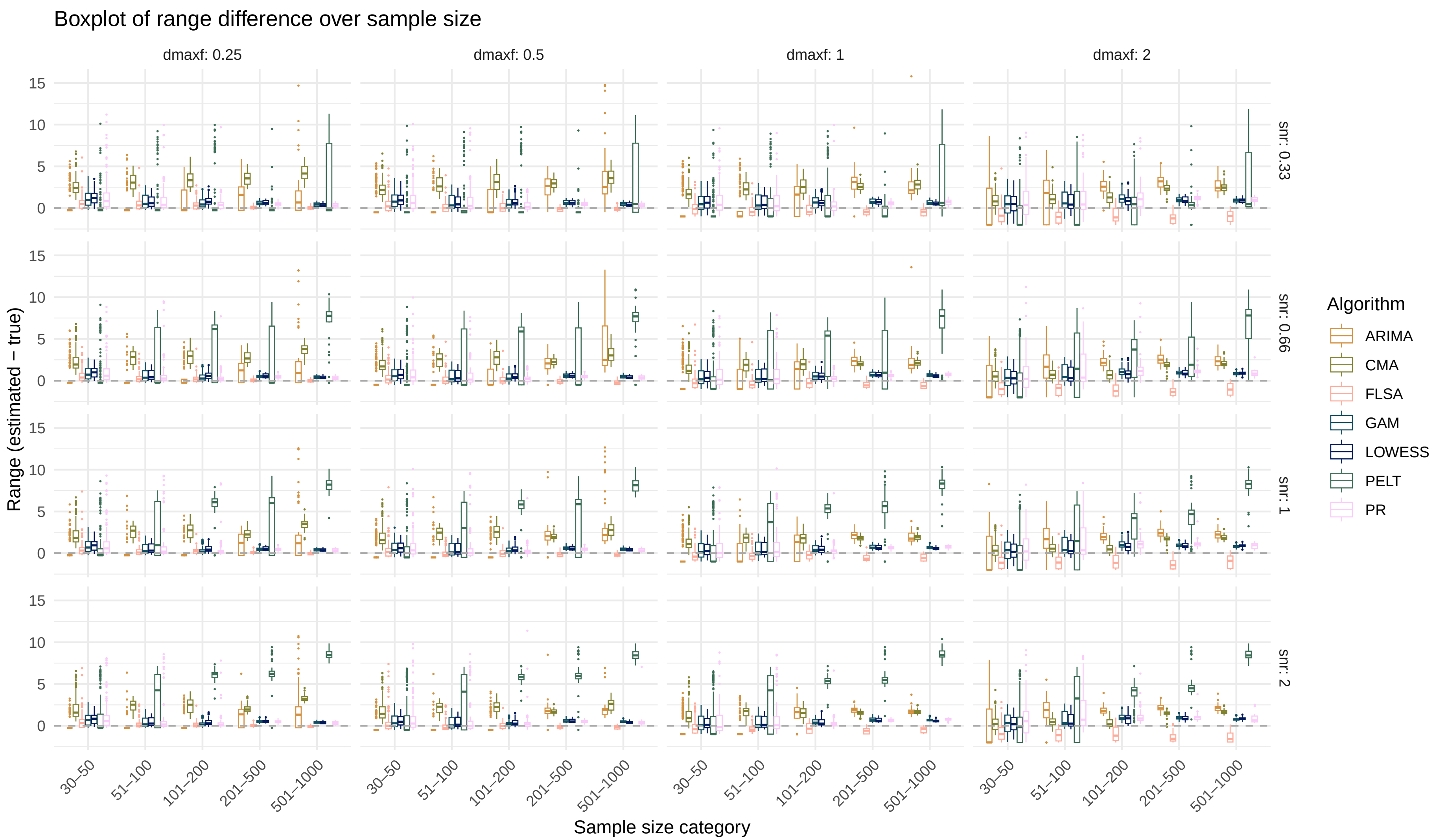
Systematic change pattern: No change




Boxplot of range difference over sample size
dmaxf: 0
Range (estimated − true)
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

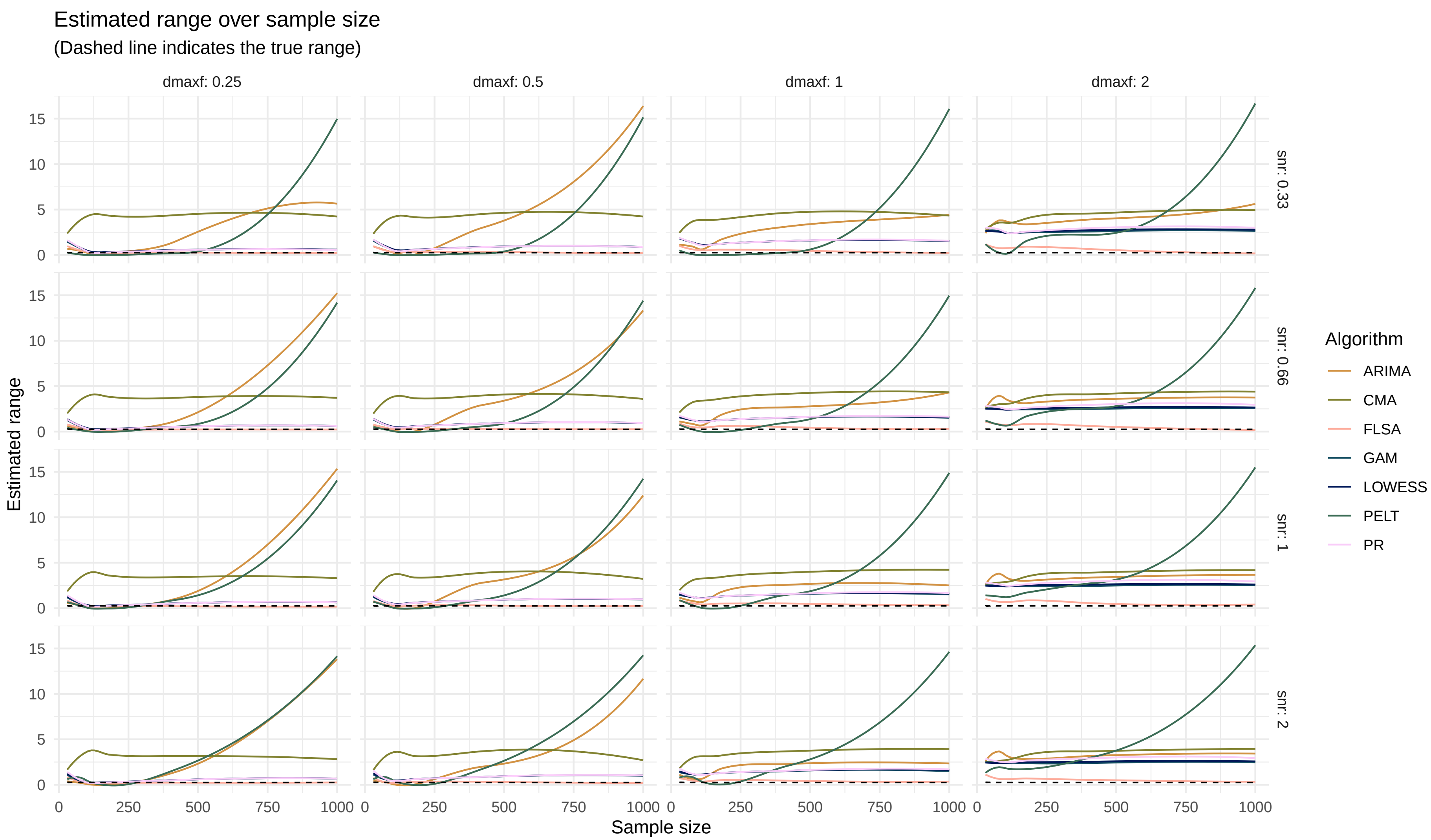
Systematic change pattern: Jump

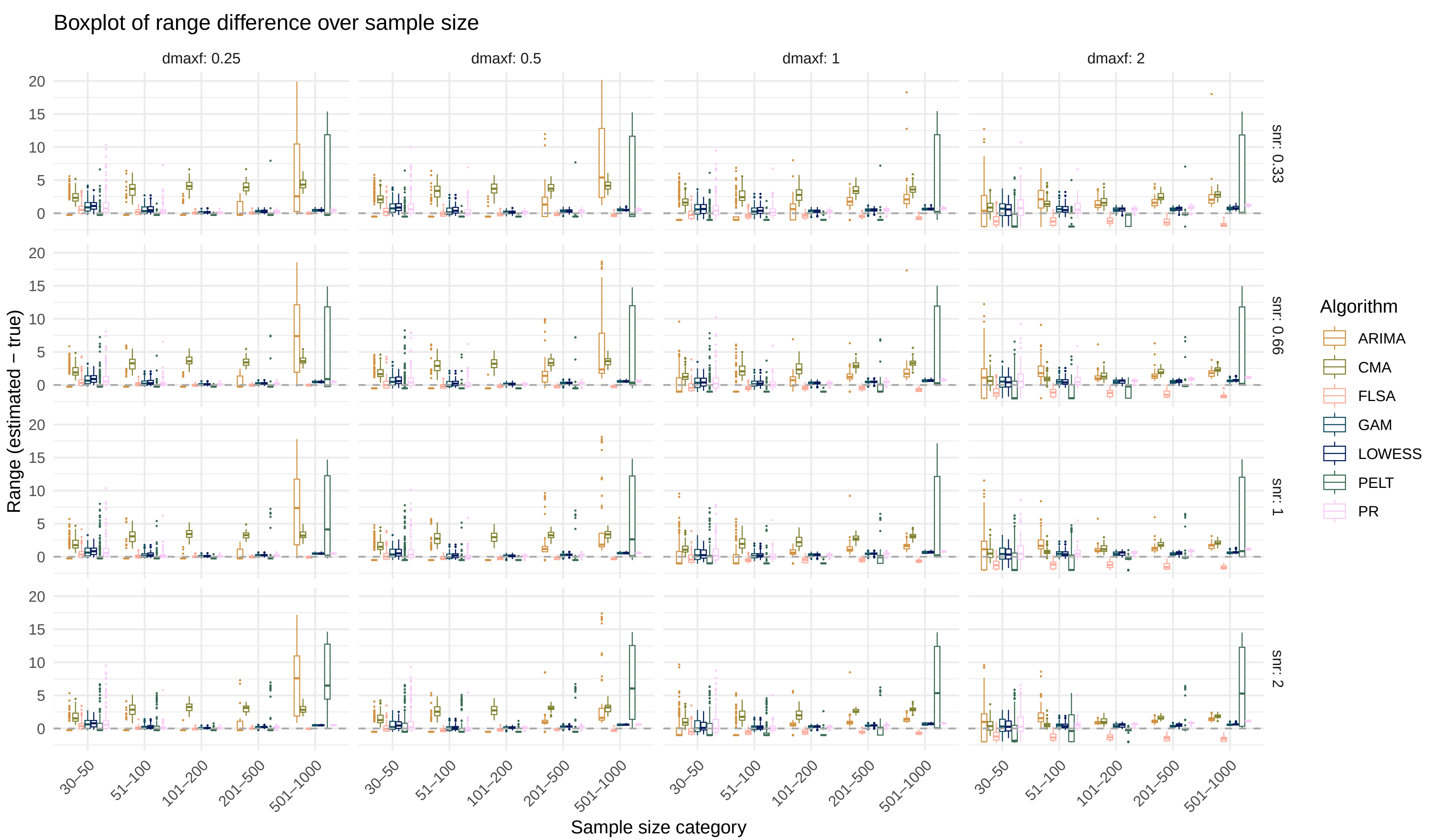




Boxplot of range difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Range (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
0
5
10
15
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

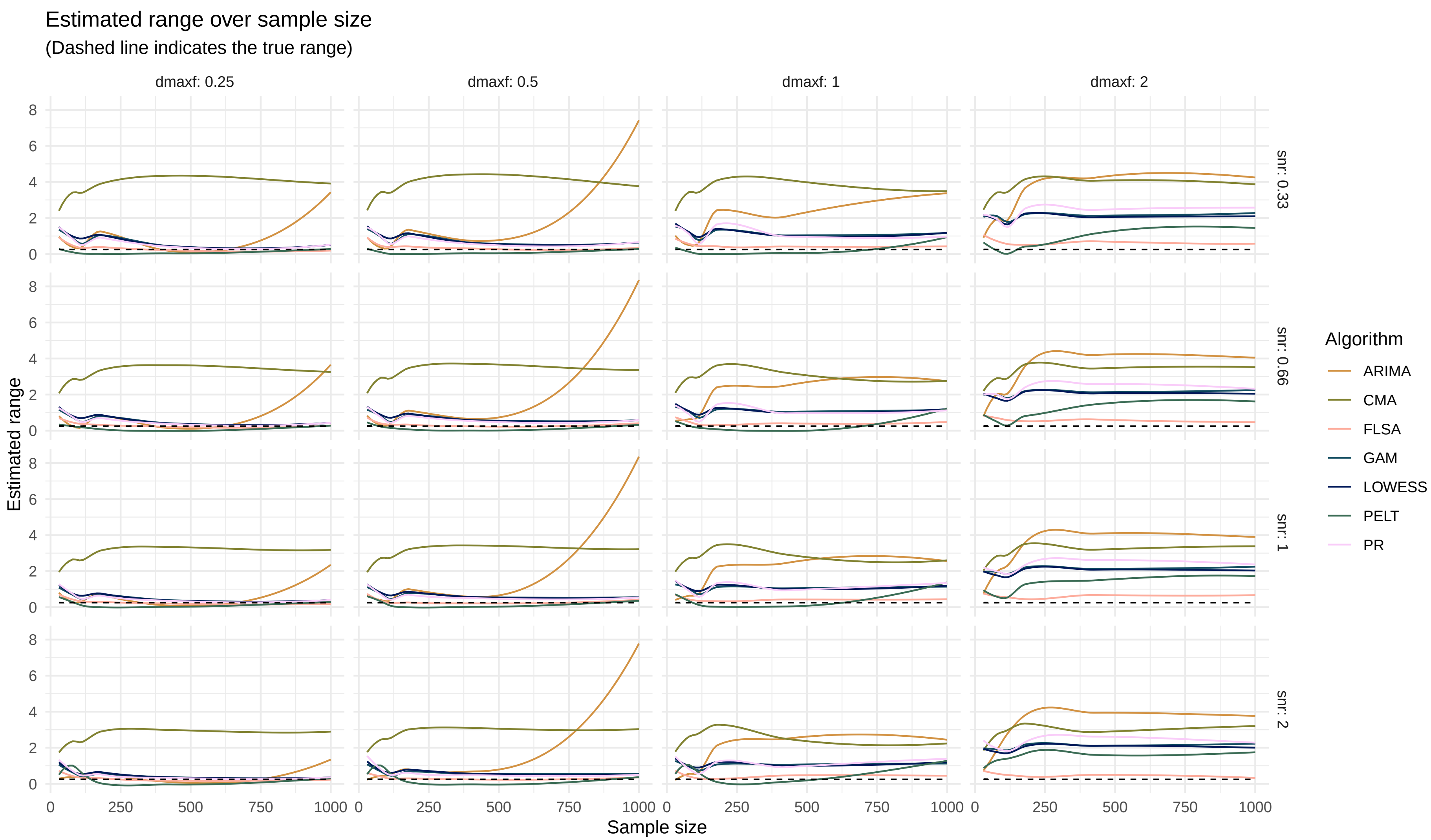
Systematic change pattern: Linear

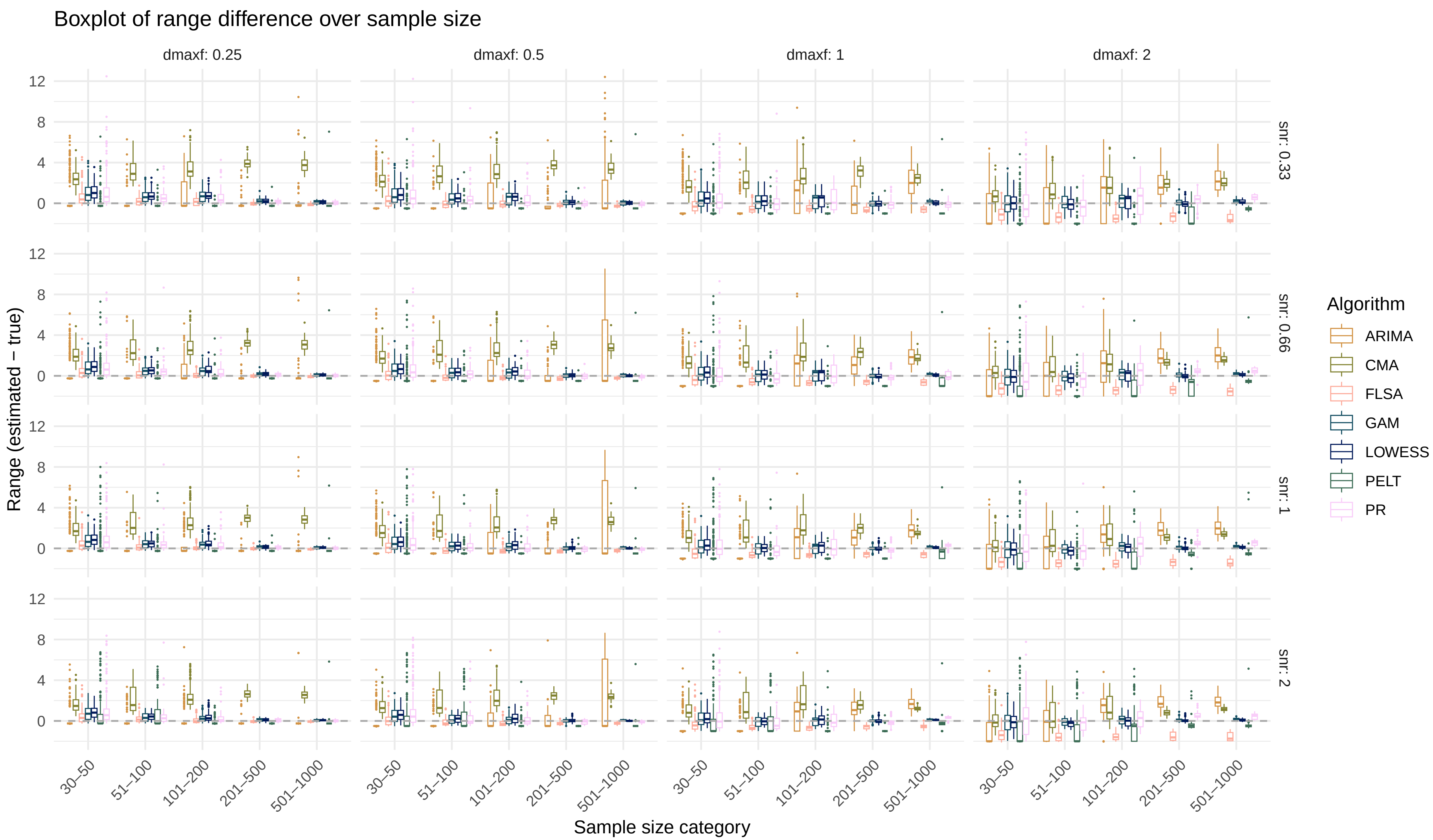




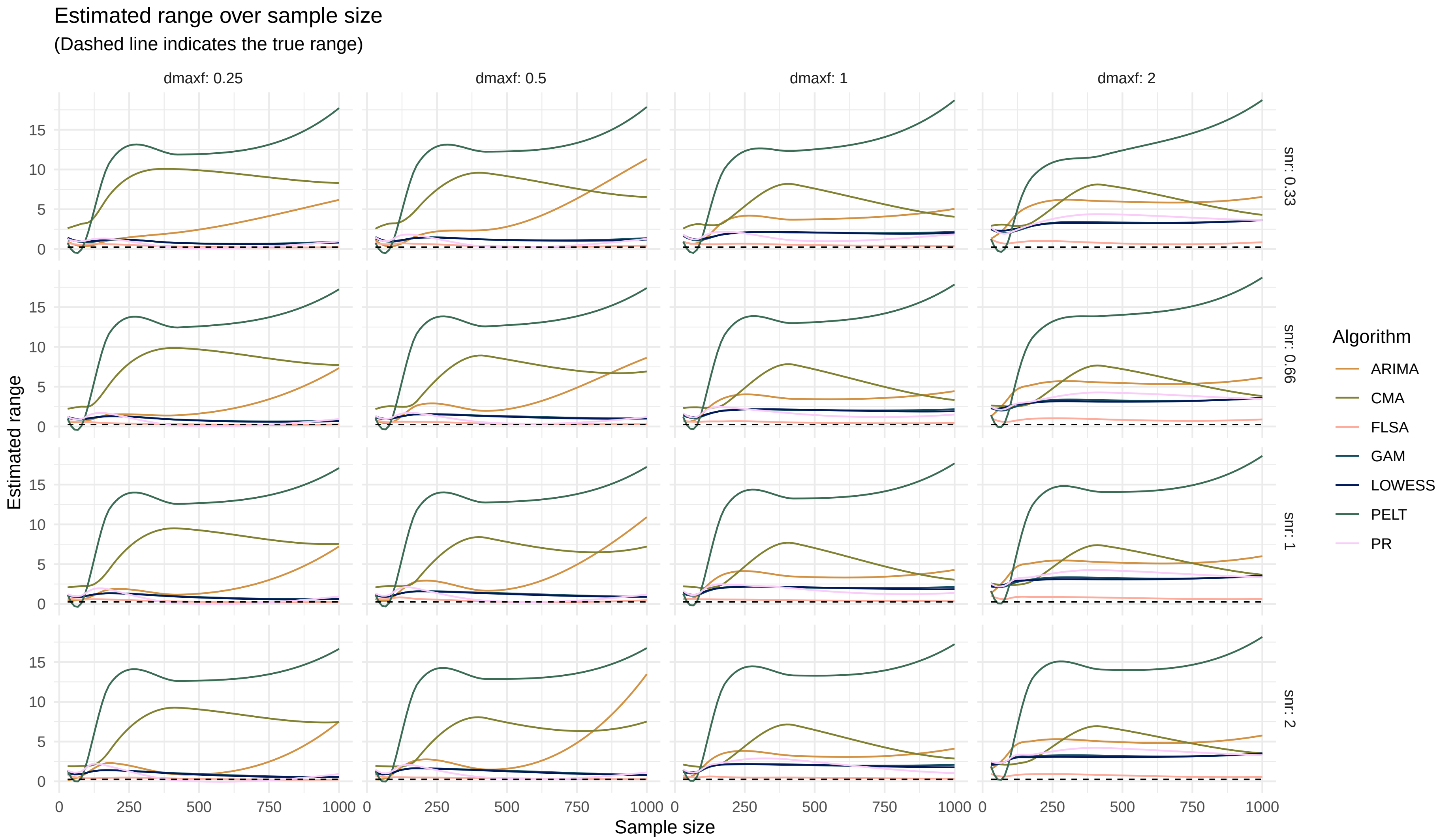
Boxplot of range difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Range (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

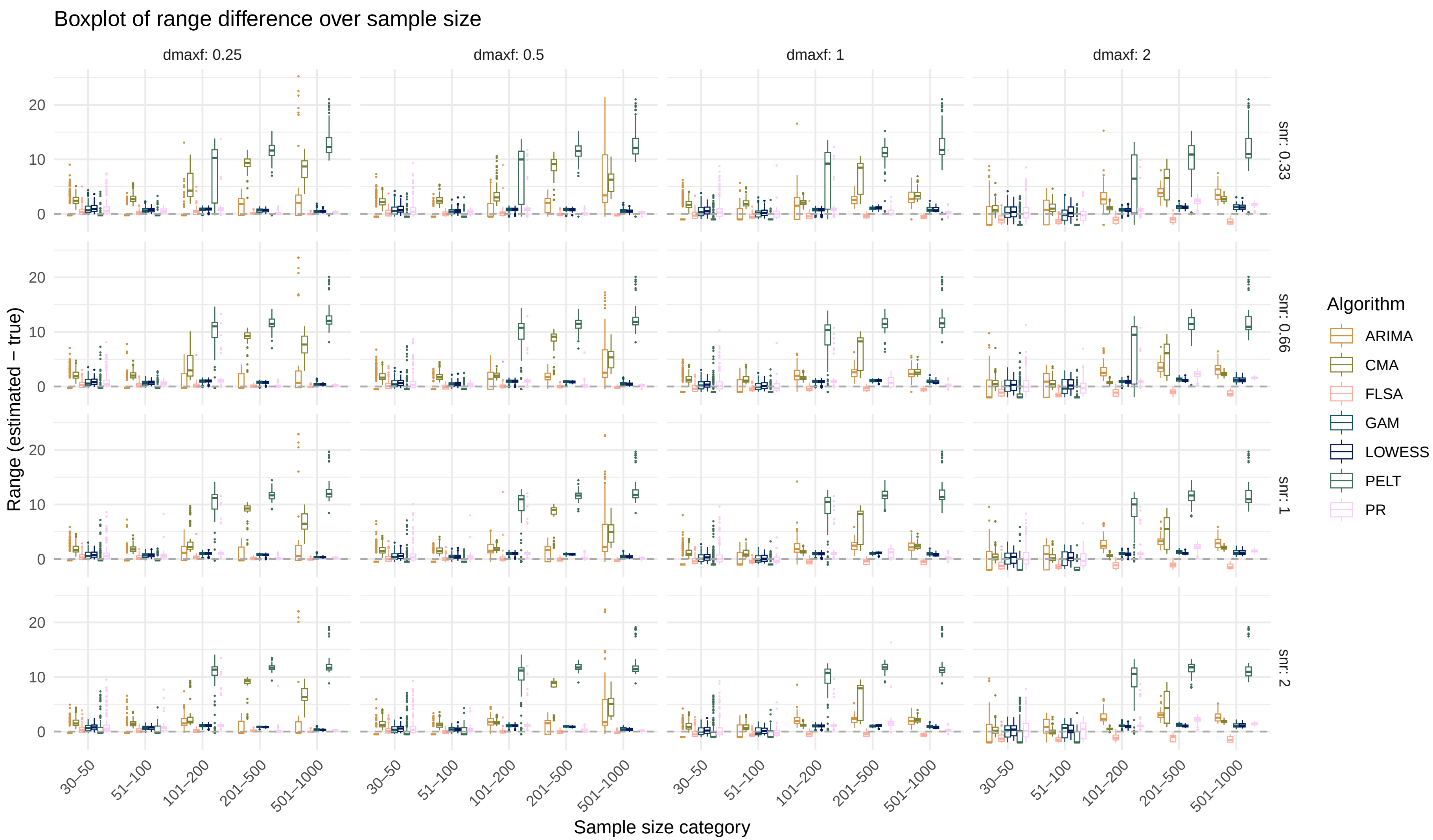
Systematic change pattern: Quadratic



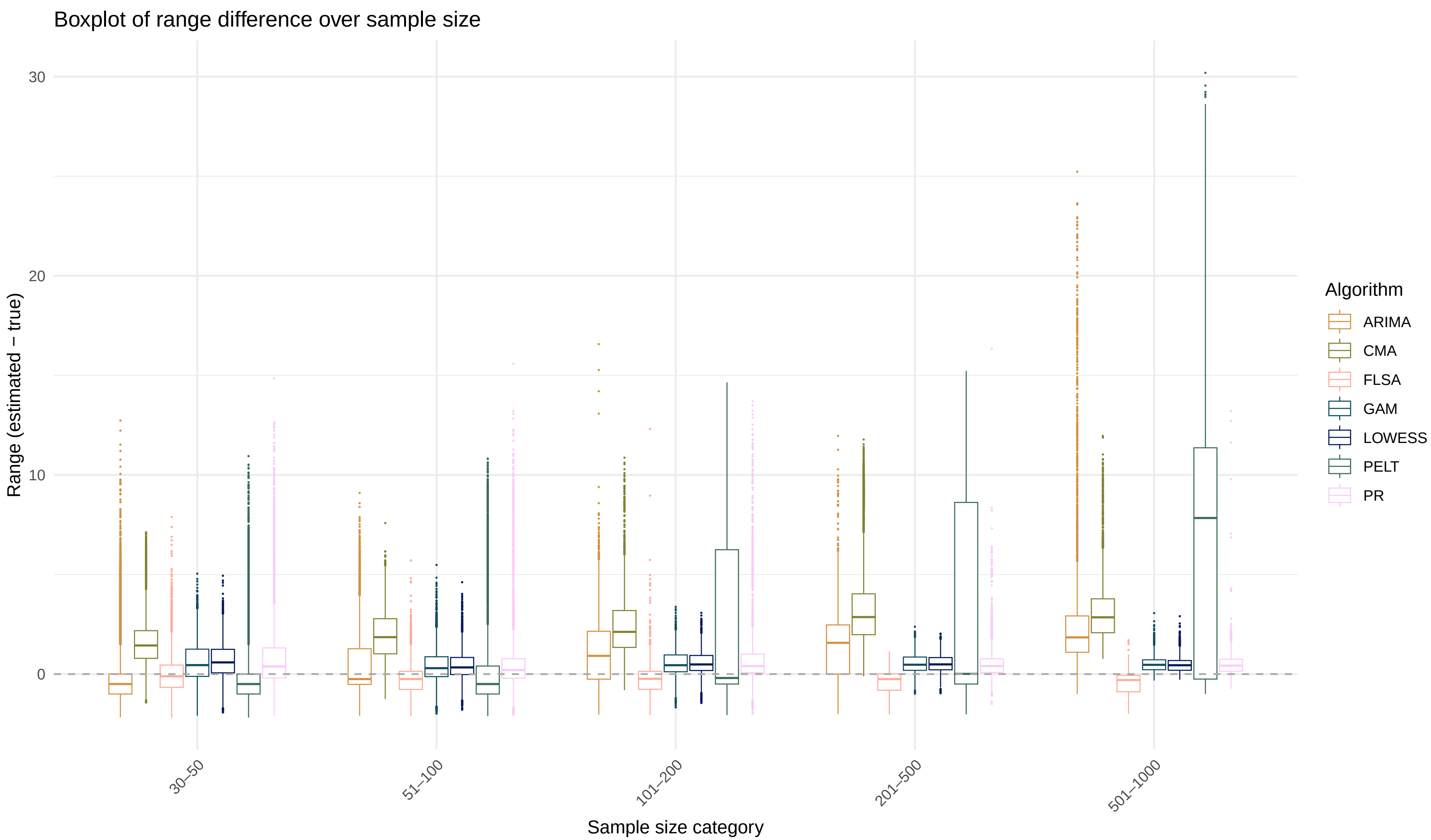

Boxplot of range difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Range (estimated – true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

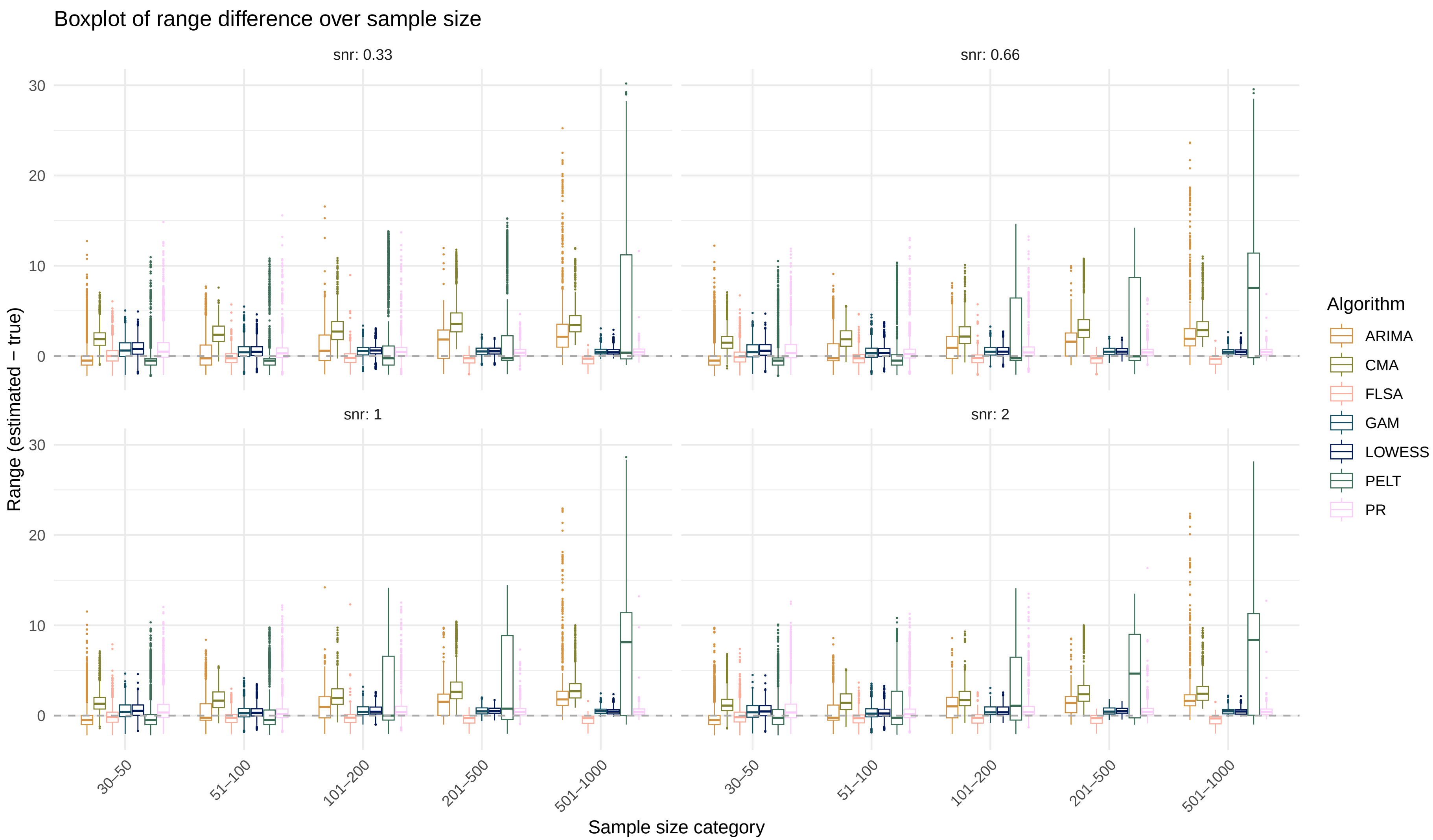
Systematic change pattern: Recurrent




Boxplot of range difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Range (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
0
10
20
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

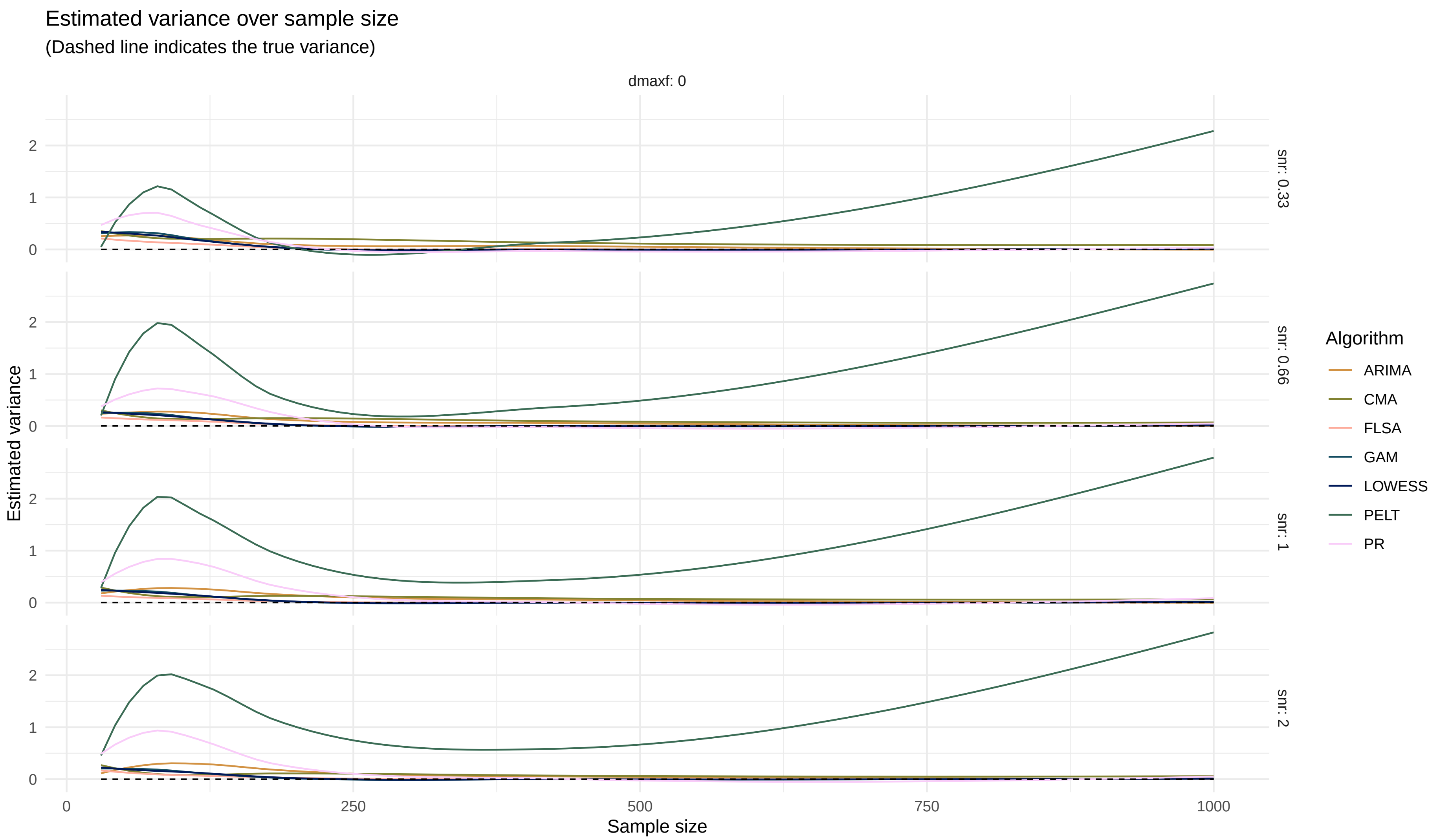
Systematic change pattern: All patterns combined

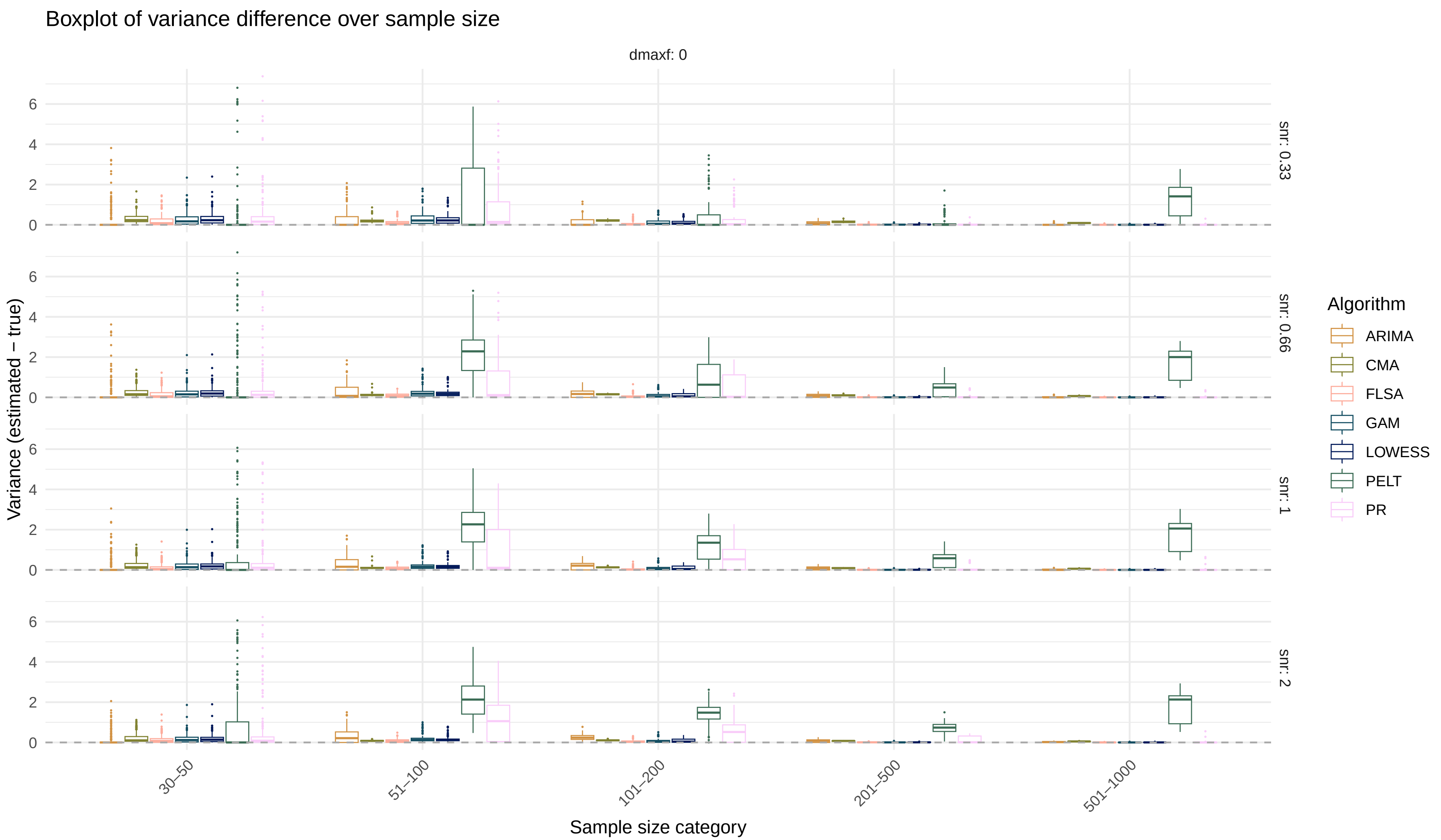



Boxplot of range difference over sample size
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Range (estimated − true)
30
20
10
0
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

Estimand: Variance

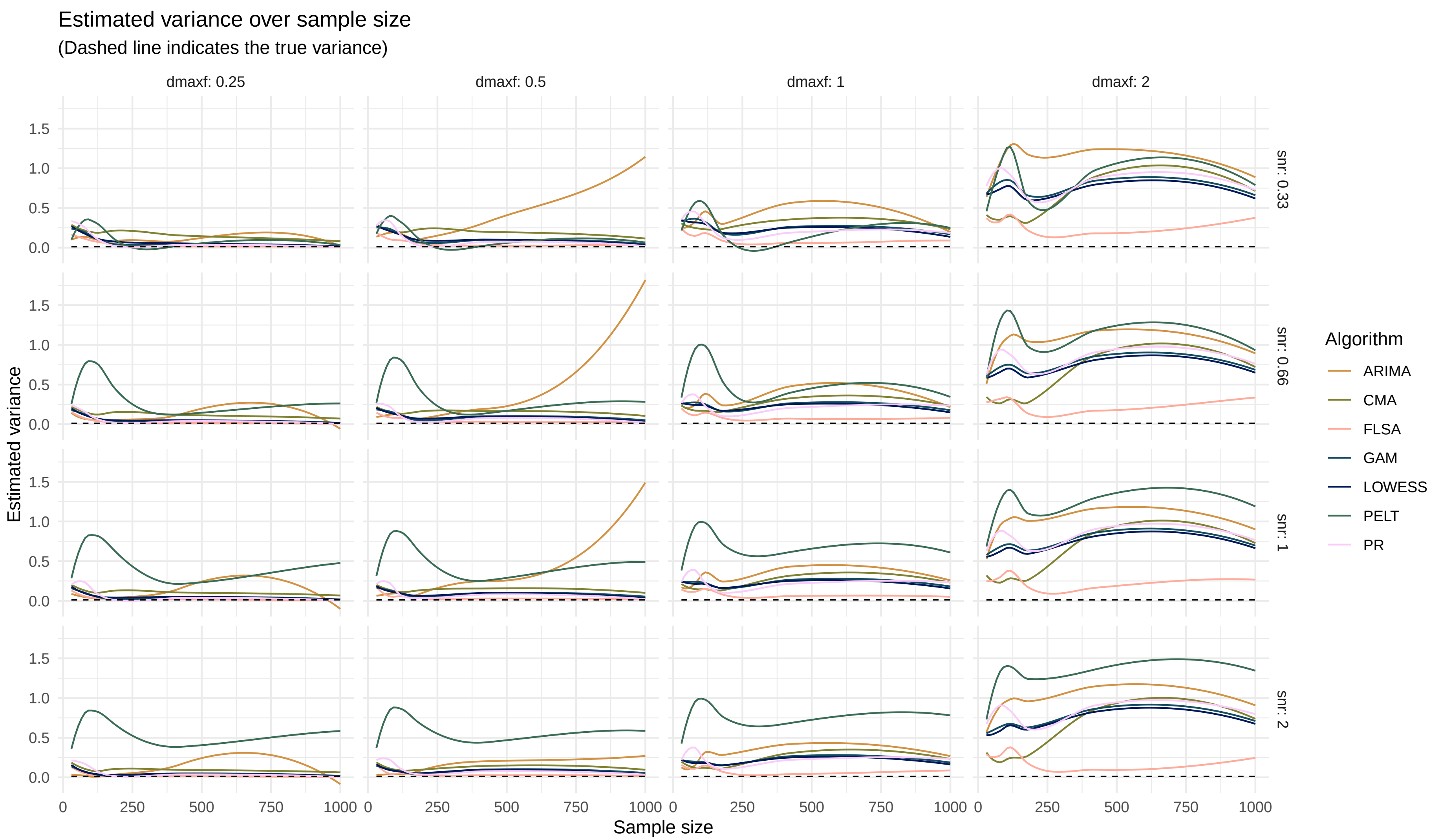
Systematic change pattern: No change

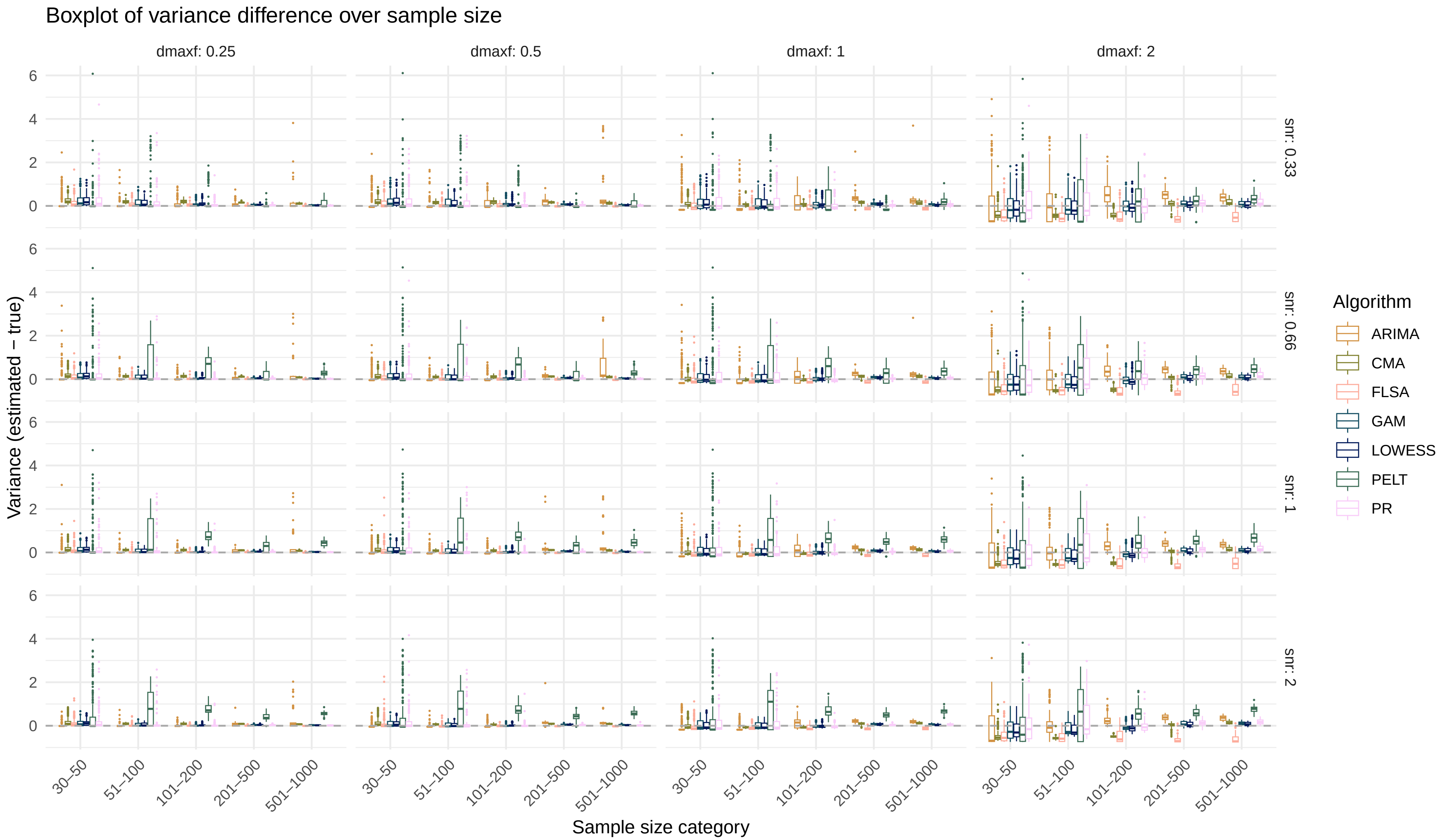




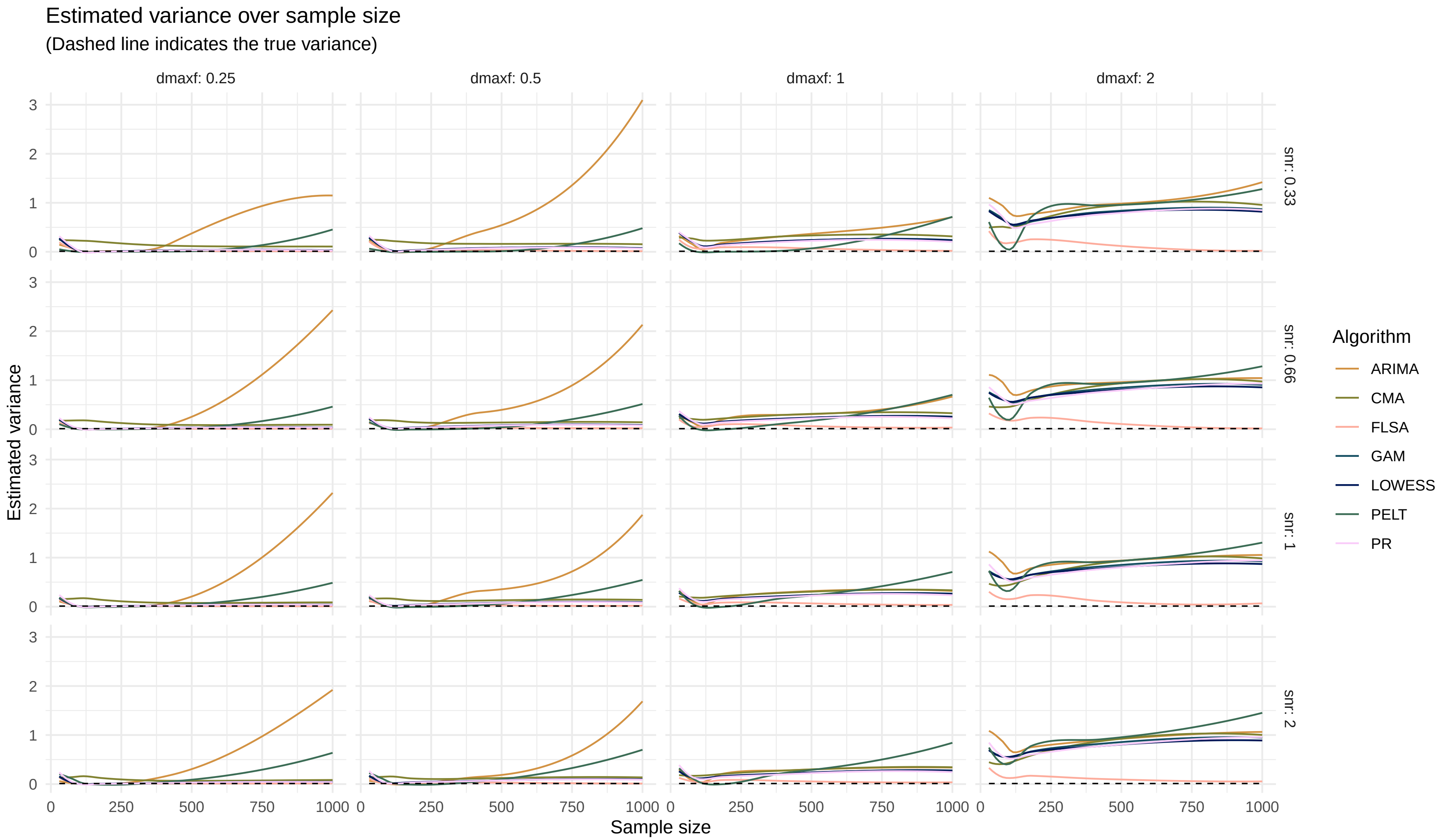
Boxplot of variance difference over sample size
dmaxf: 0
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Variance (estimated – true)
6
4
2
0
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

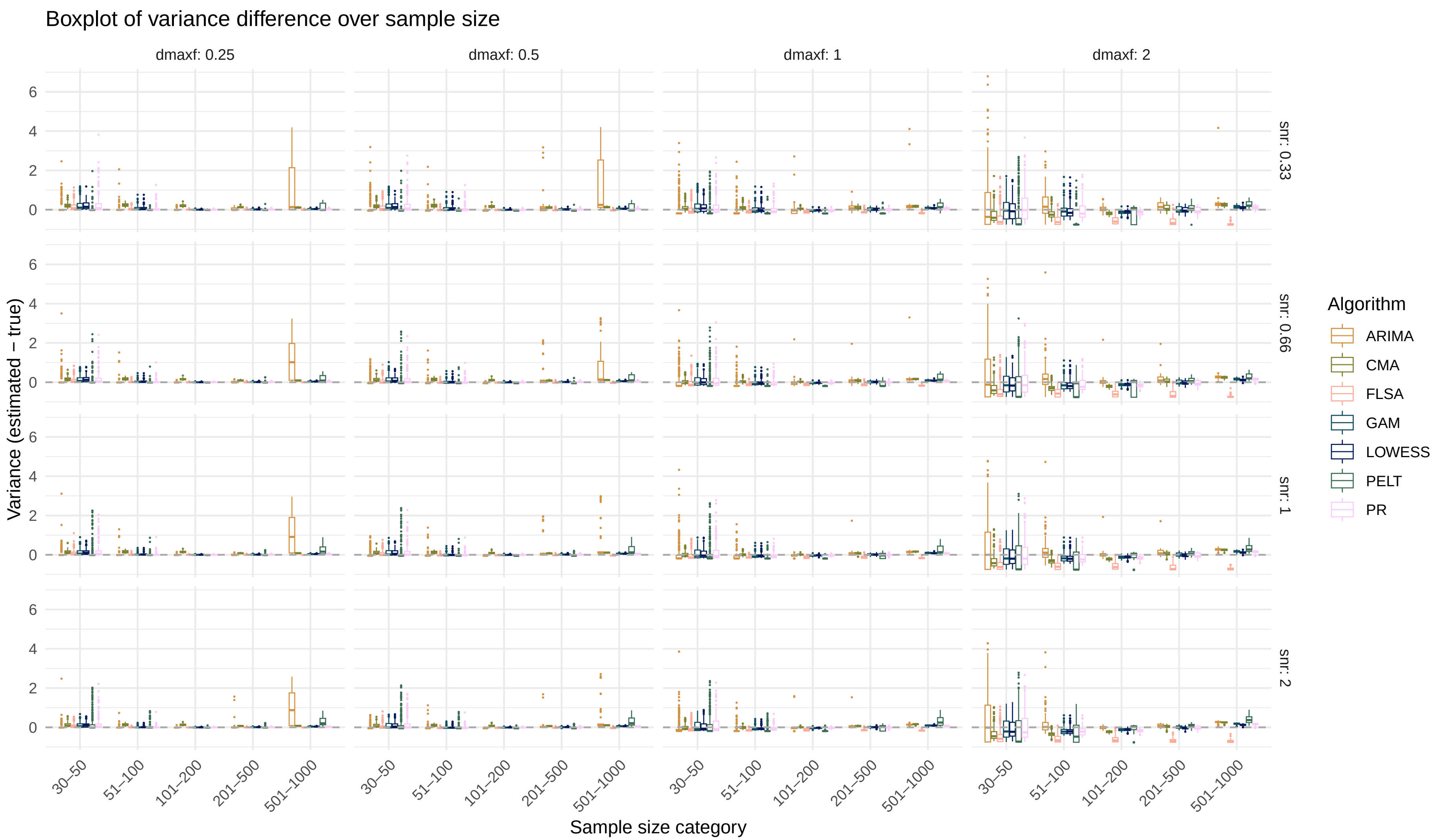
Systematic change pattern: Jump



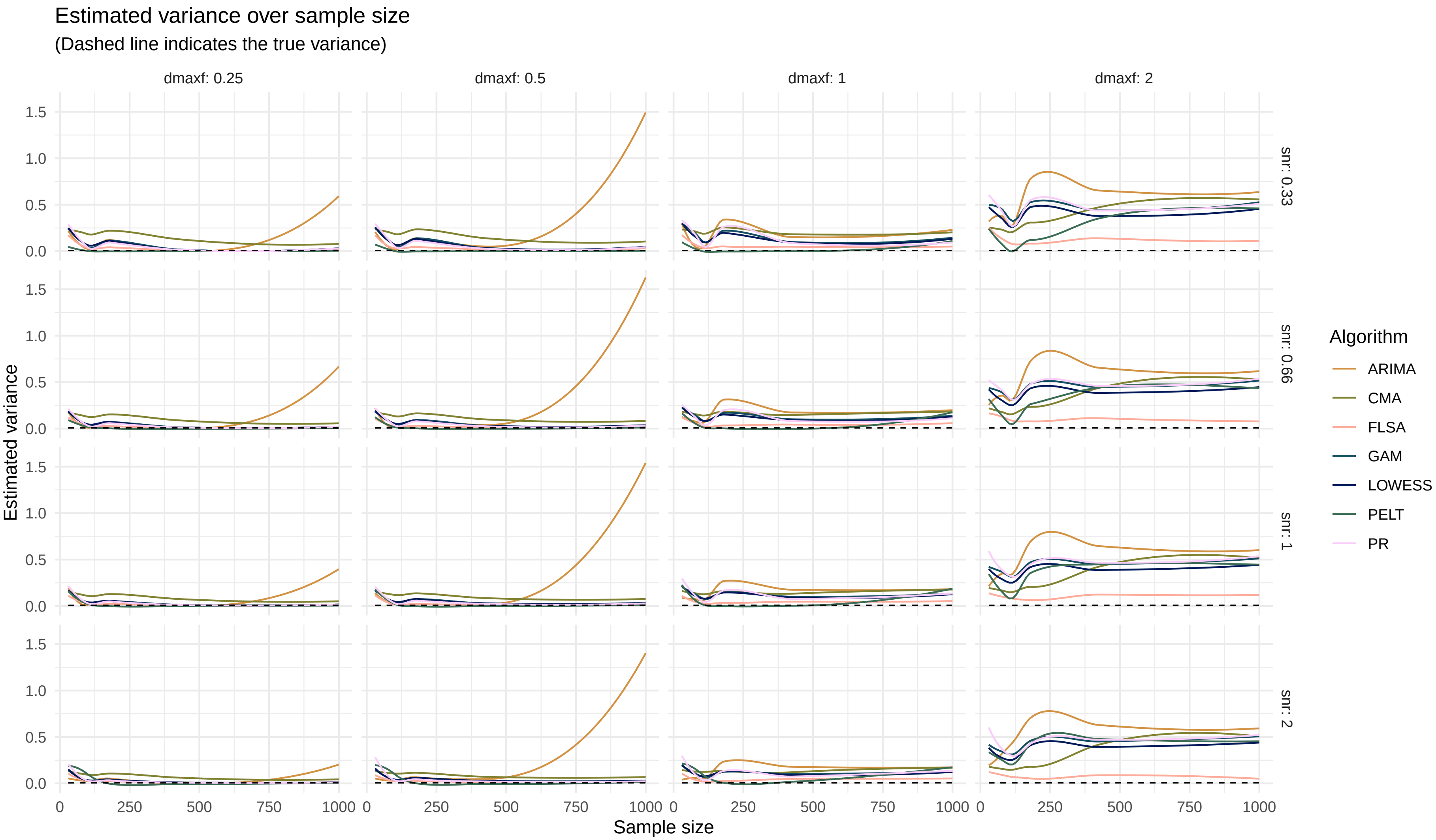

Boxplot of variance difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Variance (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

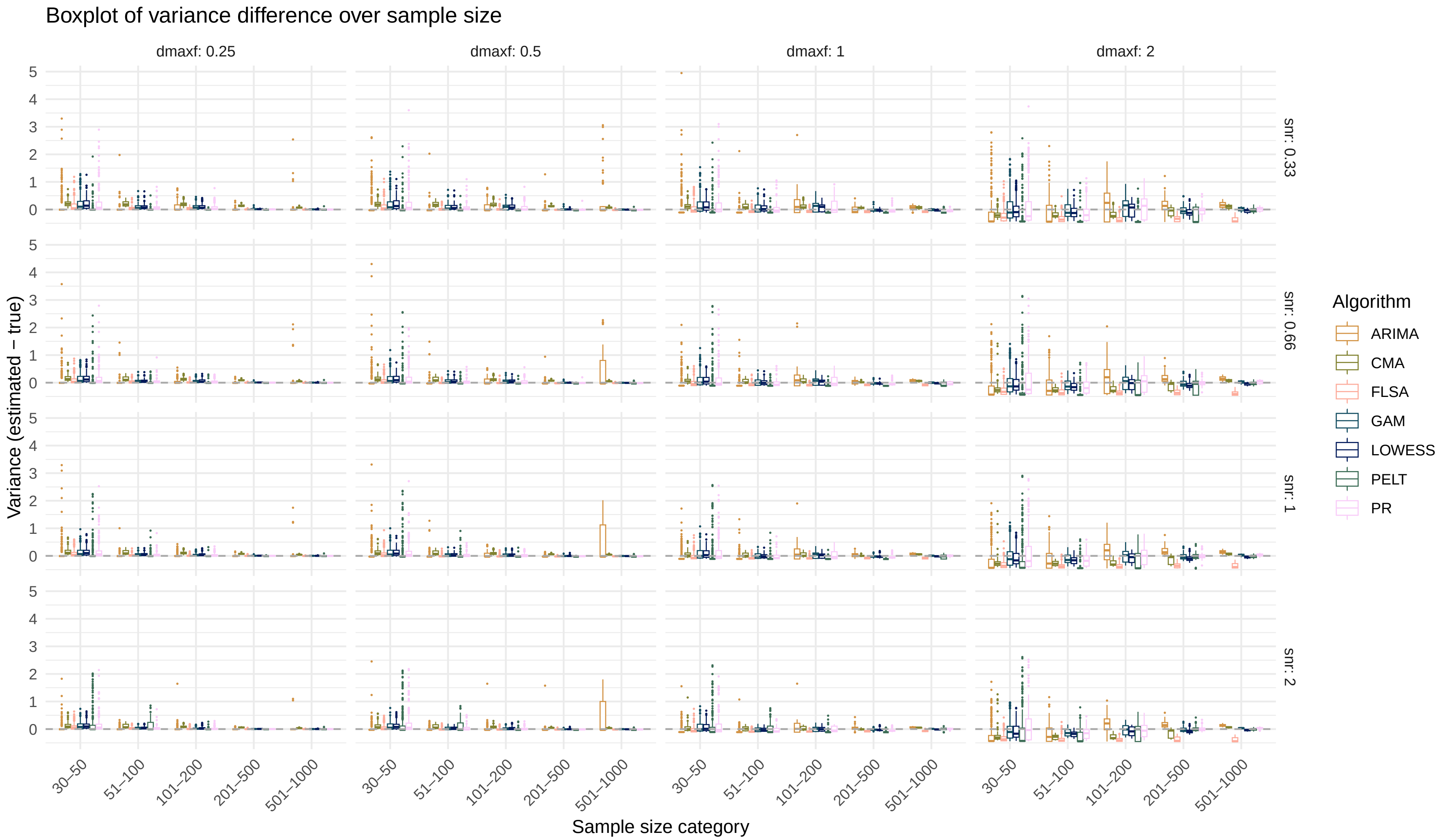
Systematic change pattern: Linear



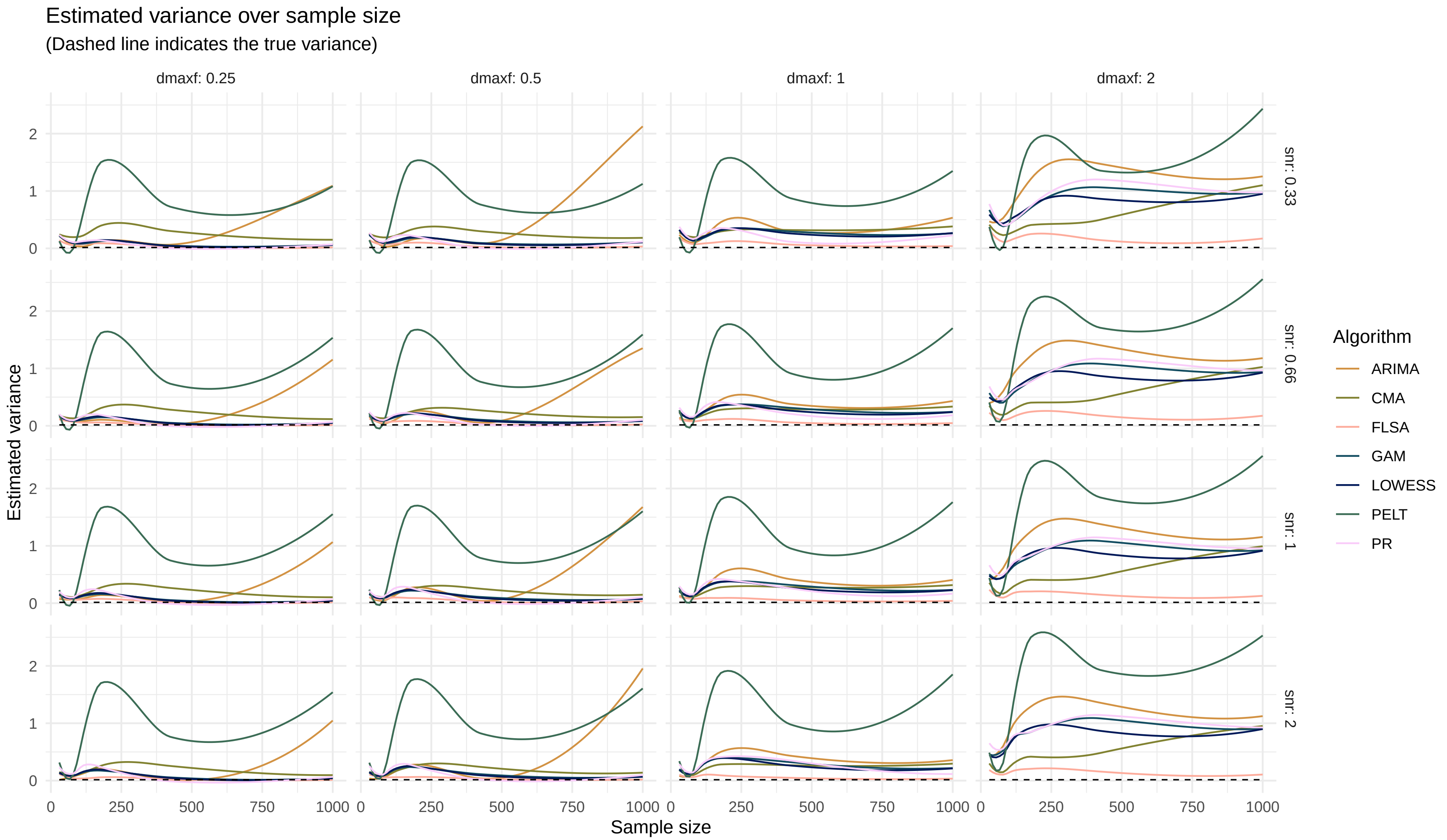

Boxplot of variance difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Variance (estimated – true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

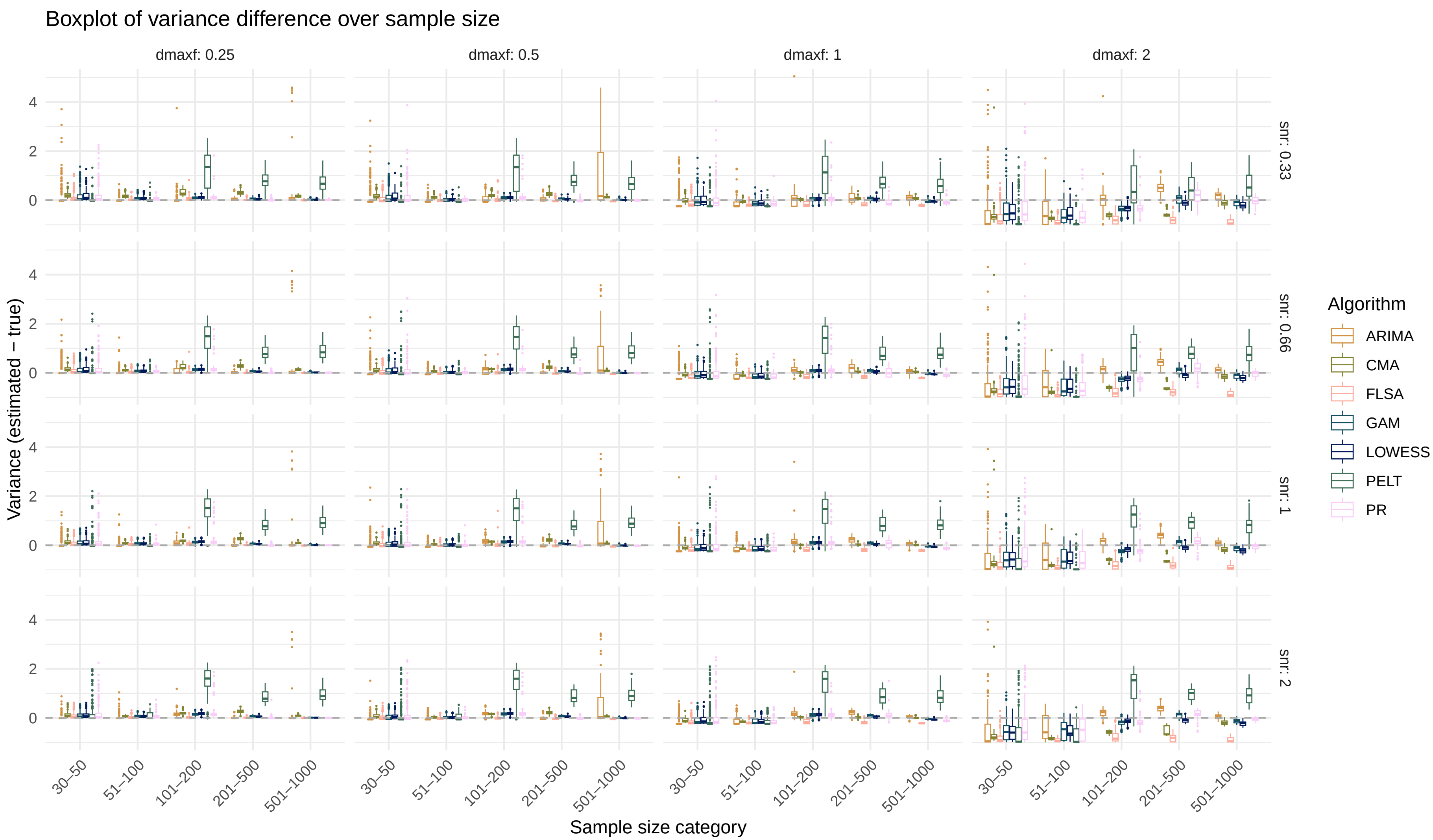
Systematic change pattern: Quadratic



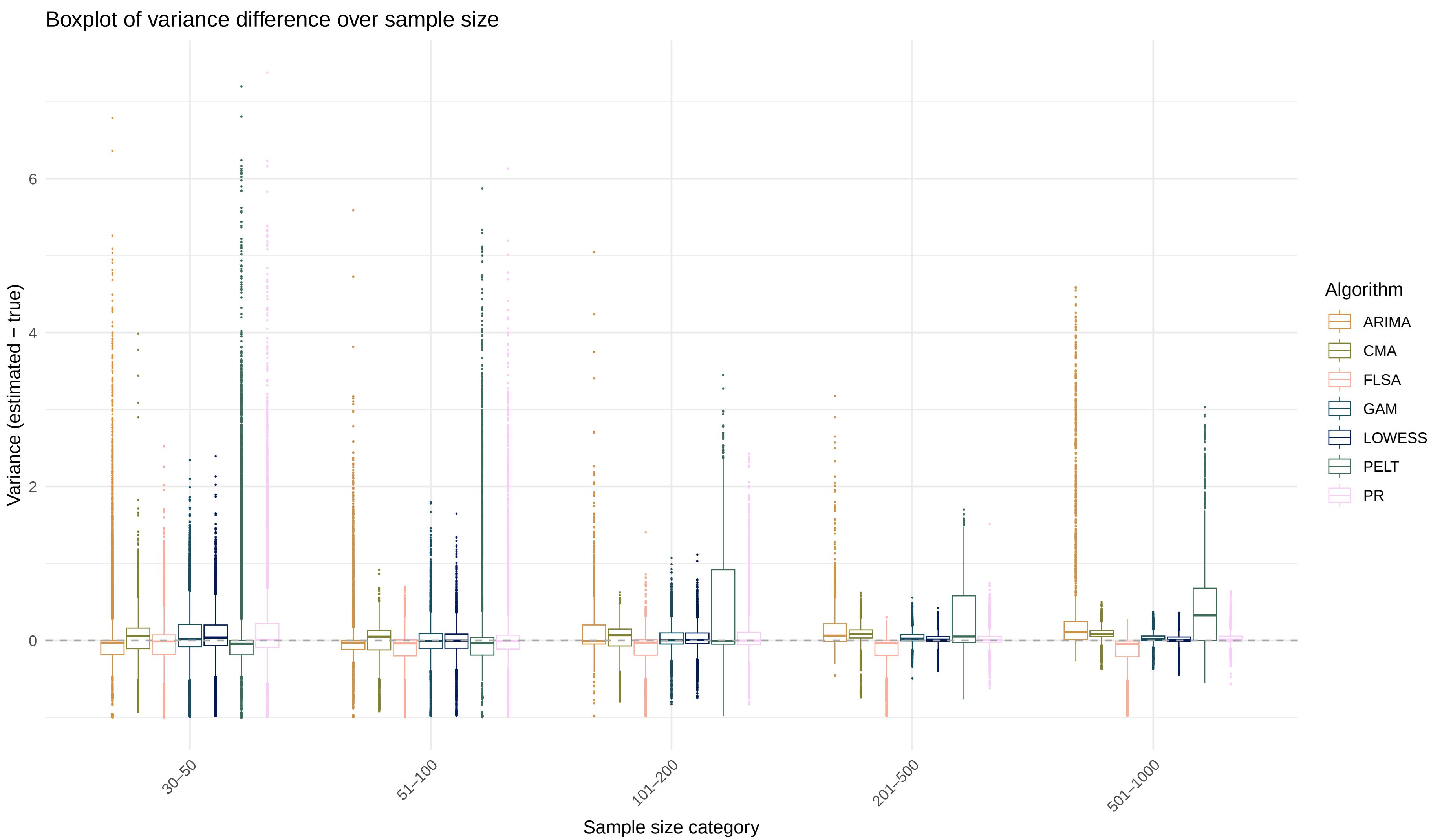

Boxplot of variance difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Variance (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

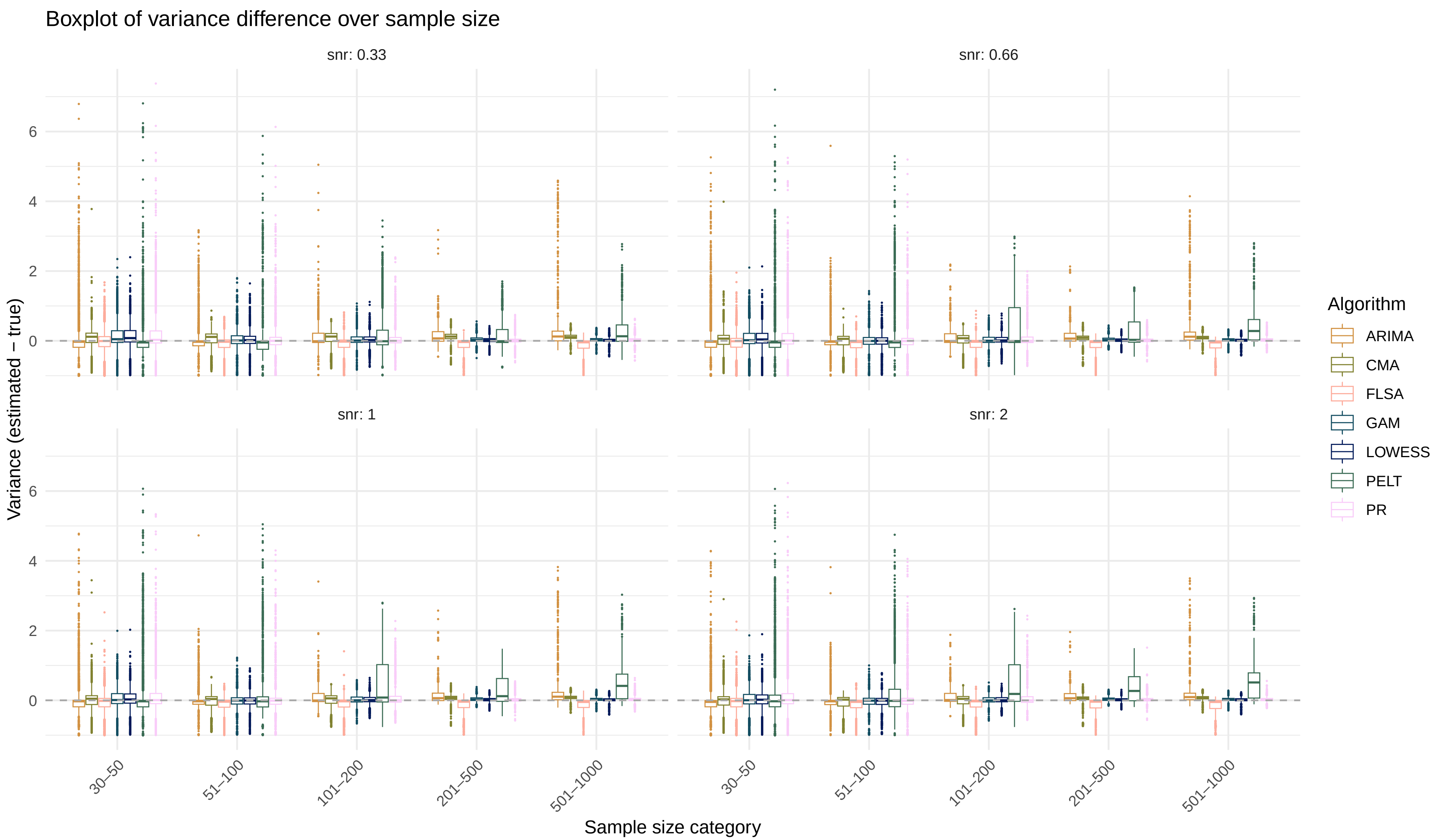
Systematic change pattern: Recurrent




Boxplot of variance difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Variance (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

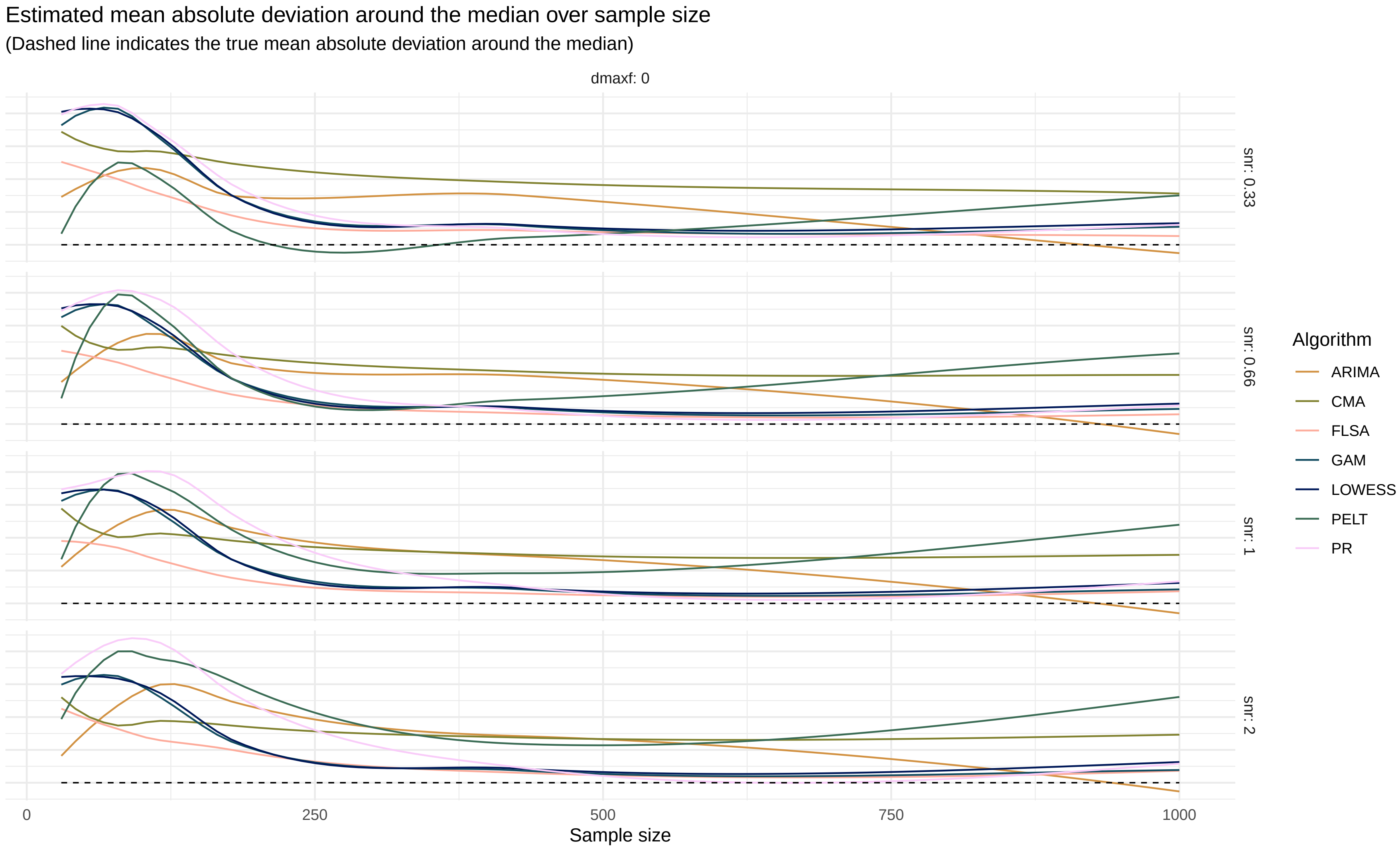
Systematic change pattern: All patterns combined

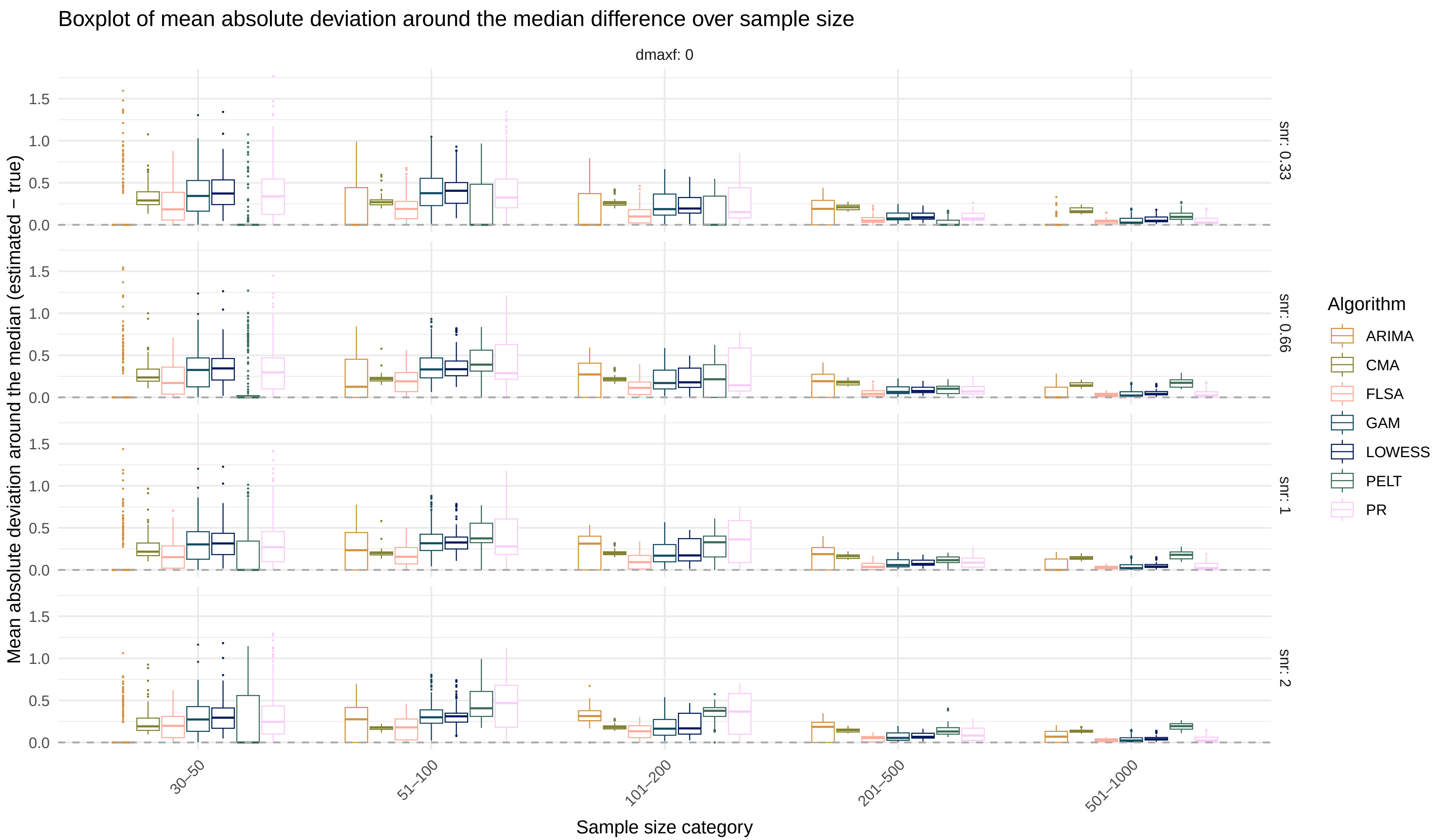



Boxplot of variance difference over sample size
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Variance (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

Estimand: Mean absolute deviation around the median

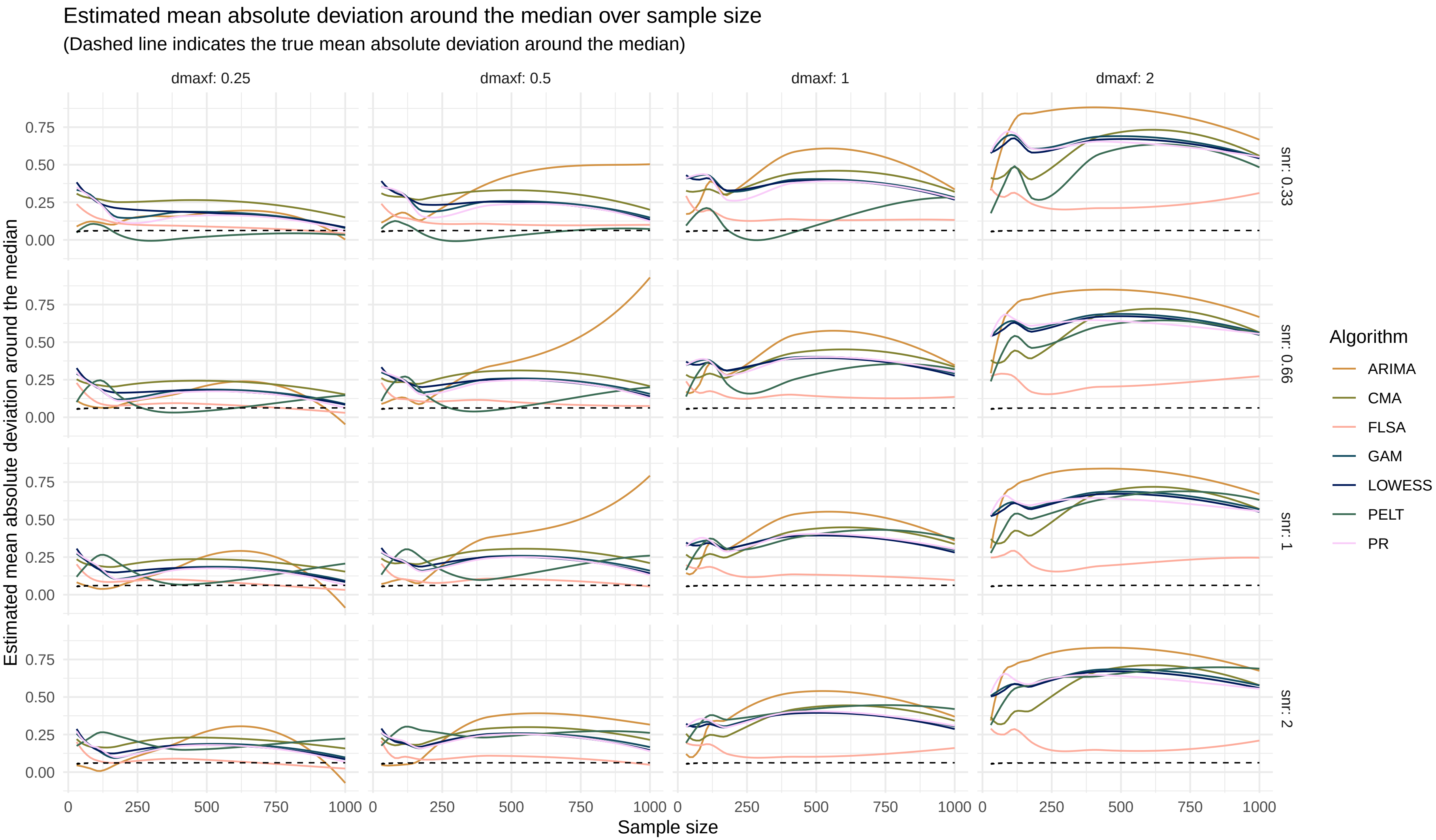
Systematic change pattern: No change

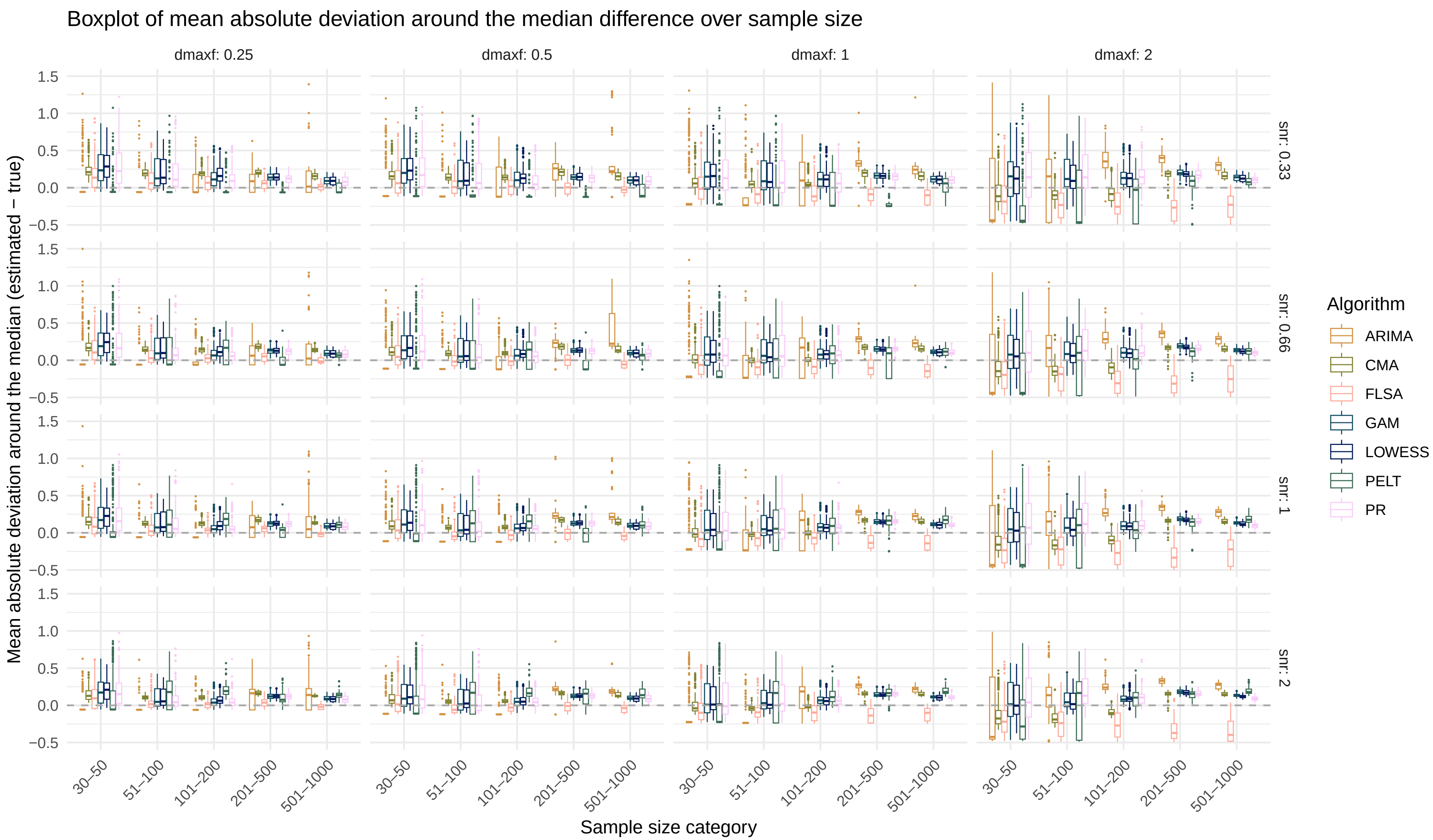




Boxplot of mean absolute deviation around the median difference over sample size
dmaxf: 0
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Mean absolute deviation around the median (estimated − true)
0.0
0.5
1.0
1.5
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

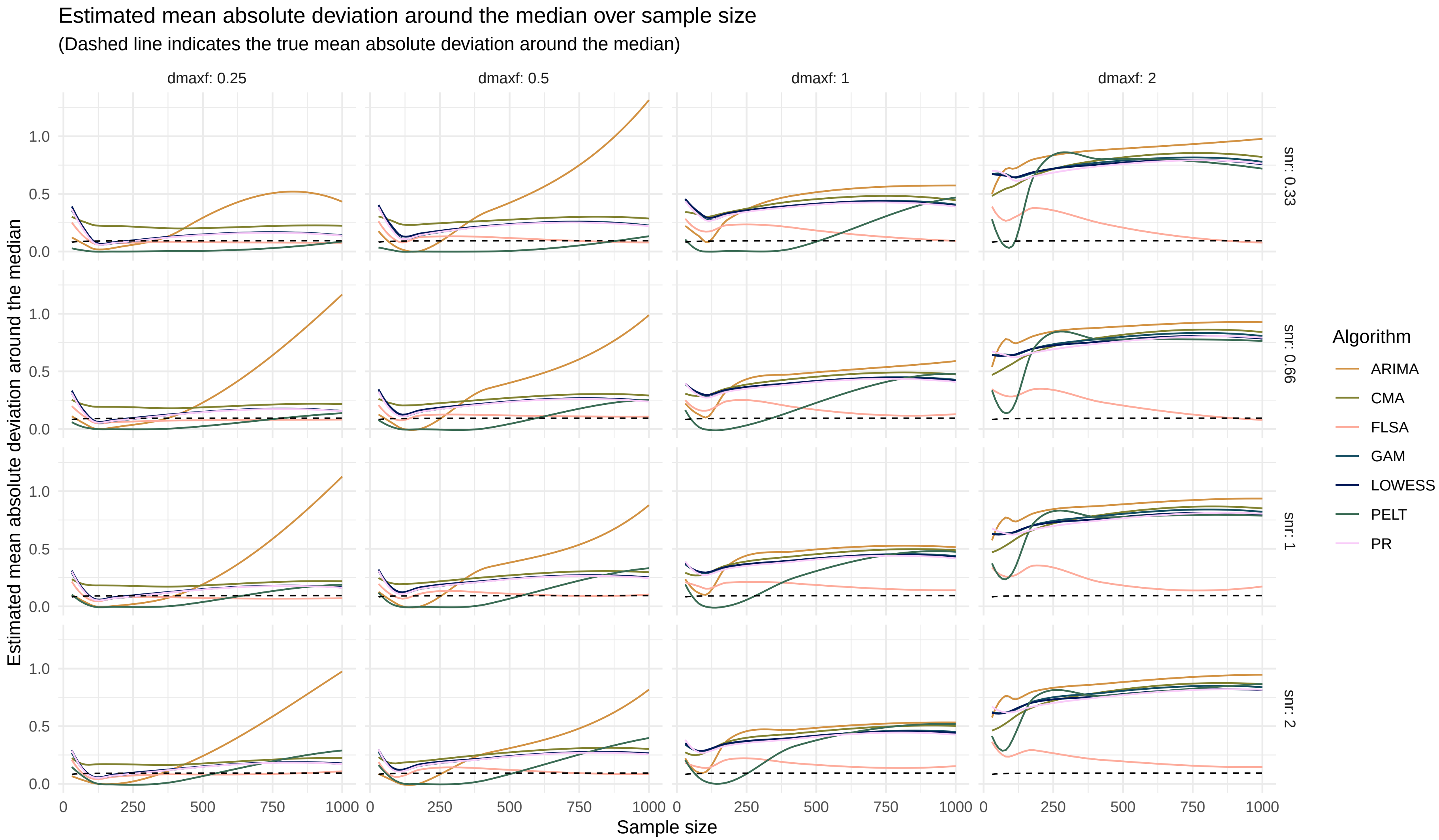
Systematic change pattern: Jump

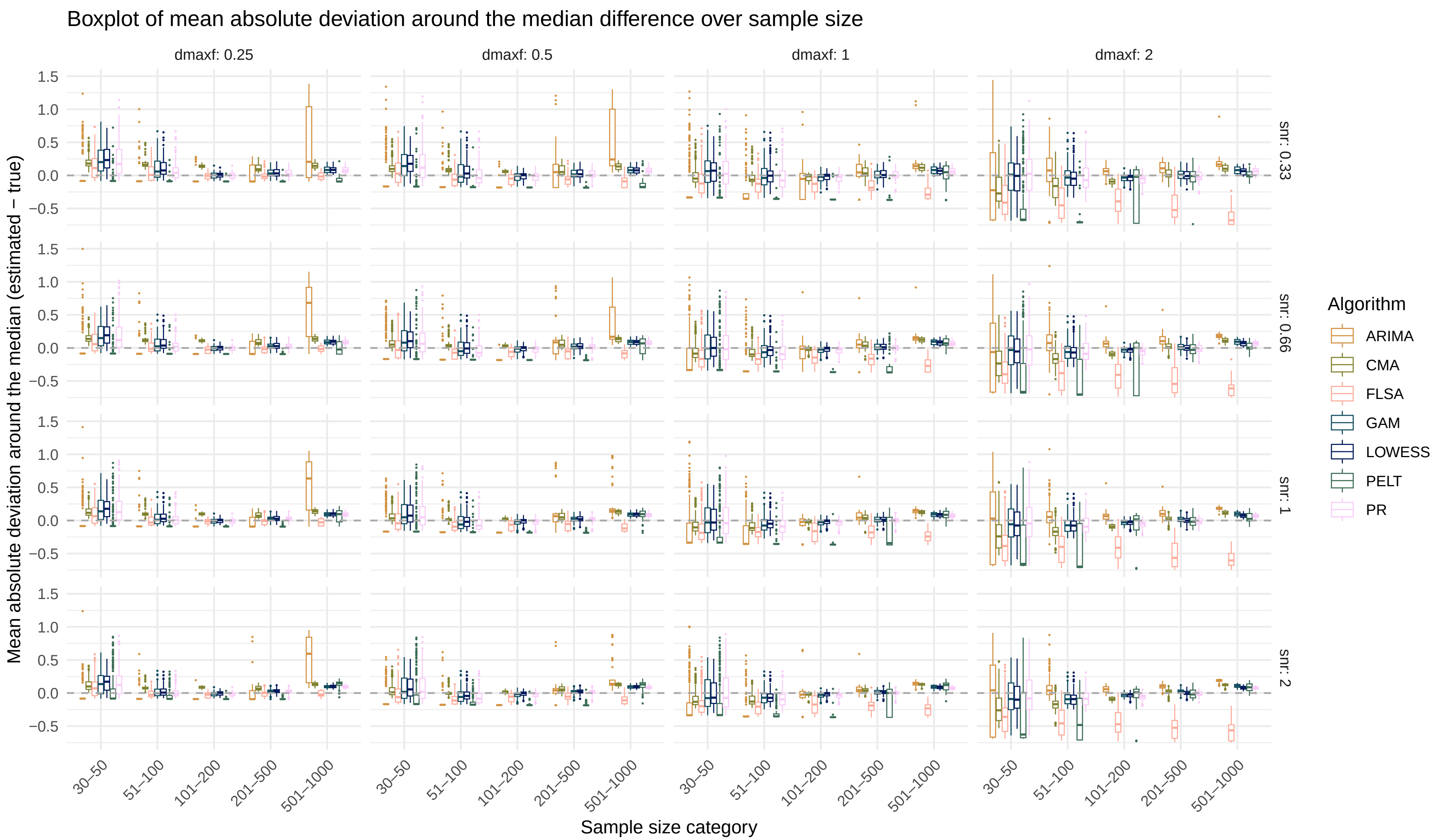




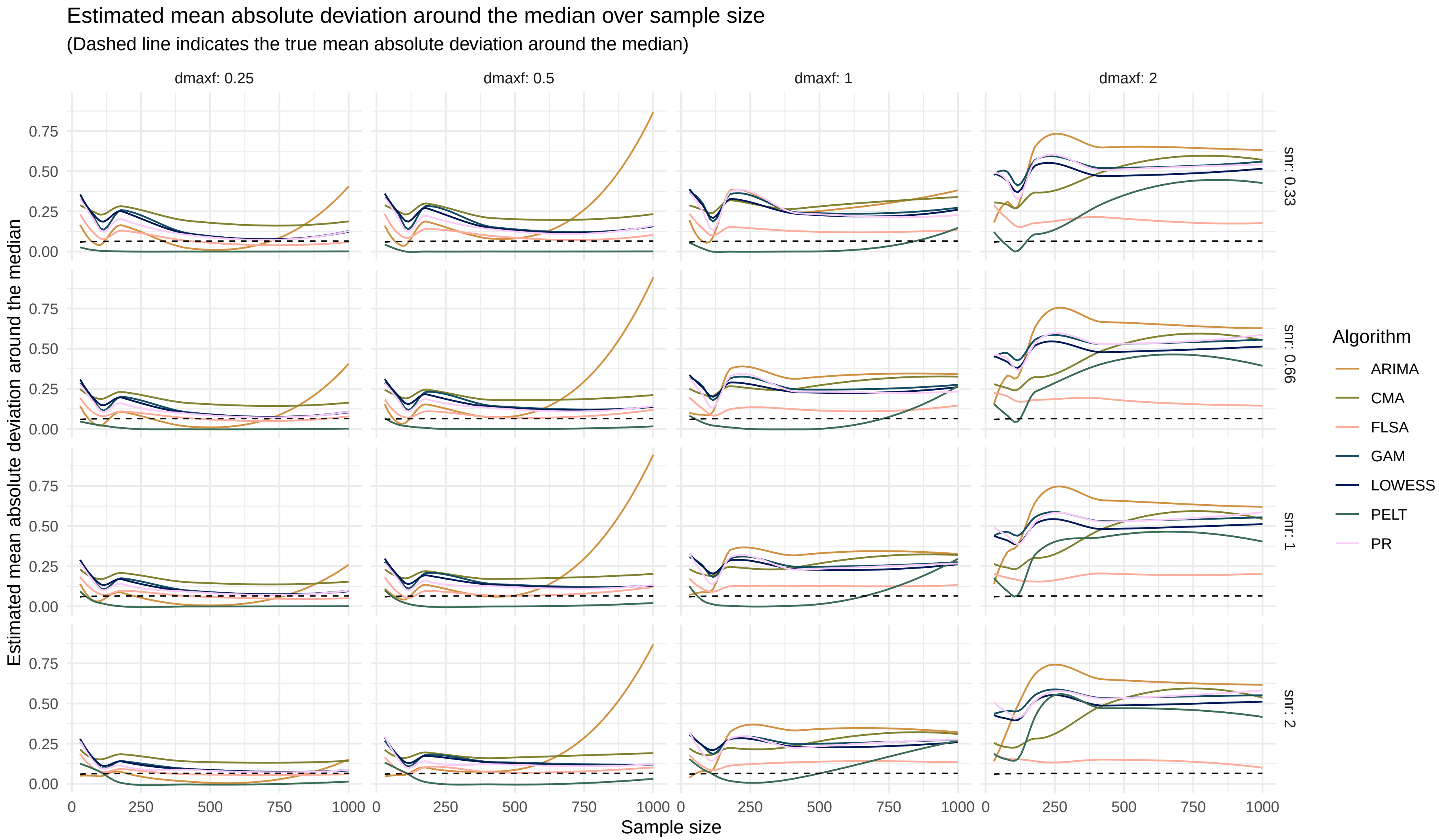
Boxplot of mean absolute deviation around the median difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Mean absolute deviation around the median (estimated − true)
1.5
1.0
0.5
0.0
−0.5
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

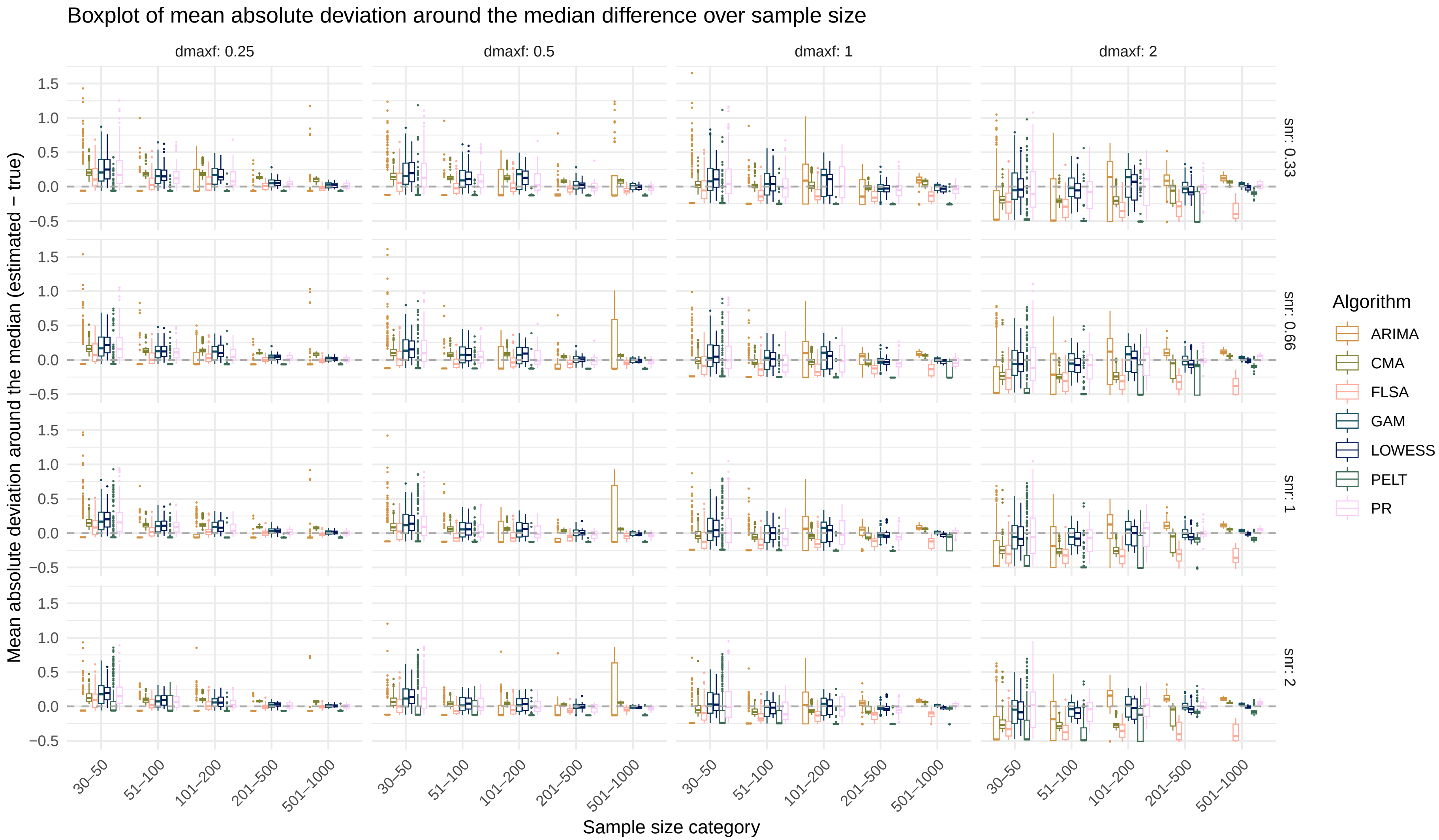
Systematic change pattern: Linear




Boxplot of mean absolute deviation around the median difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Mean absolute deviation around the median (estimated − true)
1.5
1.0
0.5
0.0
−0.5
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

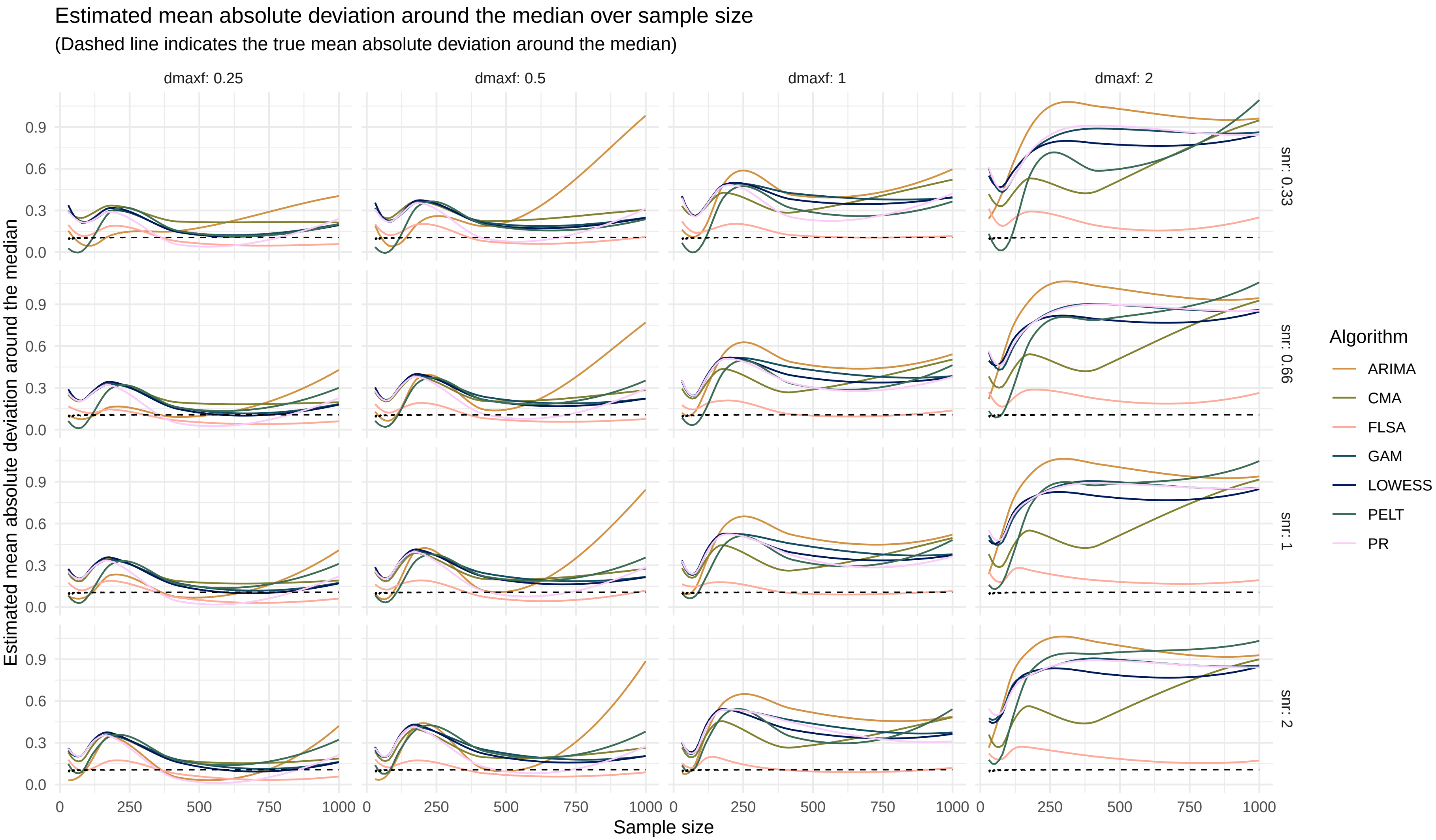
Systematic change pattern: Quadratic

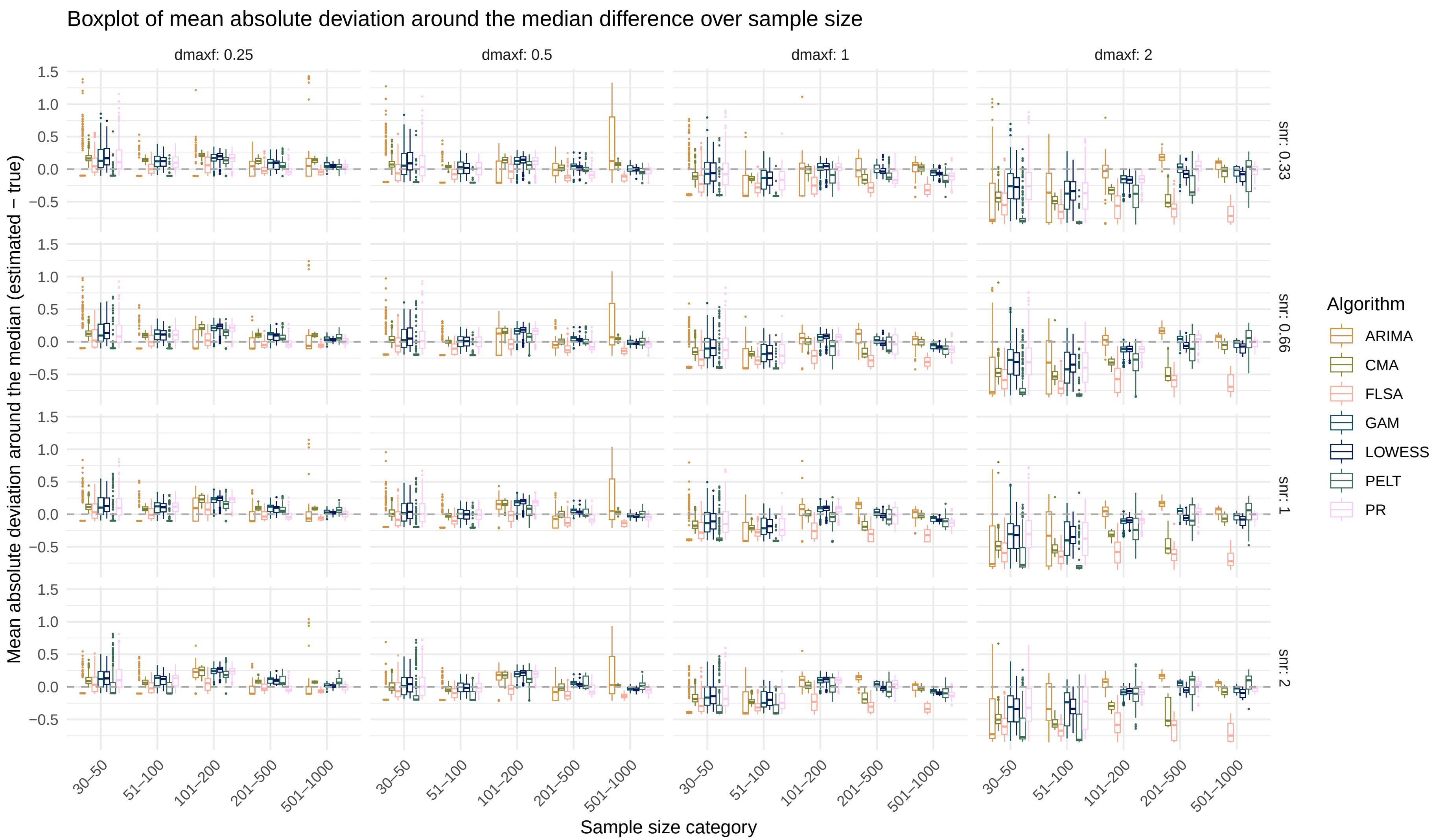




Boxplot of mean absolute deviation around the median difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Mean absolute deviation around the median (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

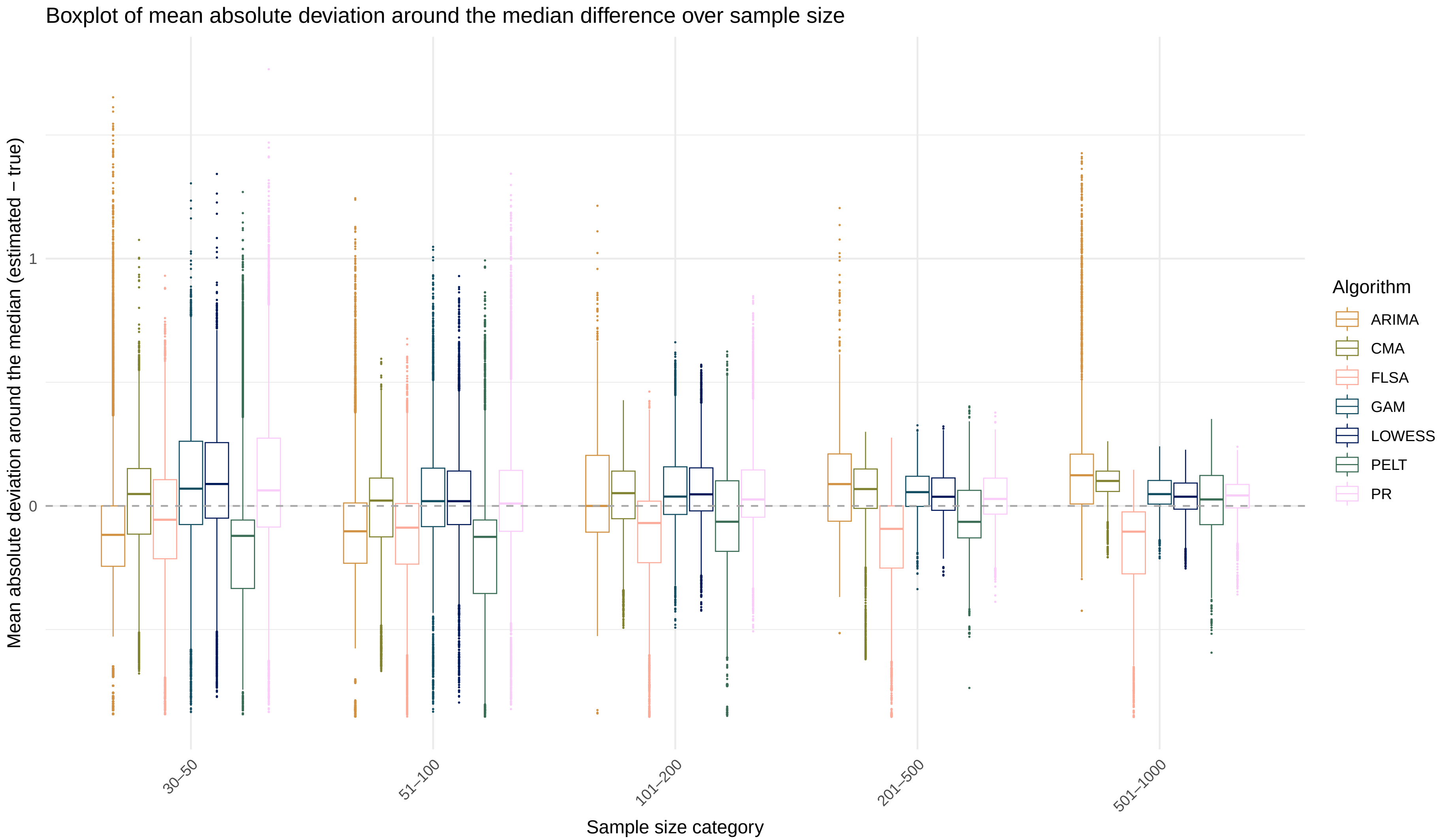
Systematic change pattern: Recurrent

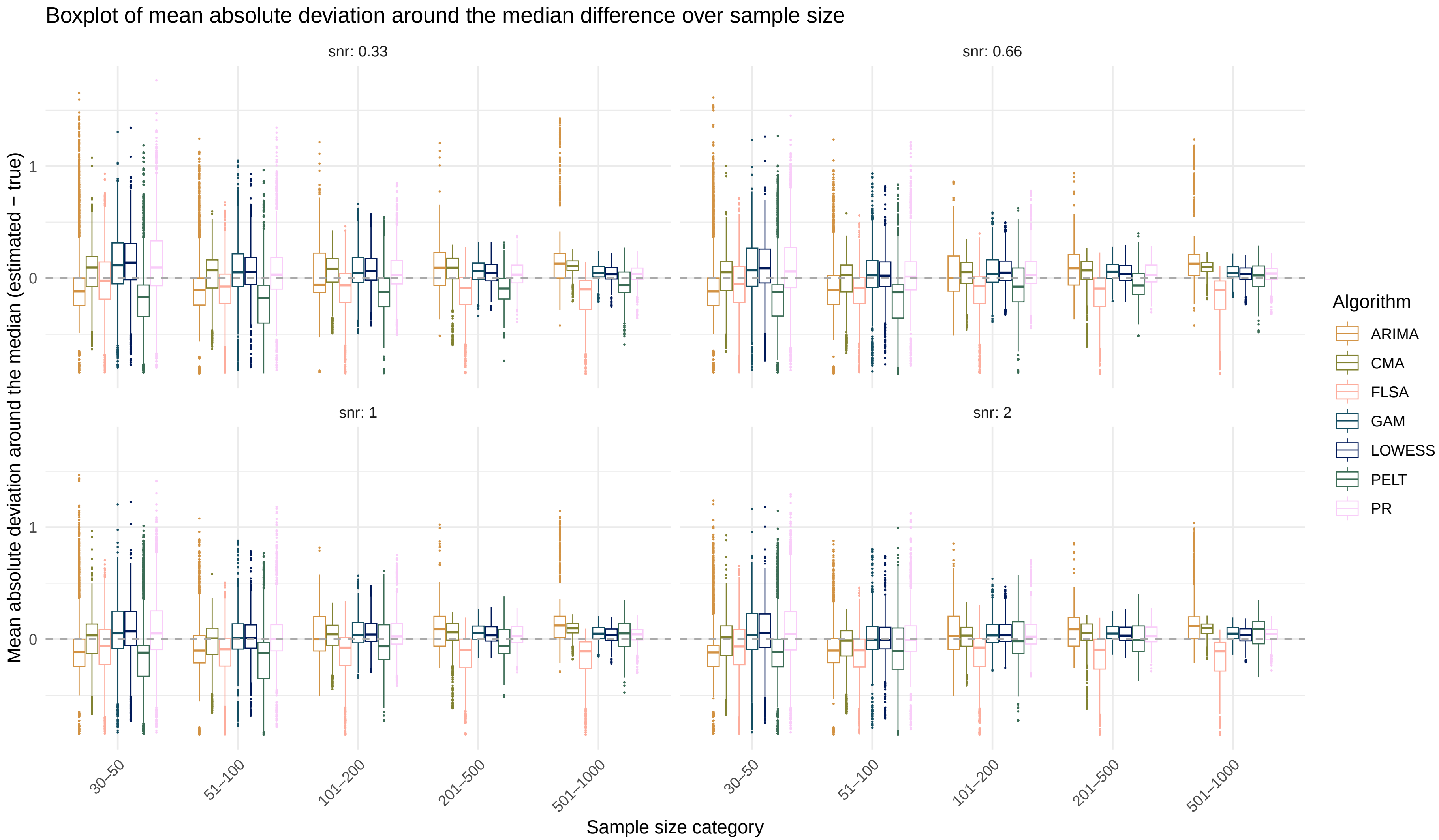




Boxplot of mean absolute deviation around the median difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Mean absolute deviation around the median (estimated − true)
1.5
1.0
0.5
0.0
−0.5
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

Systematic change pattern: All patterns combined

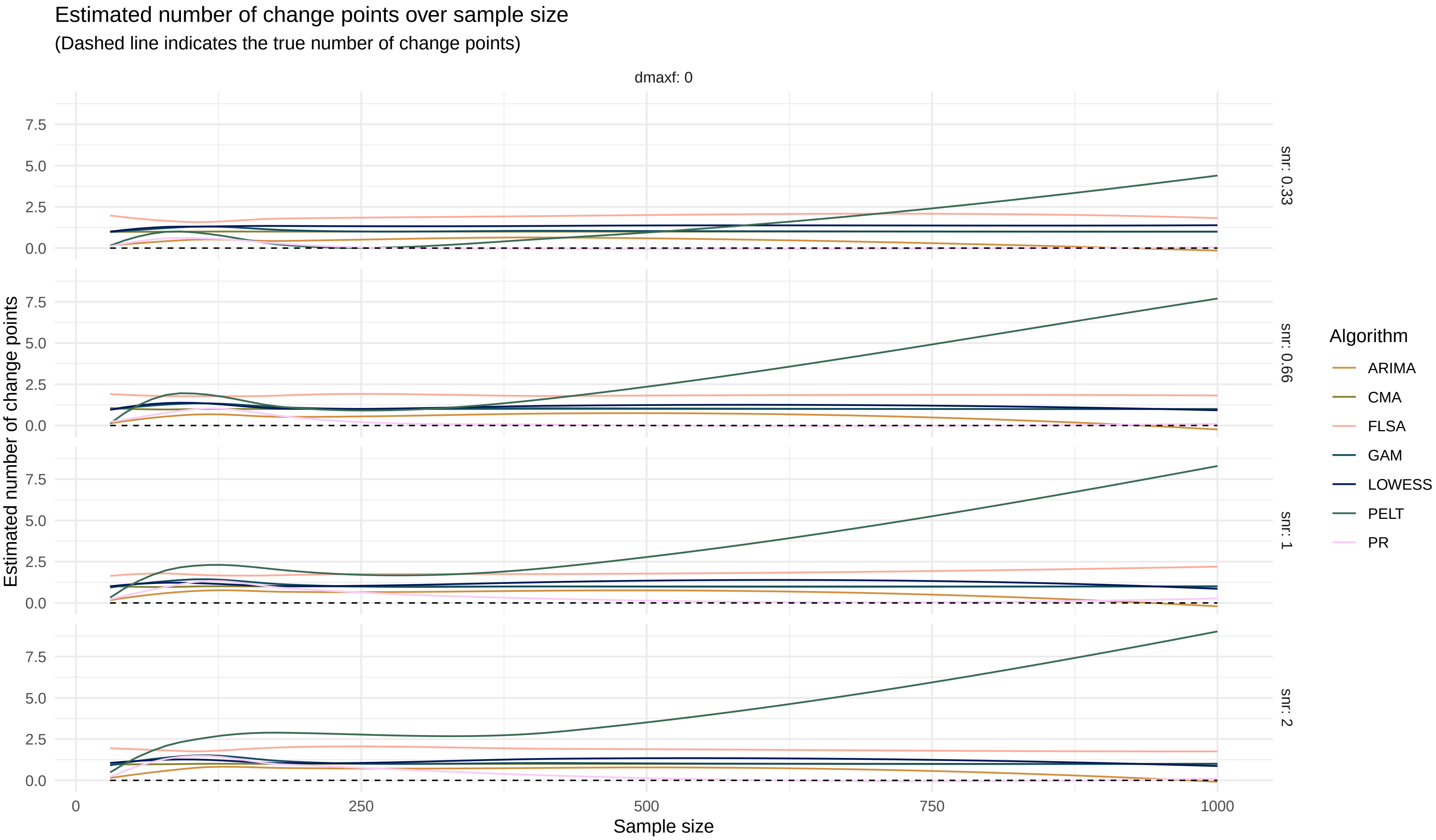


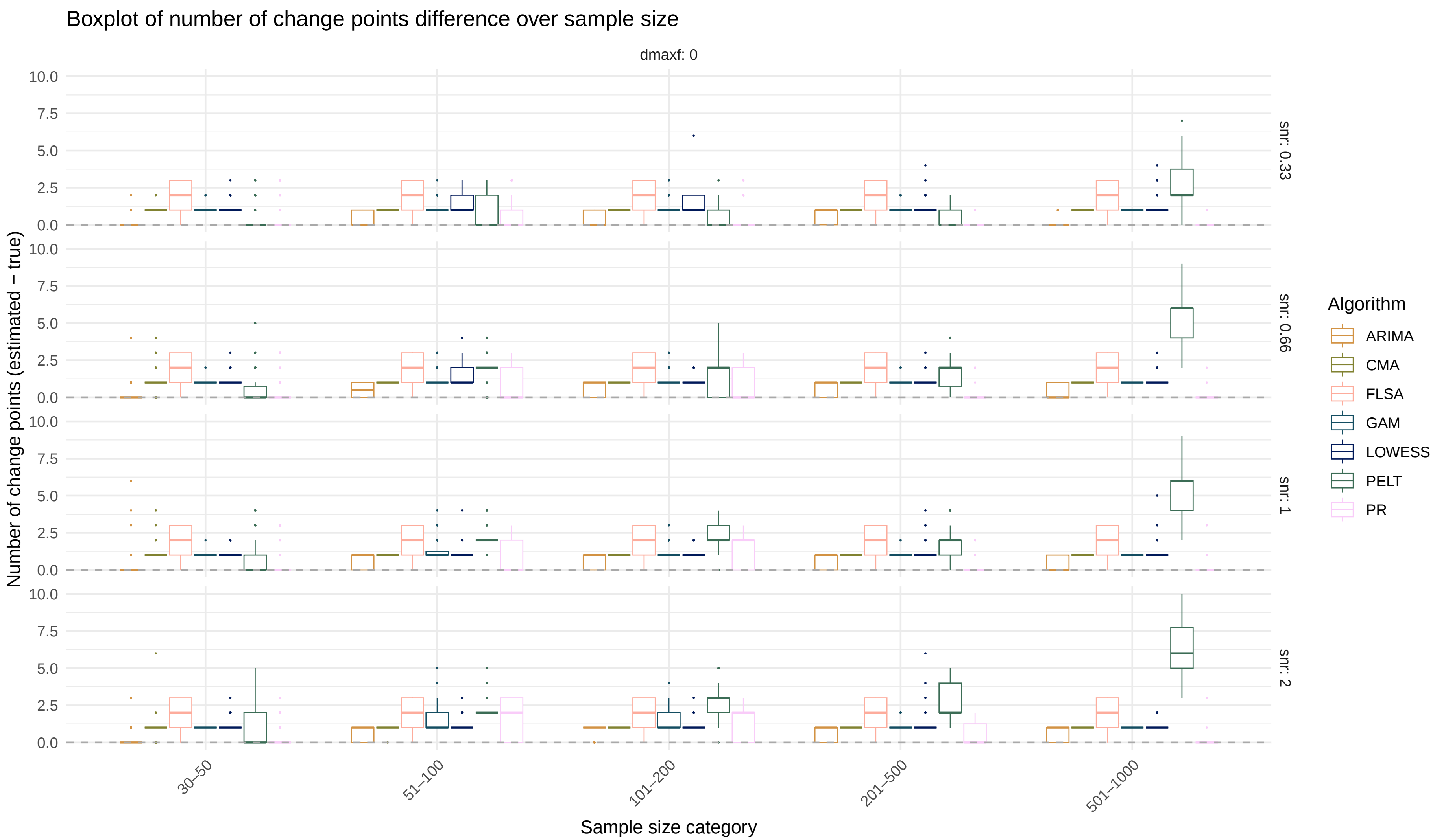


Boxplot of mean absolute deviation around the median difference over sample size
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Mean absolute deviation around the median (estimated − true)
1
0
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

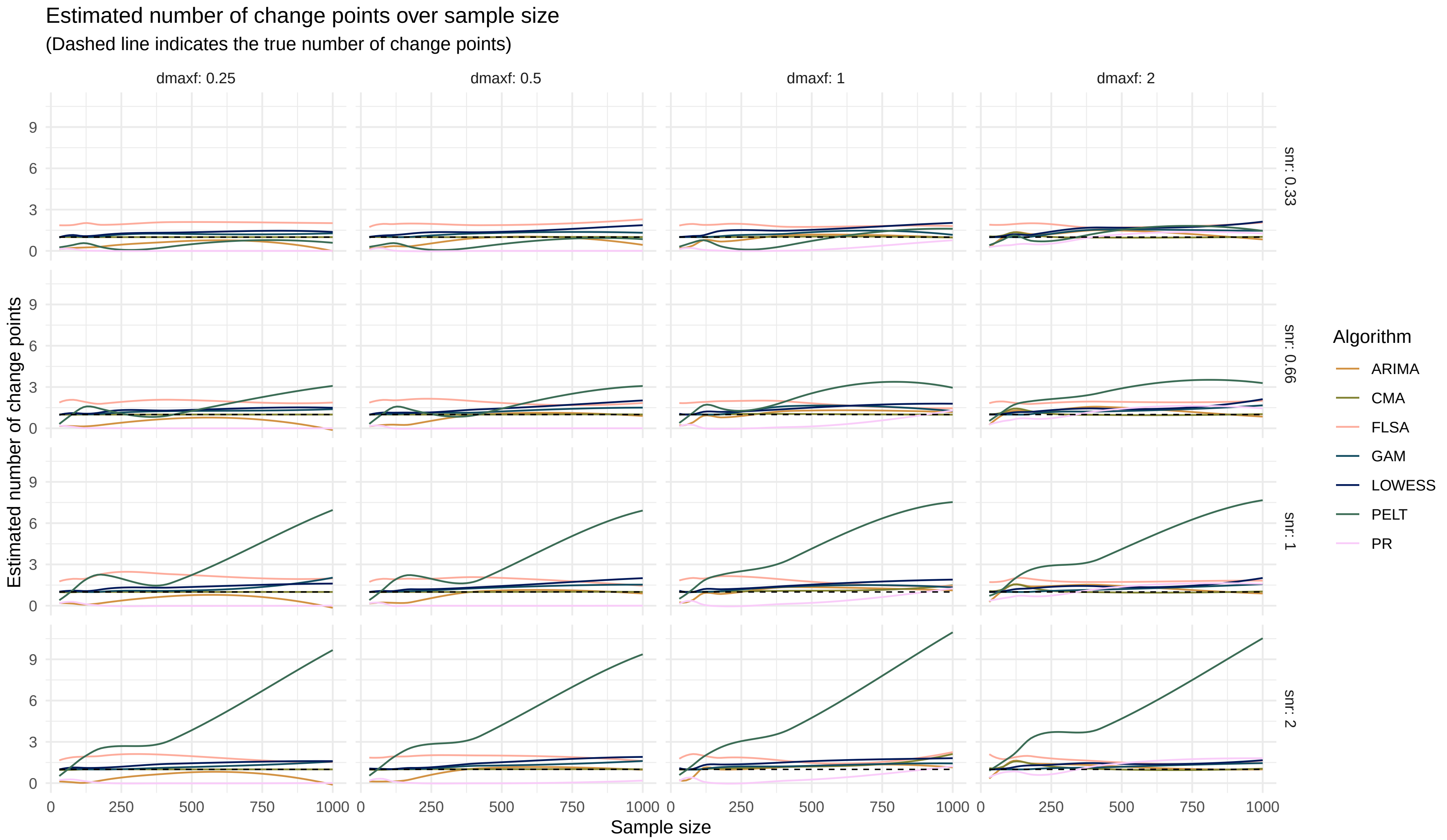
Estimand: Number of change points

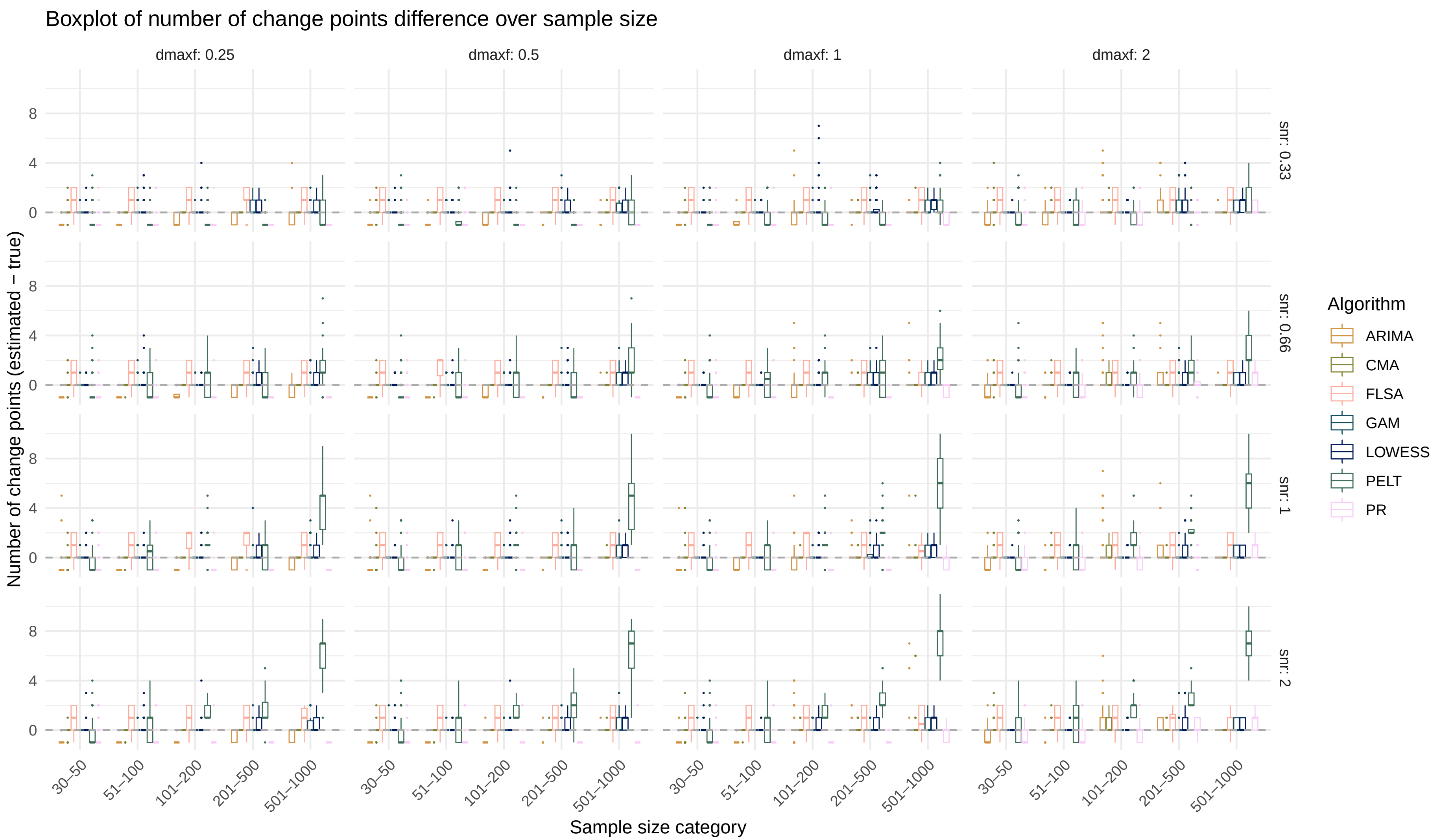
Systematic change pattern: No change



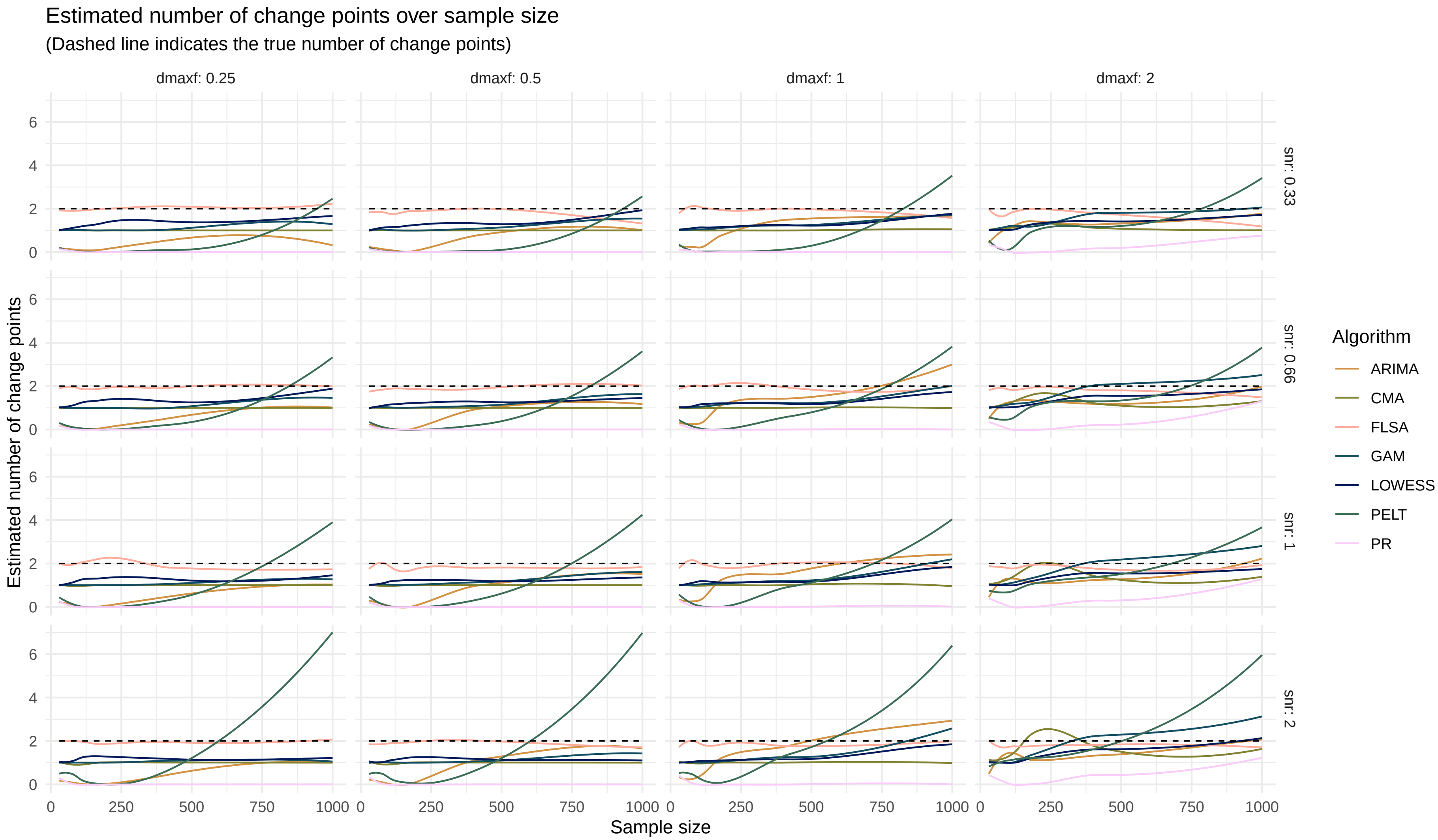

Boxplot of number of change points difference over sample size
dmaxf: 0
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Number of change points (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

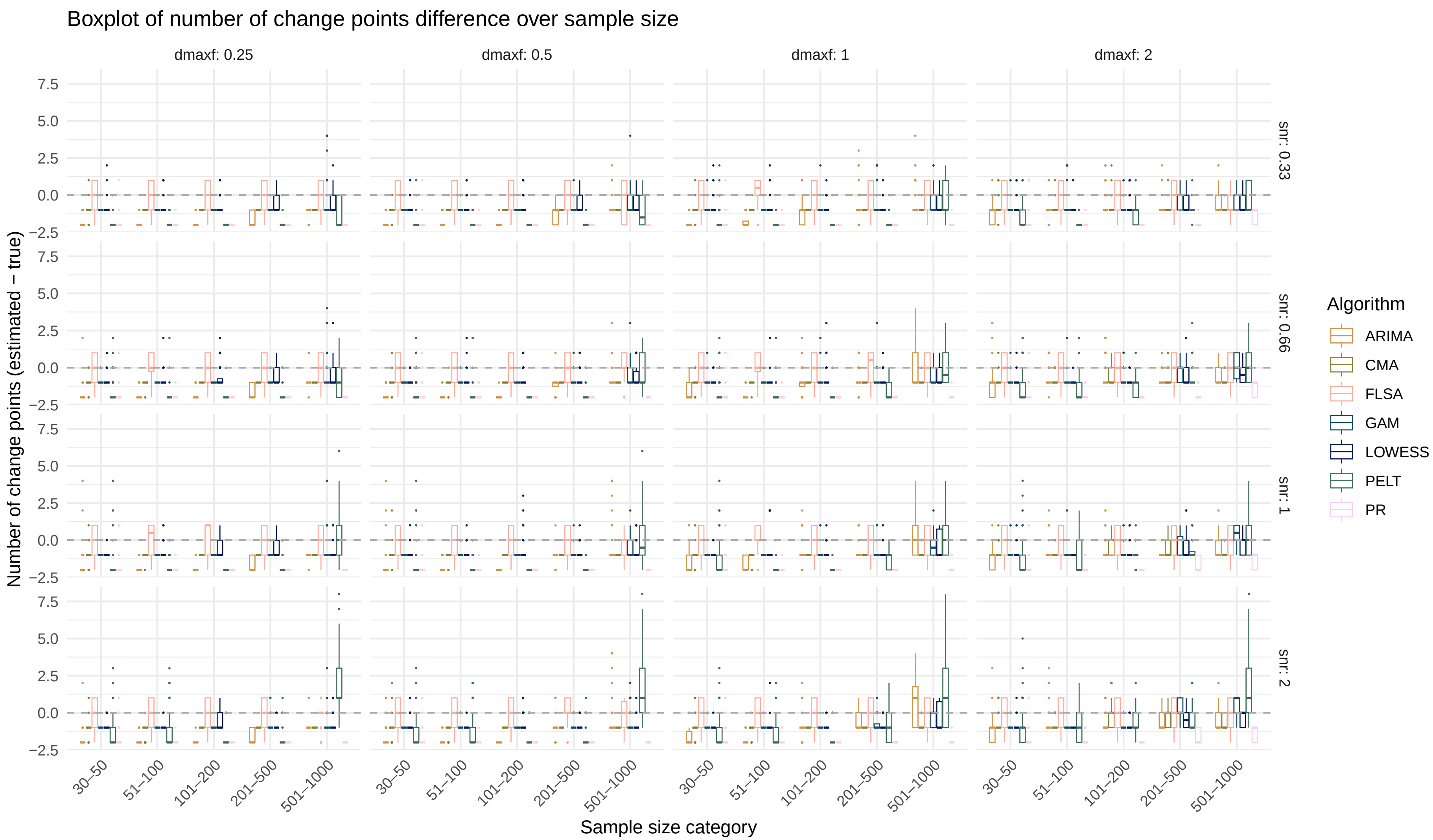
Systematic change pattern: Jump




Boxplot of number of change points difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Number of change points (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

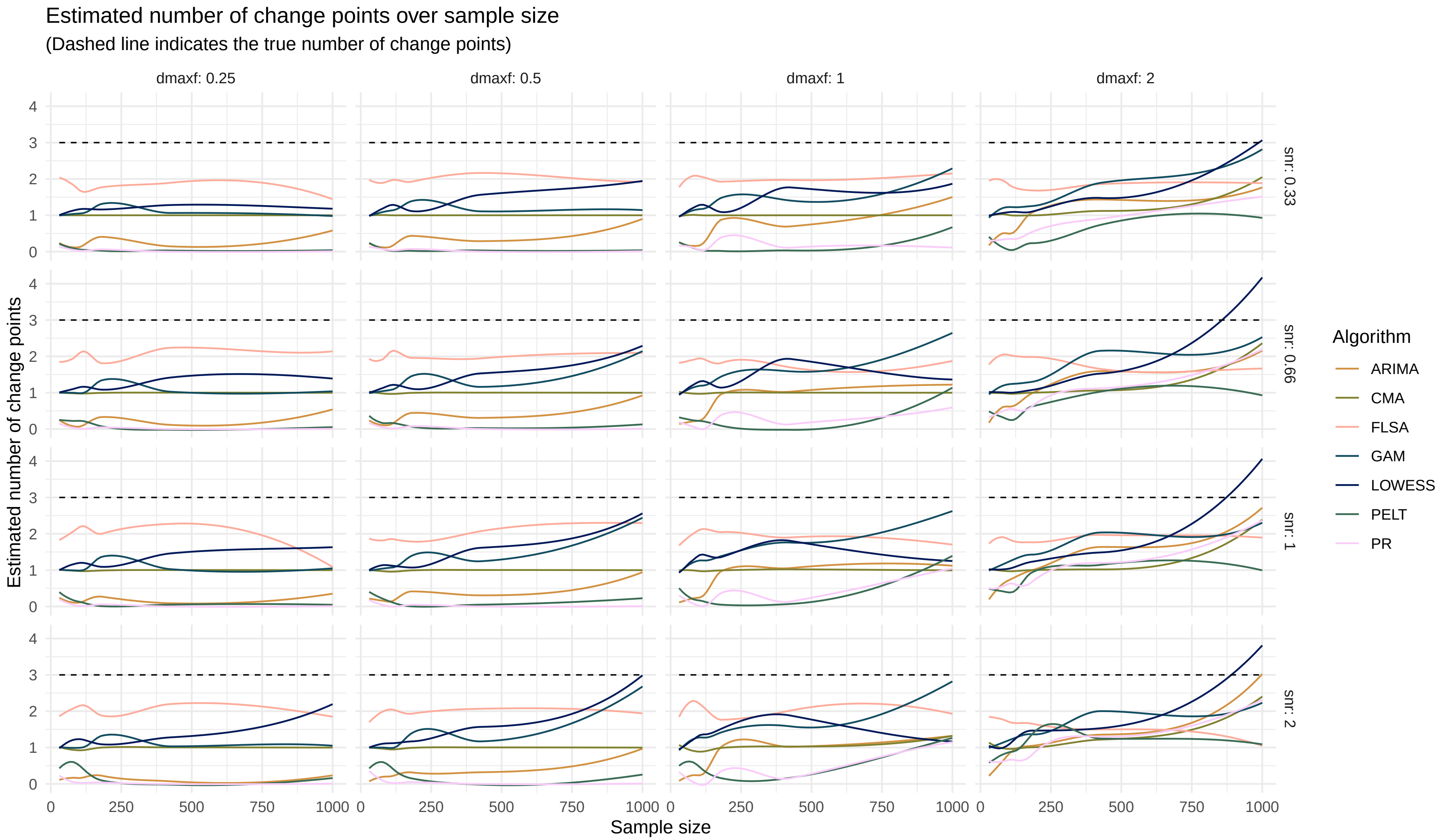
Systematic change pattern: Linear

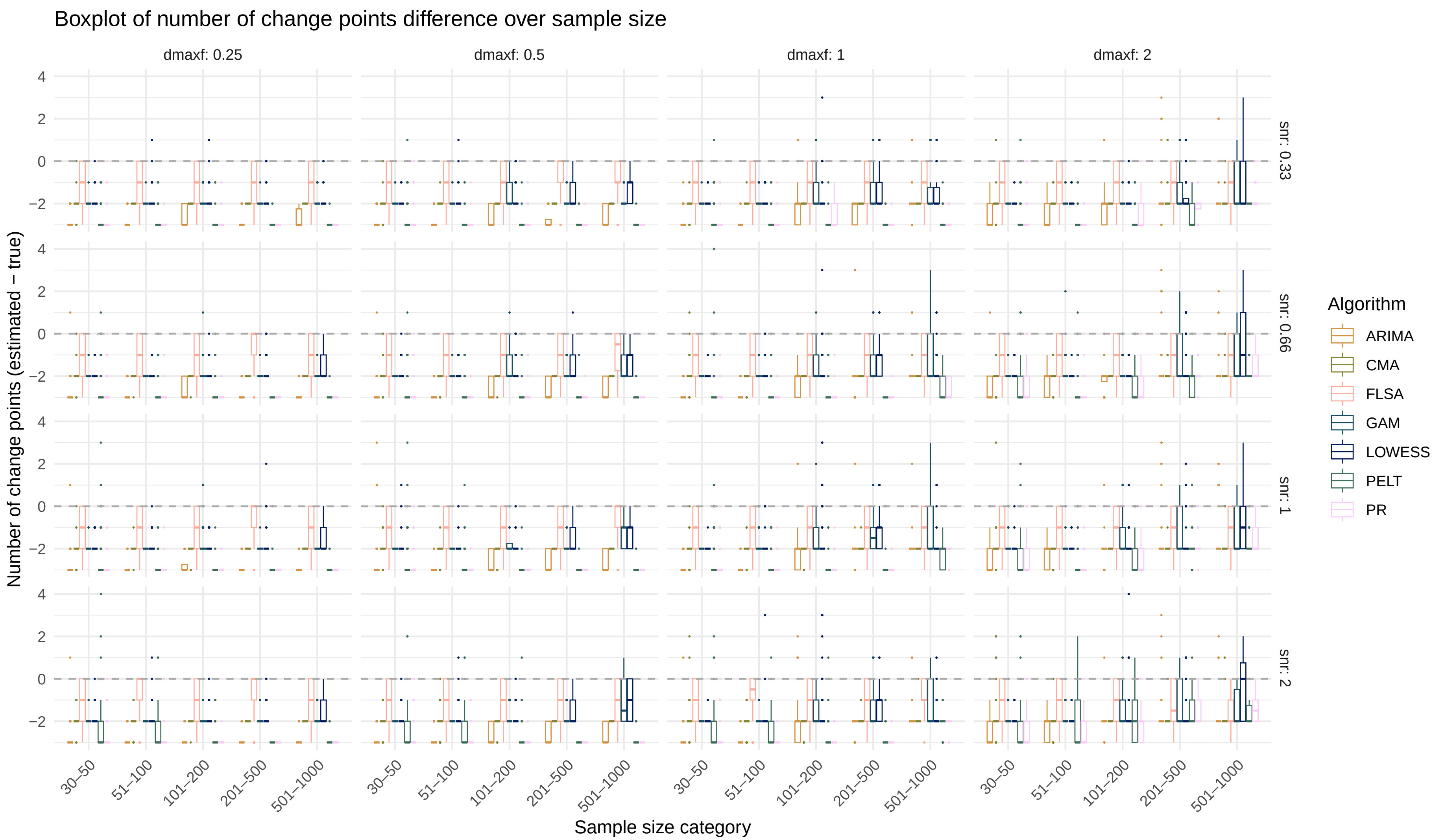




Boxplot of number of change points difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Number of change points (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
7.5
5.0
2.5
0.0
−2.5
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

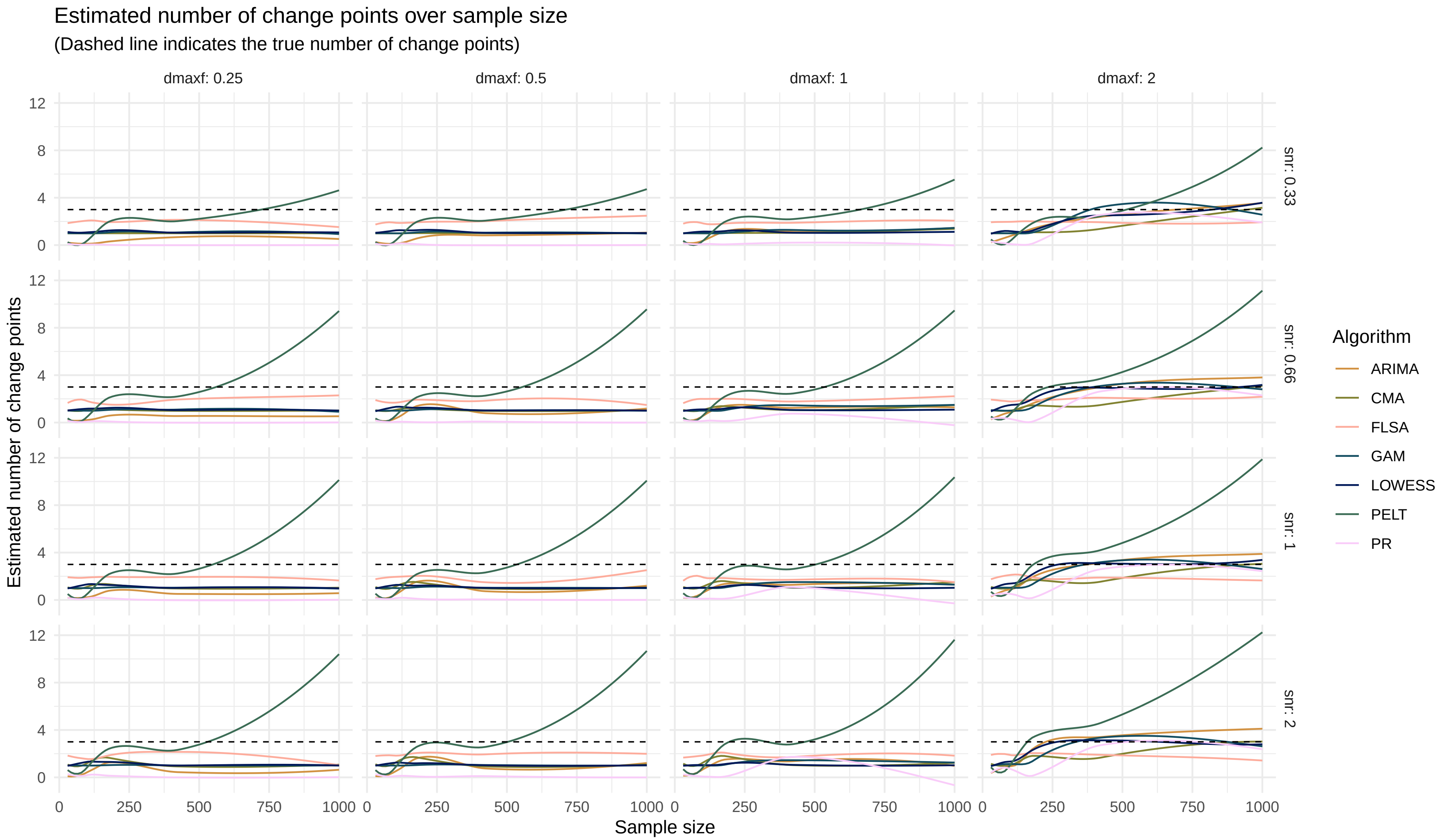
Systematic change pattern: Quadratic

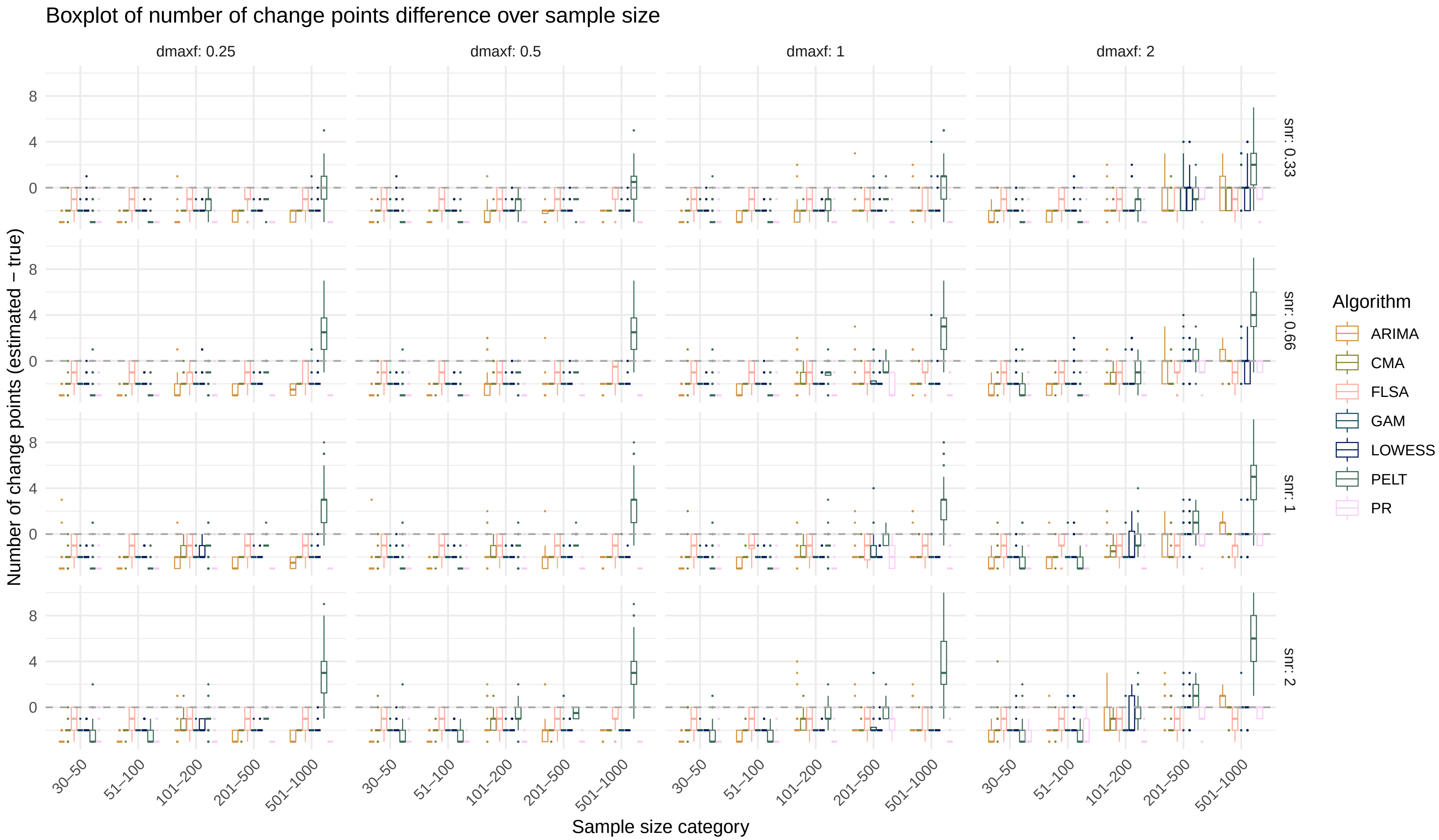




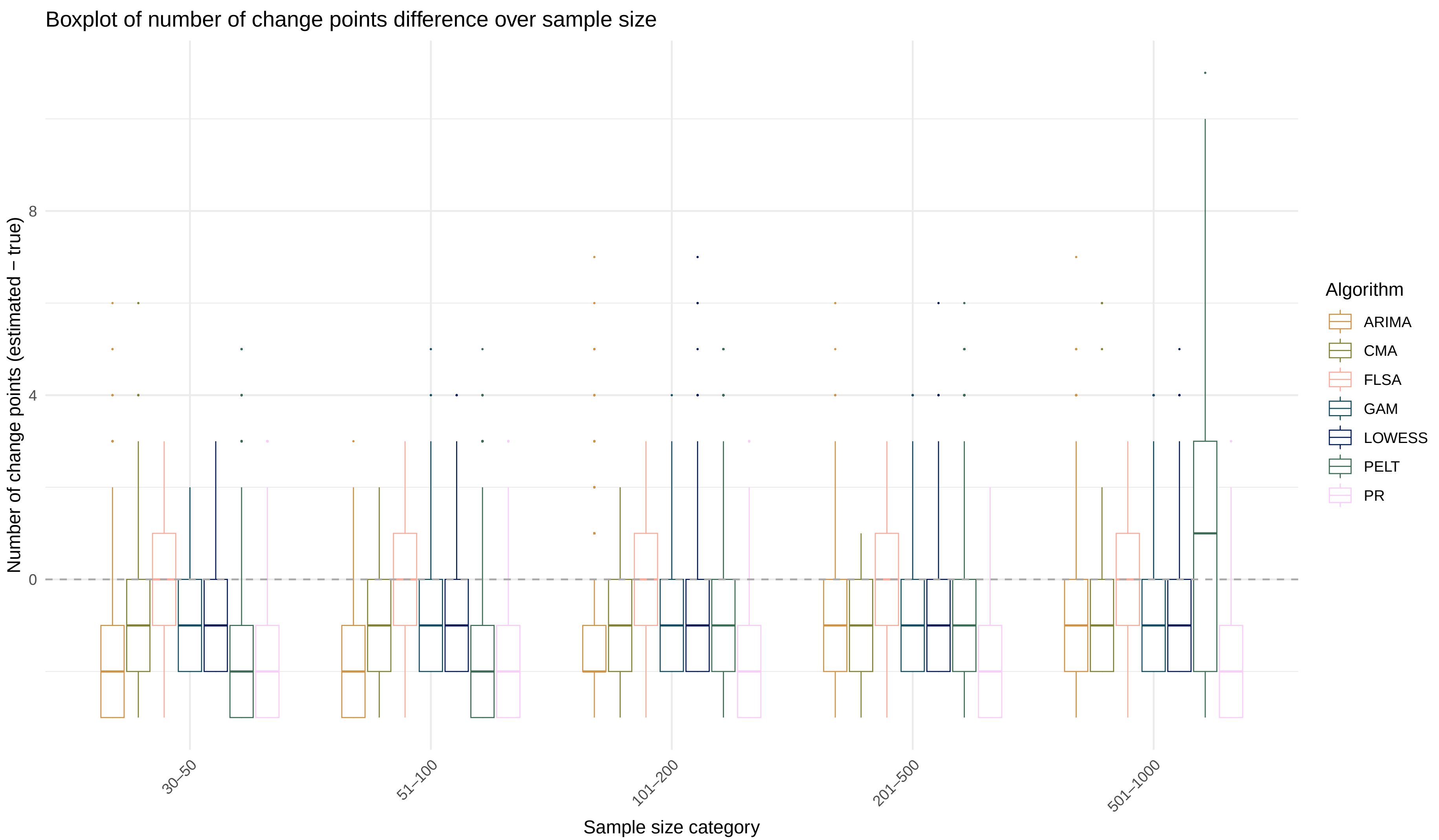
Boxplot of number of change points difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Number of change points (estimated – true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

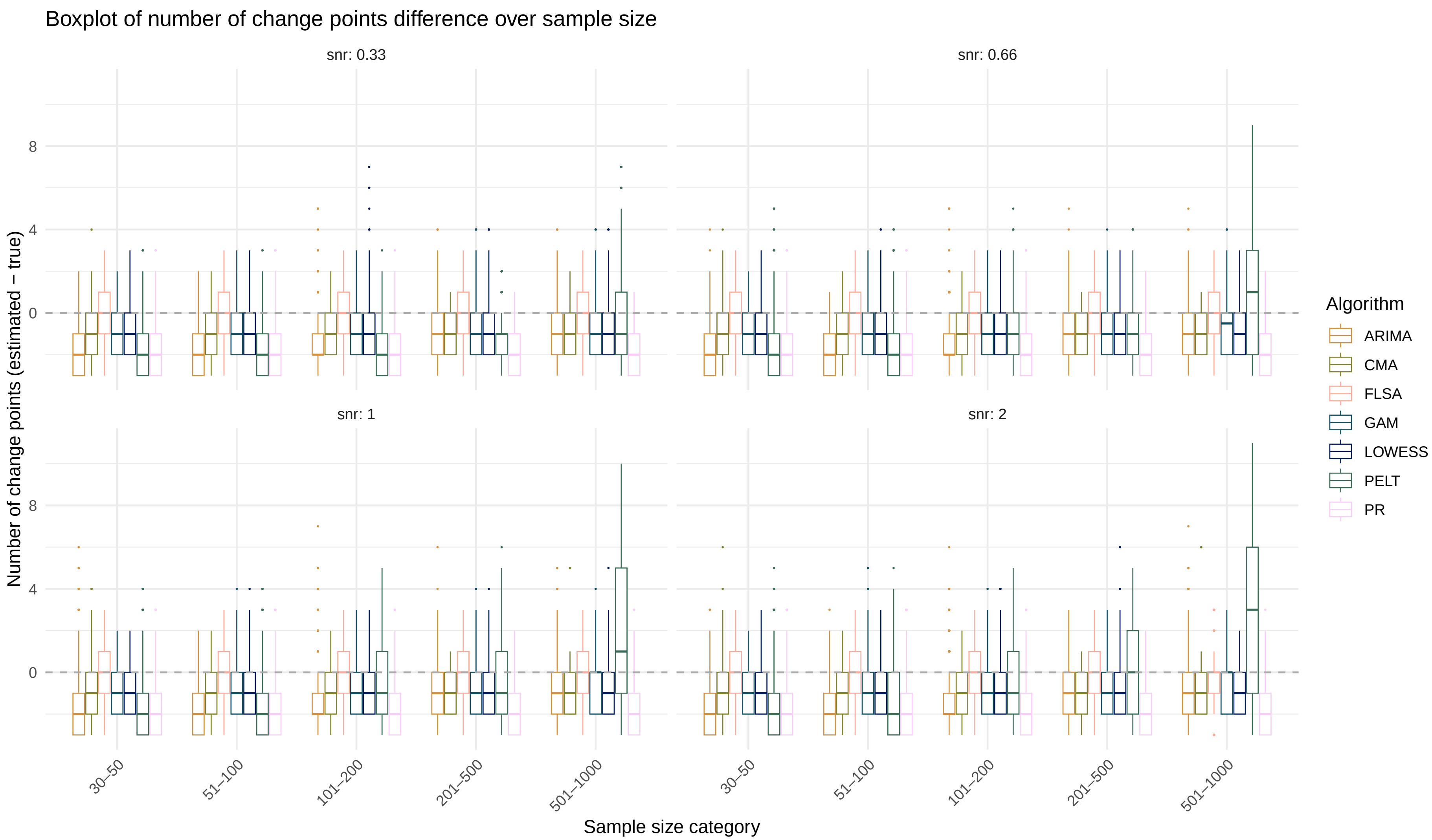
Systematic change pattern: Recurrent




Boxplot of number of change points difference over sample size
dmaxf: 0.25
dmaxf: 0.5
dmaxf: 1
dmaxf: 2
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Number of change points (estimated − true)
Sample size category
30–50
51–100
101–200
201–500
501–1000
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR

Systematic change pattern: All patterns combined





Boxplot of number of change points difference over sample size
snr: 0.33
snr: 0.66
snr: 1
snr: 2
Number of change points (estimated − true)
8
4
0
30–50
51–100
101–200
201–500
501–1000
Sample size category
Algorithm
ARIMA
CMA
FLSA
GAM
LOWESS
PELT
PR